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博士學位論文
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水稻生長相轉換之高通量表現型鑑定技術研究

High-Throughput Phenotyping of Rice Growth Phase Transition

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摘要

為追踪水稻生長相的轉換，本研究建立高通量表現型鑑定（HTP）技術。首先，為改善生育度數（GDD）在探測生殖相初期（ER）的使用，本研究透過討論環境溫度與水稻生長速度之間的關係，提出一種基於 Haun's scale（HAUN），並進行修正的新量測方式，稱為 leaf appearance scale（LAS），以解決 HAUN 的侷限性。相較於 HAUN，LAS 具有連續的每日增加量、穩定的因應氣溫變化的生長速度差異，以及在不同氣候條件下更小的數值差異。其中，每日生長率（d.LAS）被定義為兩個連續日之間 LAS 的差異。而透過不同生長階段中六種類型的每日熱量單位（DTU）與 d.LAS 的線性模式，可估測最合適的基礎溫度（T_{base}）、生長最適溫（T_{opt}）和上限溫度（T_{up}）。結果顯示，在不同生長階段之適當的 T_{base}、T_{opt} 和 T_{up} 均有所不同。而透過簡單線性回歸分析確認每個生長階段中 d.LAS 和 DTU 之間的線性關係，雖然嘗試使用每個生長階段推斷出的 T_{base}、T_{opt} 和 T_{up} 對 DTU 進行調整，但仍無法在季節之間實現 GDD 的一致性。另一方面，當直接使用 GDDs 對生長階段進行分類而不調整每個階段的參數時，兩種 GDDs（GDD_{mc2} 和 GDD_{trial}）產生了一致的值（均為 765°C）。然而，由於 GDDs 的局限性，例如缺乏特定地點的可用性，因此遙測技術被引入後續研究。透過近地高光譜數據推導出之常態化差異植被指數（NDVI）的研究結果指出，六種紅和近紅外（NIR）波長組合的斜率變化最適配 ER 的發生，分別為 NDVI (700,720)，NDVI (700, 730)，NDVI (708, 730)，NDVI (700, 760)，NDVI (708, 760) 和 NDVI (708, 800)。然而，上述分析結果係基於已發生情境分析。因此，本研究進一步嘗試高光譜全光譜分析和機器學習技術。結果顯示，基於特徵萃取的支持向量機（FE-SVM）具備有精確且可靠的結果（accuracy = 0.875）。此外，在此模型中葉片覆蓋面積（LCA）和葉片氮累積量（LNA）在不同的生長階段的階段轉換期間具有顯著的重要性，並為主要變化。雖然基於地面的高光譜方法提供了可靠的檢測結果，但儀器成本相對較高，且檢測面積不足以進行大規模篩選。因此，為克服上述限制，本研究使用無人機（UAV），藉以大規模且大面積的擷取資訊。研究結果指出，透過低成本 RGB 相機獲得的空間作物表面模型（CSMs）中提取百分位數的數據，在水稻生長

階段辨識上，是具有發展潛力的工具（F1-score＝0.85–0.96）。綜合上述，本研究透過環境氣象資料、多光譜及高光譜資料，以及無人載具的 RGB 相機，提出一系列高通量表現型鑑定技術。此技術有望於未來應用於水稻栽培管理中，實現生長階段的精確辨識，提供兼具精確度及精準度的分析技術，以利肥料及農藥的精確施作。

關鍵字：熱量單位、生育度數、高光譜、多光譜、原色相機、機器學習

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Abstract

This study established high-throughput phenotyping (HTP) techniques for tracking the rice growth phase transition. Firstly, the relationship between air temperature and the appearance of rice leaves was investigated to improve the use of growing degree days (GDD) in detecting early reproductive phase (ER). A modified measure called the leaf appearance scale (LAS), based on Haun's scale (HAUN), was proposed to address the issue of non-continuity in HAUN. LAS had advantages over HAUN, including stable daily increment and consistent scale range across different weather conditions. The daily growth rate (d.LAS) was defined as the difference in LAS between two consecutive days. The base temperature (T_{base}), optimum temperature (T_{opt}), and upper limit temperature (T_{up}) were evaluated by analyzing d.LAS using six types of daily thermal units (DTU) during different growth stages. The result showed that the appropriate T_{base}, T_{opt}, and T_{up} were differed across different growth stages. The linearity between d.LAS and DTU in each growth stage was confirmed through simple linear regression analysis. Although the DTUs were attempted to be fine-tuned using the inferred T_{base}, T_{opt}, and T_{up} for each growth stage, they were unable to achieve consistency in GDDs between cold and warm seasons. On the other hand, when the growth stages were directly classified using GDDs without adjusting parameters for each stage, two types of GDDs (GDD_{mc2} and GDD_{trial}) yielded consistent values (both are 765°C) across growing seasons. Due to the limitations of GDDs, such as lack of site-specific availability, the remote sensing technology was introduced to the subsequent works. By applying normalized difference vegetation index (NDVI) derived from ground-based hyperspectral data, the slope changes of six combinations of red and near-infrared (NIR) wavelengths, which are NDVI (700, 720), NDVI (700, 730), NDVI (708, 730), NDVI (700, 760), NDVI (708, 760), and NDVI (708, 800), were found best fitted with ER occurrence. However, this phenomenon was only useful for historical inferences. To encounter with the limitation of real-time detection, further progress was conducted by implementing hyperspectral

full-spectrum analysis and machine learning technology. The result showed that the feature extraction-based support vector machine (FE-SVM) was accurate and reliable (accuracy = 0.875, sensitivity = 0.917, specificity = 0.833, precision = 0.846). Besides, the leaf cover area (LCA) and leaf nitrogen accumulation (LNA) exhibited significant importance in this model, indicating that the LCA and LNA of an individual are the predominant changes during phase transition. Although the ground-based hyperspectral approach provides reliable detection result, the instrumental cost is rather high, and the detection area is insufficient for large-scale screening. Consequently, the unmanned aerial vehicle (UAV) platform was introduced to overcome the limitations. By extracting percentile data from a spatial crop surface model (CSMs) obtained from a low-cost RGB camera is a reliable approach for growth phase discrimination (F1-score = 0.85–0.96).

Keywords: heat units, growing degree days, hyperspectral, multispectral, RGB, machine learning

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Abbreviation

Avg	Average data extraction
B	Blue band
CS	Crop season
CSMs	Spatial crop surface model
CSMt	Temporal crop surface model
d.HAUN	Growth rate (HAUN-based)
d.LAS	Growth rate (LAS-based)
DAT	Days after transplanting
DSM	Digital surface model
DTM	Digital terrain model
DTU	Daily thermal unit
DTU_{mc1}	Method 1 in McMaster and Wilhelm (1997)
DTU_{mc2}	Method 2 in McMaster and Wilhelm (1997)
$DTU_{nielsen}$	Method in Nielsen and Hinkle (1996)
DTU_{trial}	Method in this study
DTU_{zhou3}	The average of HTU_{zhou3} in each day
DTU_{zhou4}	The average of HTU_{zhou4} in each day
ER	Early reproductive stage
FE	Feature extraction
FFE	Flag leaf fully expanded stage
FN	False negative
FP	False positive
FS	Feature selection
G	Green band
GBM	Stochastic Gradient Boosting
GCPs	Ground control points
GDD	Growing degree days
GDD_{mc1}	Accumulation of DTU_{mc1} from 1 DAT
GDD_{mc2}	Accumulation of DTU_{mc2} from 1 DAT

$GDD_{nielsen}$	Accumulation of $DTU_{nielsen}$ from 1 DAT
GDD_{trial}	Accumulation of DTU_{trial} from 1 DAT
GDD_{zhou3}	Accumulation of DTU_{zhou3} from 1 DAT
GDD_{zhou4}	Accumulation of DTU_{zhou4} from 1 DAT
GLM	Generalized linear model
GNDVI	Green NDVI
GP	Gaussian process with radial kernel
HAUN	Haun's scale
HN	200% of conventional nitrogen treatment
HTU	Hourly thermal unit
HTU_{zhou3}	HTU formula in Zhou and Wang (2018) (Method 3)
HTU_{zhou4}	HTU formula in Zhou and Wang (2018) (Method 4)
ICR	Independent Component Regression
KH147	A japonica rice cultivar – Kaohsiung 147
KNN	k-Nearest neighbors
LAS	Leaf appearance scale
LCA	Leaf cover area
LDW	Leaf dry weight
LN	50% of conventional nitrogen treatment
LNA	Leaf nitrogen accumulation
LNC	Leaf nitrogen content
LOOCV	Leave-one-out cross validation
LV	Late vegetative phase
MaxTill	Maximum tillering stage
MLP	Multiple layer perceptron
MLP	Multilayer perceptron
MN	100% of conventional nitrogen treatment
MS	Multispectral
mtry	Number of variables
NB	Naïve bayes
NDRE	Normalized differential red-edge

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NDSI	Normalized difference spectral index
NDTI	Normalized difference texture index
NDVI	Normalized difference vegetation index
NExG	Normalized excess green index
NGRDI	Normalized green-red difference vegetation index
NIR	Near-infrared band
ntree	Number of trees
PC	Principal component
PCA	Principal component analysis
PCR	Principal component regression
PD	Panicle differentiation
Perc	Percentile data extraction
PI	Panicle initiation
PLR	Penalized Logistic Regression
PLSR	Partial least square regression
P_n	The nth percentile
PVE	Proportion of variance explained
R	Red band
RDA	Regularized Discriminant Analysis
RE	Relative error
RE	Red edge band
Rep	Reproductive phase
RF	Random forest
RIDGE	Ridge regression
RRMSE	Relative root-mean-square error
SPAD	Leaf chlorophyll readings
SVM	Support vector machine
SVM-RBF	Support vector machine (radial basis kernel)
T_{base}	Base temperature
T_h	Hourly temperature records
T_{max}	Maximum air temperature

T_{mean}	Average air temperature
T_{min}	Minimum air temperature
TN	True negative
T_{opt}	Optimal temperature
TP	True positive
T_{up}	Upper threshold temperature
UAV	Unmanned aerial vehicle
VARI	Visible atmospherically resistant index
Veg	Vegetative phase
VI	Vegetation index
XGB	eXtreme gradient boosting

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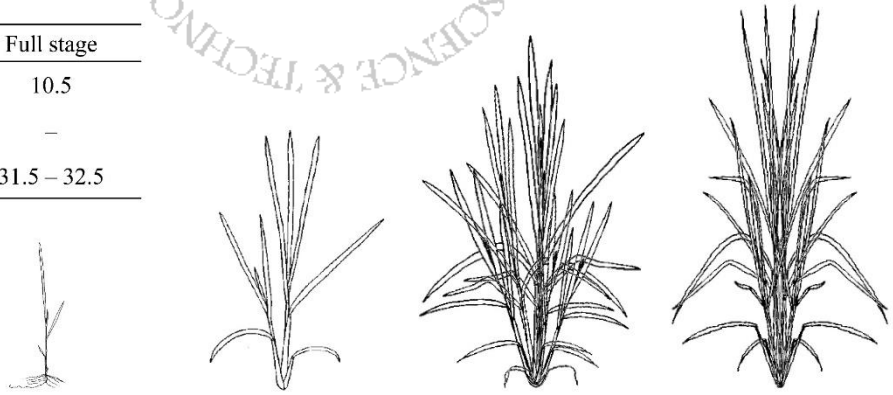
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Chapter 1. Fundamental study on growing degree days (GDD) and rice leaf appearance

Highlights

- Leaf appearance scale (LAS) is more suitable for rice growth rate measurement than the Haun's scale (HAUN).
- Base temperature (T_{base}) for seedling, tillering, maximum tillering (MaxTill), and early reproductive (ER) stage are respectively 13.0℃, 8.5℃, 17.0℃, and 10.0℃.
- Optimal temperature (T_{opt}) for seedling, tillering, MaxTill, and ER are respectively 28.5℃, 40.0℃, 30.5℃, and 32.0℃.
- Upper limit temperature (T_{up}) for seedling, tillering, MaxTill, and ER are respectively 50.0℃, 50.0℃, 34.5℃, and 32.5℃.
- Optimal T_{base} and T_{up} for all stages calculated with DTU_{mc2} or DTU_{trial} are respectively 10.5℃ and 31.5 – 32.5℃.

	Full stage
T_{base} (℃)	10.5
T_{opt} (℃)	–
T_{up} (℃)	31.5 – 32.5



	Seedling	Tillering	MaxTill	ER
T_{base} (℃)	13.0	8.5	17.0	10.0
T_{opt} (℃)	28.5	40.0	30.5	32.0
T_{up} (℃)	50.0	50.0	34.5	32.5

(Plant sketches were adopted and modified from Counce et al., 2000)

1.1. Introduction

From an agricultural perspective, rice growth can be defined as the irreversible increase in physical dimensions of either an individual or a specific organ (Wilhelm and McMaster, 1995). On an individual scale, growth could be measured by the permanent increase in volume and dry weight. On an organ scale, growth could be measured by the irreversible expansion of leaf blade tissue. The volume and dry weight of an individual are usually measured using destructive approaches while leaf blade measurement is typically non-destructive. When resources are limited, the latter can be a competent way to quantify rice growth.

The growth and development of rice are usually defined with several key features that are agronomically important, such as the beginning of tillering, the end of tillering, the initiation of panicle, and the end of leaf exertion. Several growth staging systems had been proposed to clarify these critical events in cereal plants, such as Zadok scale (Zadoks et al., 1974), IRRI scale (IRRI, 1980), BBCH system (Lancashire et al., 1991), and Counce system (Counce et al., 2000). However, these scaling systems are qualitative. To better understanding and analyzing plant growth and development, it is suggested to involve the calculation of growth rate at more frequent interval of time to obtain the continuous trace of plant changes (Newman and Beard, 1962; Radford, 1967; Haun, 1973).

The Haun's scale (HAUN) is one of the most commonly used measures to quantify the growth and development of cereal crop (Haun, 1964). The basis of this scaling is that the development of cereal plants is highly coordinated with shoot apex development, panicle initiation, and stem elongation (Kirby, 1990). The integration of external appearance and internal changes provides possibilities to predict plant development non-destructively. However, applying HAUN on rice growth quantification seemed to be inappropriate due to two limitations. The first one is, there were a portion of preceding leaves still expanding when the latter ones started emerging. In this situation, the HAUN is substantially insensitive to the actual growth amount. The second is, when

the leaf length of the leading leaf exceeds the preceding one, the carry digit could be confounding and misleading. When investigating the growth rate of rice in daily basis, these limitations raise unnecessary noises. Therefore, the primary step of this work is to establish a more suitable scale for rice growth measurement.

On the other hand, using leaf scaling to precisely measure rice growth is inefficient for field practices. To simplify the decision-making process for farmers, a simple and effective method called calendar-days was invented by predecessors. By tracking weather changes, farmers can easily predict crop development and manage their cultivation schedules year on year. However, this type of cultivation model highly depends on stable weather conditions (Gilmore and Rogers, 1958). Due to climate change, applying calendar-days in rice cultivation may now be ineffective.

Since 1730 when the concept of heat units was introduced by Reaumur (McMaster and Wilhelm, 1997), the description and prediction of crop development progress have been changed from the solid calendar-days to a relatively abstract measure. This concept has been widely used in agricultural sciences and has proven to be satisfactory for practical needs. The heat units of rice are usually expressed in growing degree days (GDD), which is a cumulative expression of daily thermal unit (DTU). The canonical form of the DTU is the average of the maximum and minimum daily air temperature (McMaster and Wilhelm, 1997; Zhou and Wang, 2018). Upon this basis, the base temperature (T_{base}) and upper threshold temperature (T_{up}) at which plant stops growing are also implemented into the equation to depict the floor and ceiling of plant growth rate in nature.

The relationship between air temperature and DTU within the interval of T_{base} and T_{up} could be linear or nonlinear, depends on different formulae (Nielsen and Hinkle, 1996; McMaster and Wilhelm, 1997; Zhou and Wang, 2018). By calculating DTU with an appropriate formula designation, it is assumed to be linearly related with the rice growth rate. The primary objective of this work is to investigate the relationship between rice growth rates and

different derivatives of DTU. By retrieving the growth rate with DTU, it is possible to estimate the suitable DTU formula associating with the optimal T_{base} and T_{up} with the best fit of the simple linear regression model. Furthermore, the feasibility of classifying growth stages with GDD is also tested to acquire a single set of optimal temperature parameters for convenient in-field application.

1.2. Materials and method

1.2.1. Study site and materials

An outdoor pot experiment was conducted at the National Pingtung University of Science and Technology during warm season in 2019 and during cold season in 2020. A japonica rice cultivar, Kaohsiung147 (KH147), was planted in 30 pots, with five seedlings per pot. Ten of these pots were used for data acquisition, while the remaining 20 were used for destructive sampling to examine the ER stage.

1.2.2. Meteorological data collection

Air temperature data was obtained from a meteorological station located approximately seven kilometers from the experimental site. The maximum (T_{max}), minimum (T_{min}), and average (T_{mean}) air temperature values were calculated from the station's hourly temperature records.

1.2.3. Growth stage determination

The growth stages of the plant are divided into five segments: seedling, tillering, maximum tillering (MaxTill), early reproductive (ER), and flag leaf fully expanded (FFE) stages. The seedling stage occurs between transplanting and tillering, while the tillering stage is reached when 50% of the tillers have emerged from the transplanted seedlings. The MaxTill stage is determined when the number of tillers stops increasing. The ER stage is defined through anatomical observation referring to the preceding report (Lai and Lin, 2021b). Finally, the FFE stage is reached when the leaf age stops increasing, and the flag leaf is fully expanded.

1.2.4. Leaf age and growth rate

The Haun's scale (HAUN) is the most commonly used method for counting the leaf age of rice plants. The leaf appearance scale (LAS), which is a modification of the HAUN, is defined as follows (Equation 1-1):

$$LAS = \frac{\sum_{i=1}^{i_{max}} \frac{L_i}{max(L_i)}}{i_{max}} \qquad \text{(Equation 1-1)}$$

The i is the leaf enumerator and the i_{max} is the leaf enumerator of the flag leaf. The L_i represents the current leaf length of the ith leaf. The $max\ (L_i)$ represents the maximum leaf length of the ith leaf.

The growth rate is defined as below (Equation 1-2):

$$d.LAS_j = \frac{LAS_{j+1} - LAS_j}{DAT_{j+1} - DAT_j} = LAS_{j+1} - LAS_j \qquad \text{(Equation 1-2)}$$

The $d.LAS_j$ is the growth rate in j^{th} DAT, which is calculated by taking the difference in leaf length between two consecutive days. Since the LAS is measured daily, the denominator is consistently equal to one.

1.2.5. Growing degree days (GDD)

The daily thermal unit (DTU) is a measure of the heat-driven forces that drive daily plant growth. It is calculated as the average of the maximum and minimum temperatures for each day. The growing degree days (GDD) is defined as follow (Equation 1-3):

$$GDD = \sum_{j=1}^{j_{max}} DTU = \sum_{j=1}^{j_{max}} \frac{(T_{max} + T_{min})_j}{2} \qquad \text{(Equation 1-3)}$$

The $[(T_{max} + T_{min})_j / 2]$ is the average of the maximum and minimum temperature in the jth DAT. Based on the fundamental DTU, five derived methods were evaluated as follows:

1.2.5.1. Daily based methods

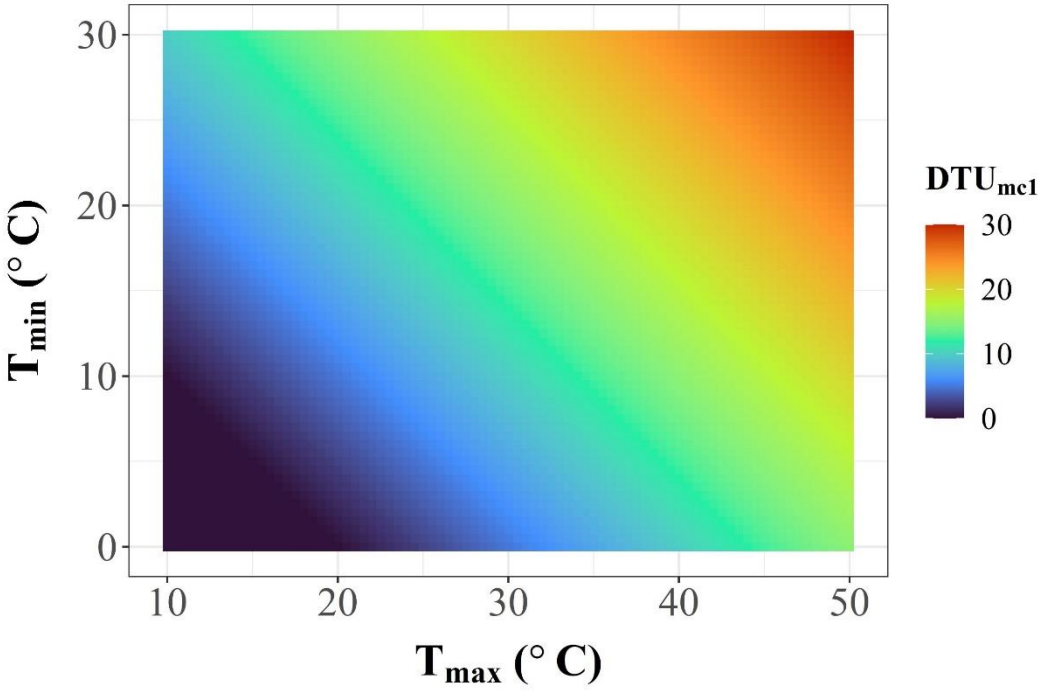
McMaster and Wilhelm (1997) proposed two methods for calculating the DTU, both of which consider the limits of the base temperature and the upper threshold temperature. The first method, known as DTU_{mc1}, sets a limit on the overall DTU (as shown in Equation 1-4), while the second method, DTU_{mc2}, handles the maximum and minimum temperatures more precisely before calculating the DTU (as shown in Equation 1-5). Both DTU_{mc1} and DTU_{mc2} remain linear until the threshold is met. (Figure 1-1).

$$DTU_{mc1} = \begin{cases} 0 & ;\ DTU < T_{base} \\ DTU - T_{base} & ;\ T_{base} < DTU < T_{up} \\ T_{up} - T_{base} & ;\ T_{up} < DTU \end{cases} \qquad \text{(Equation 1-4)}$$

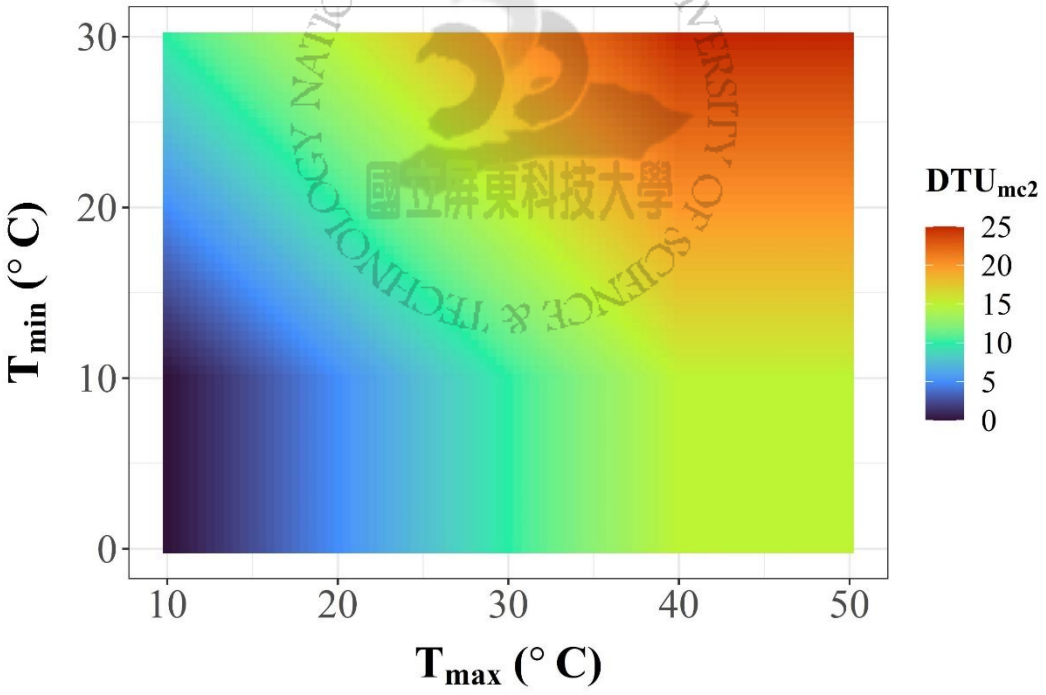
$$DTU_{mc2} = \begin{cases} 0 & ;\ DTU' < T_{base} \\ DTU' - T_{base} & ;\ T_{base} < DTU' < T_{up} \\ T_{up} - T_{base} & ;\ T_{up} < DTU' \end{cases} \qquad \text{(Equation 1-5)}$$

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The DTU_{mc1} and DTU_{mc2} are referring to the method 1 and method 2 in the report of McMaster and Wilhelm (1997). The T_{base} and T_{up} are referring to the base temperature and upper threshold temperature that cause the rice plant stops growing. Moreover, the DTU' is calculated by $[(T_{max}' + T_{min}') / 2]$, where $T_{max}' = min\ (T_{max}, T_{up})$ and $T_{min}' = max\ (T_{min}, T_{base})$.



(a)



(b)

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Figure 1-1. Illustration of (a) DTU_{mc1} and (b) DTU_{mc2} calculated with different combinations of T_{min} and T_{max}. The T_{base} and T_{up} were set at 10°C and 40°C, respectively.

The $DTU_{nielsen}$ (Equation 1-6) is referring to the 'heat stress method' that was proposed by Nielsen and Hinkle (1996). For this method, the amount that the T_{max} exceeds the upper threshold was subtracted from the upper threshold value (Nielsen and Hinkle, 1996). This manipulation results in non-linear changes of the $DTU_{nielsen}$ when T_{up} was met (Figure 1-2a).

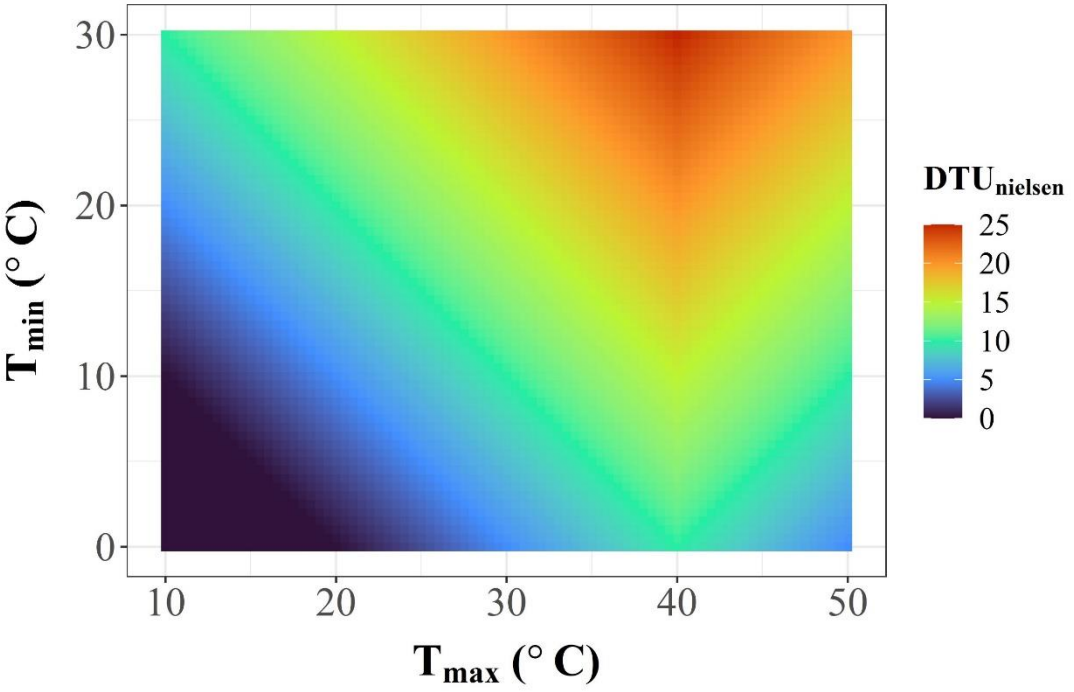
$$DTU_{nielsen} = \begin{cases} 0 & ;\ DTU^{\S} < T_{base} \\ DTU^{\S} - T_{base} & ;\ T_{base} < DTU^{\S} < T_{up} \end{cases} \quad \text{(Equation 1-6)}$$

The $DTU^{\S}$ is calculated by $[(T_{max}{}^{\S} + T_{min}) / 2]$, where $T_{max}{}^{\S} = min\ [T_{up} - (T_{max} - T_{up}), T_{max}]$. When the $DTU^{\S}$ is lesser than T_{base}, the $DTU_{nielsen}$ is directly denoted as zero; when the $DTU^{\S}$ is between T_{base} and T_{up}, the $DTU_{nielsen}$ is calculated via subtracting T_{base} from the $DTU^{\S}$.

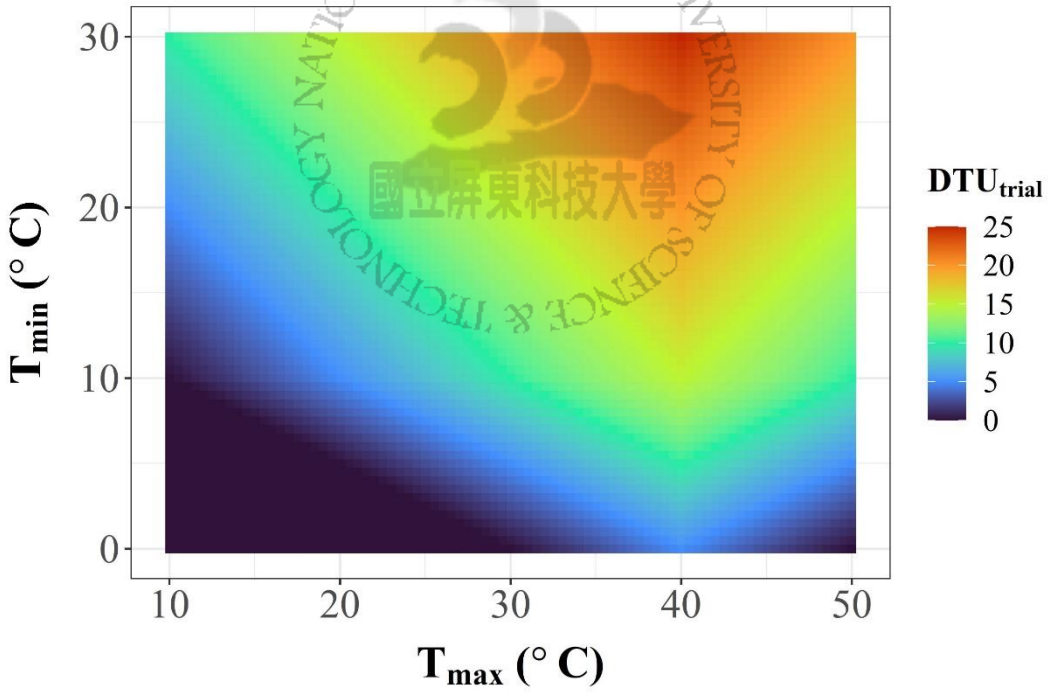
The DTU_{trial} (Equation 1-7) is an extension of the $DTU_{nielsen}$. This method implements 'cold stress' response into the formula. The only difference from the $DTU_{nielsen}$ is that the amount where the T_{min} below the T_{base} was subtracted from the T_{base} value. The nonlinearity changes of the DTU_{trial} occur when either the T_{base} or T_{up} was met (Figure 1-2b).

$$DTU_{trial} = \begin{cases} 0 & ;\ DTU^{\dagger} < T_{base} \\ DTU^{\dagger} - T_{base} & ;\ T_{base} < DTU^{\dagger} < T_{up} \end{cases} \quad \text{(Equation 1-7)}$$

The $DTU^{\dagger}$ is calculated by $[(T_{max}{}^{\S} + T_{min}{}^{\S}) / 2]$, where $T_{max}{}^{\S} = min\ [T_{up} - (T_{max} - T_{up}), T_{max}]$ and $T_{min}{}^{\S} = min\ [T_{min} - (T_{base} - T_{min}), T_{min}]$.



(a)



(b)

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Figure 1-2. Illustration of (a) $DTU_{nielsen}$ and (b) DTU_{trial} calculated with different combinations of T_{min} and T_{max}. The T_{base} and T_{up} were set at 10℃ and 40℃, respectively.

1.2.5.2. Hourly based methods

The hourly based methods investigate the changes of temperature in hourly basis, which is termed as hourly thermal unit (HTU). An assumption of optimal temperature (T_{opt}), which is the temperature point that growth rate is maximized, is introduced into the formulae (Ritchie and Nesmith, 1991; Aiken, 2005; Wang et al., 2017). The HTU_{zhou3} (Equation 1-8), which is the 'method 3' in Zhou and Wang (2018), introduces linearity assumption between T_{base} and T_{opt} and between T_{opt} and T_{up} (Figure 1-3a). Similarly, the HTU_{zhou4} introduces T_{opt} yet assuming nonlinearity relationship using a beta-distribution function to describe the value changes between T_{base} and T_{opt}, and between T_{opt} and T_{up} (Figure 1-3b) (Zhou and Wang, 2018).

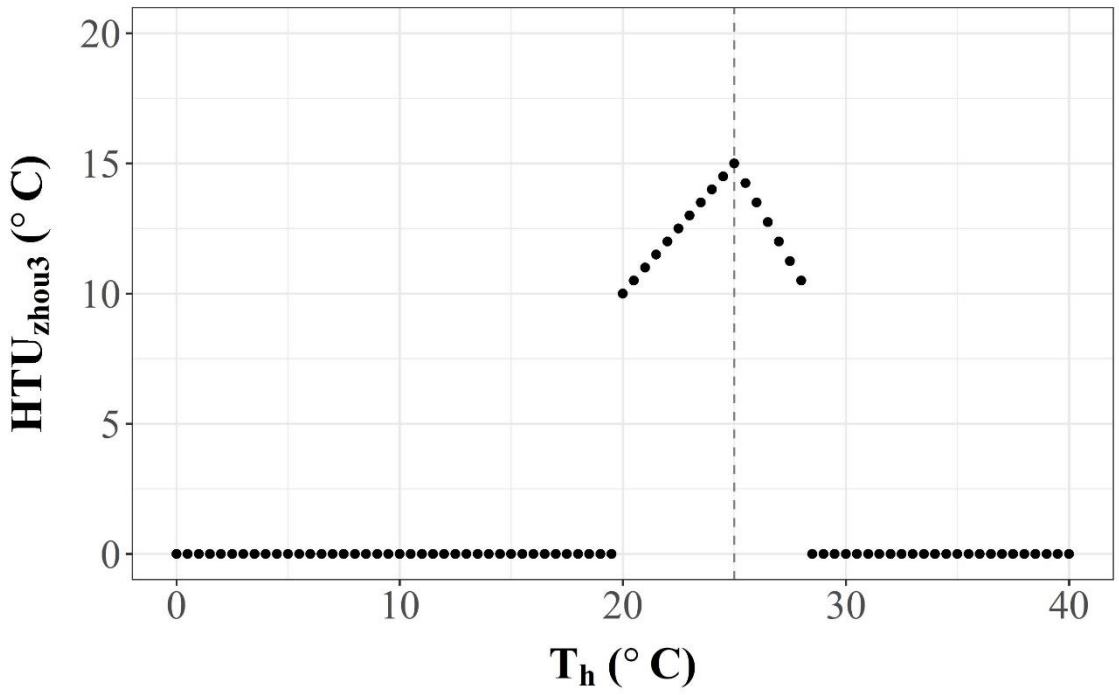
$$HTU_{zhou3} = \begin{cases} 0 & ;\ T_h < T_{base} \\ T_h - T_{base} & ;\ T_{base} \le T_h \le T_{opt} \\ \frac{T_{opt}-T_{base}}{T_{up}-T_{opt}}\left(T_{up} - T_h\right) & ;\ T_{opt} < T_h \le T_{up} \\ 0 & ;\ T_{up} < T_h \end{cases} \quad \text{(Equation 1-8)}$$

$$HTU_{zhou4} = \begin{cases} 0 & ;\ T_h < T_{base} \\ \left(\frac{T_h-T_{base}}{T_{opt}-T_{base}}\right)\left(\frac{T_{up}-T_h}{T_{up}-T_{opt}}\right)^{\frac{T_{up}-T_{opt}}{T_{opt}-T_{base}}}\left(T_{opt} - T_{base}\right) & ;\ T_{base} < T_h \le T_{up} \\ 0 & ;\ T_{up} < T_h \end{cases}$$

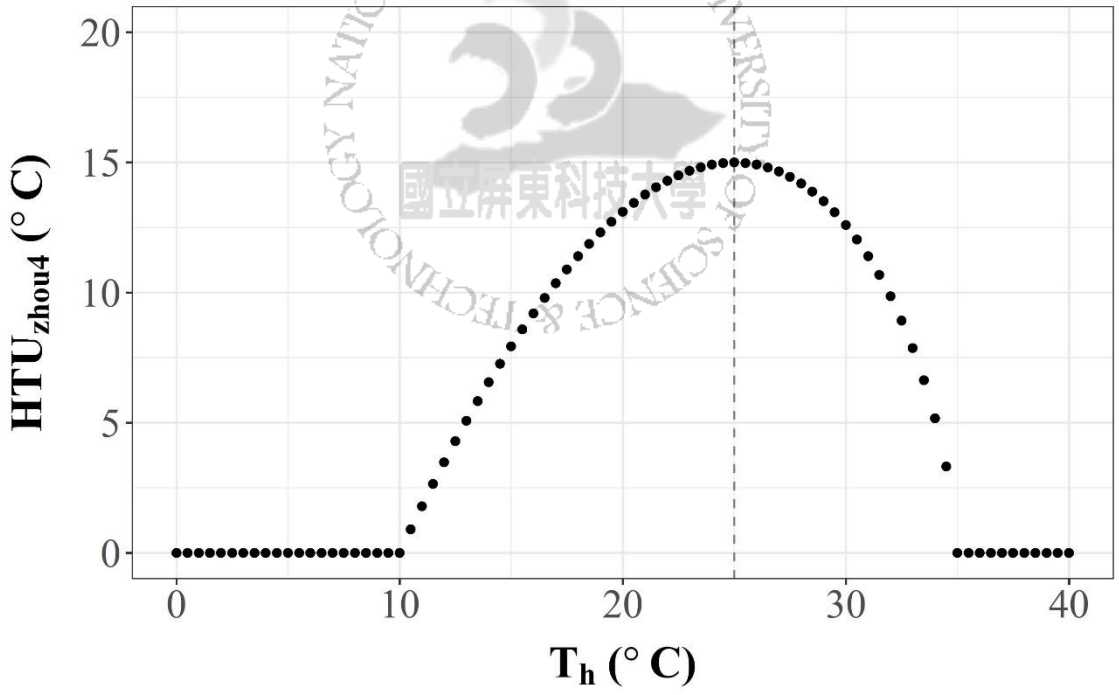
(Equation 1-9)

$$DTU = \frac{1}{24} \cdot \sum_{h=1}^{24} HTU_h \quad \text{(Equation 1-10)}$$

The T_h is the hourly temperature records, T_{base} is the base temperature, T_{opt} is the optimal temperature, T_{up} is the upper threshold temperature. The DTU is the average of the HTU.



(a)



(b)

Figure 1-3. Illustration of (a) HTU_{zhou3} and (b) HTU_{zhou4} calculated with different T_h. The T_{base}, T_{opt}, and T_{up} were set at 10℃, 25℃, and 35℃, respectively.

1.2.6. Data analysis process

The data analysis and modelling procedures were conducted in R language (R Core Team, 2022). The data separated randomly by nine pots were primarily split into two parts: internal dataset (the 1st, 3rd, 5th, 6th, 8th, and 10th pot of samples) and external dataset (the 2nd, 7th, and 9th pot of samples) (Figure 1-4). The 4th pot was excluded because of the abnormal growth of the rice plant, which was probably caused by the water leakage of the 4th pot. The internal dataset was partitioned into a training dataset and a testing dataset with a 6:4 ratio. Next, the training dataset was utilized for modelling, either with simple linear regression (retrieve d.LAS with DTU) or support vector machine with radial basis kernel (SVM-RBF) (classify growth stage with GDD). During the training process, the temperature parameters (T_{base}, T_{opt}, and T_{up}) were tuned with a ten-fold cross-validation cycle. The ten-fold cross validation was automatically done with the *train* function that is implemented in the caret package (Kuhn, 2008). The trained models were then validated by the testing dataset to obtain internal testing result and the optimal T_{base}, T_{opt}, and T_{up}. Subsequently, the DTUs of the internal dataset were re-calculated by applying the optimal temperature parameters. Consequently, the entire internal dataset was re-trained by applying the selected parameters and finally tested by the external dataset.

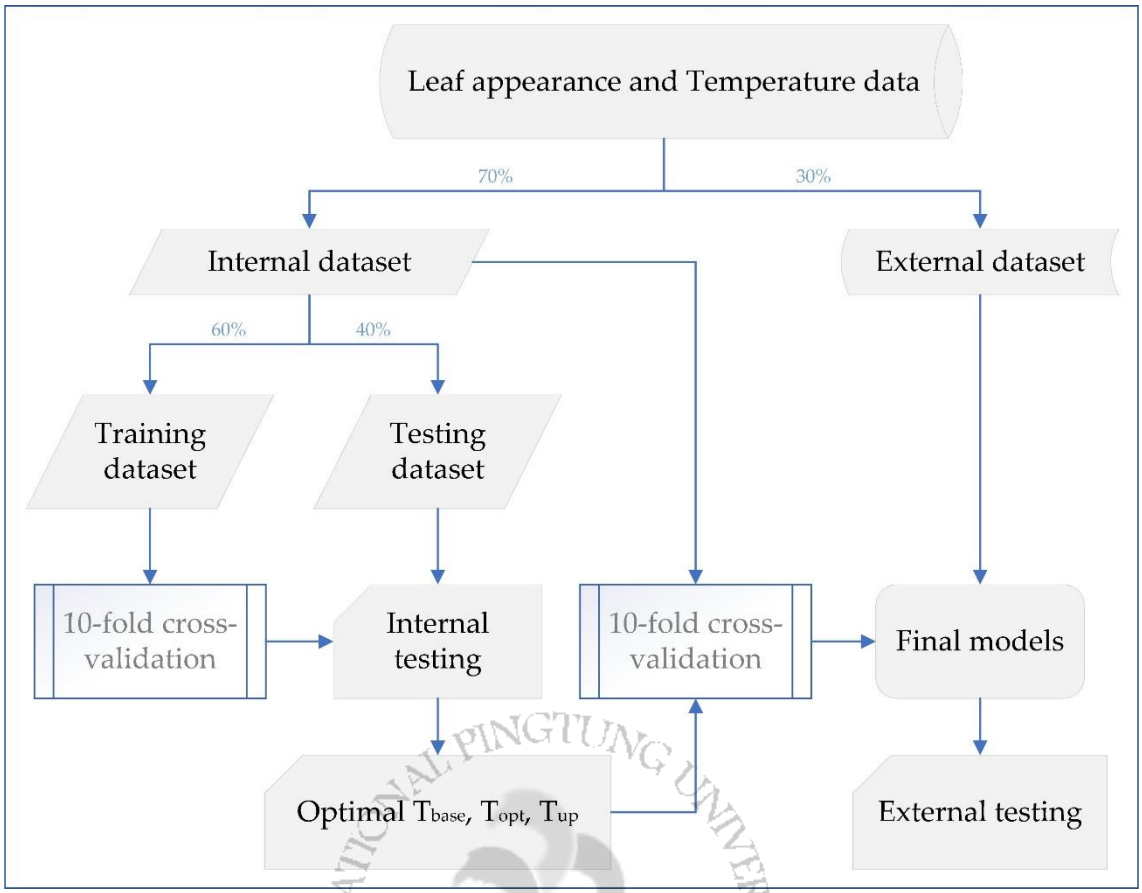


Figure 1-4. Data analysis process.

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1.3. Results and discussion

1.3.1. Air temperature records

The maximum (T_{max}), minimum (T_{min}), and average (T_{mean}) value of the air temperature were recorded in Appendix 1-A and summarized in Figure 1-5. In the cold season, the air temperature was range from 6.7°C to 31.1°C, with an average of 21.1°C ± 0.3°C. In the warm season, the air temperature was range from 19.7°C to 36.4°C, with an average of 28.0°C ± 0.3°C.

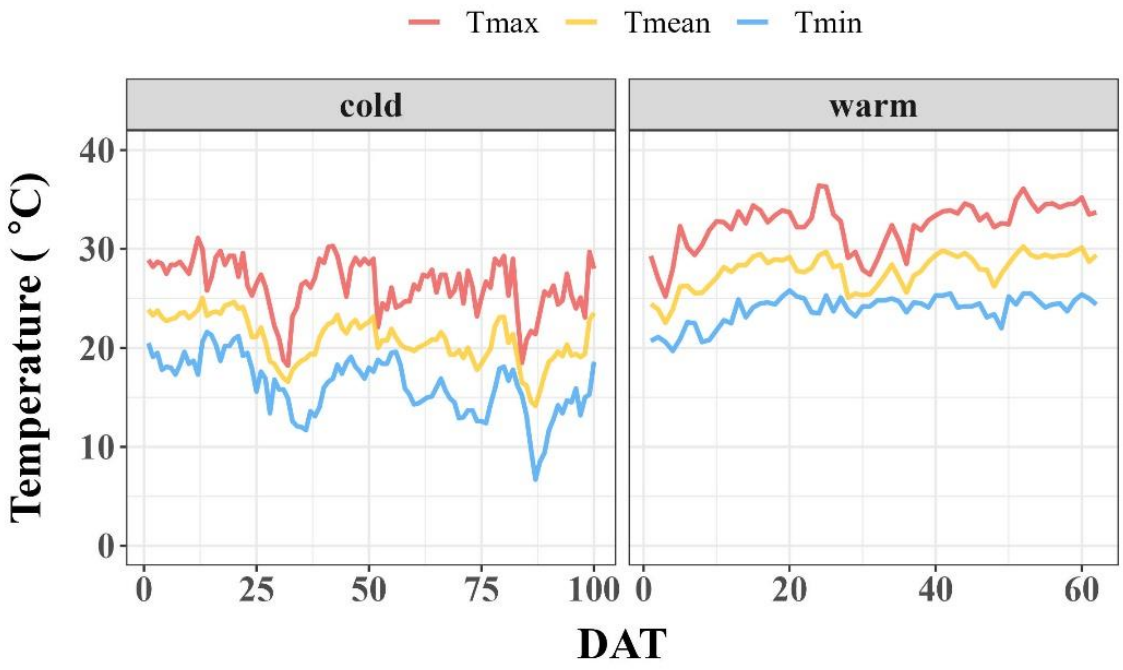
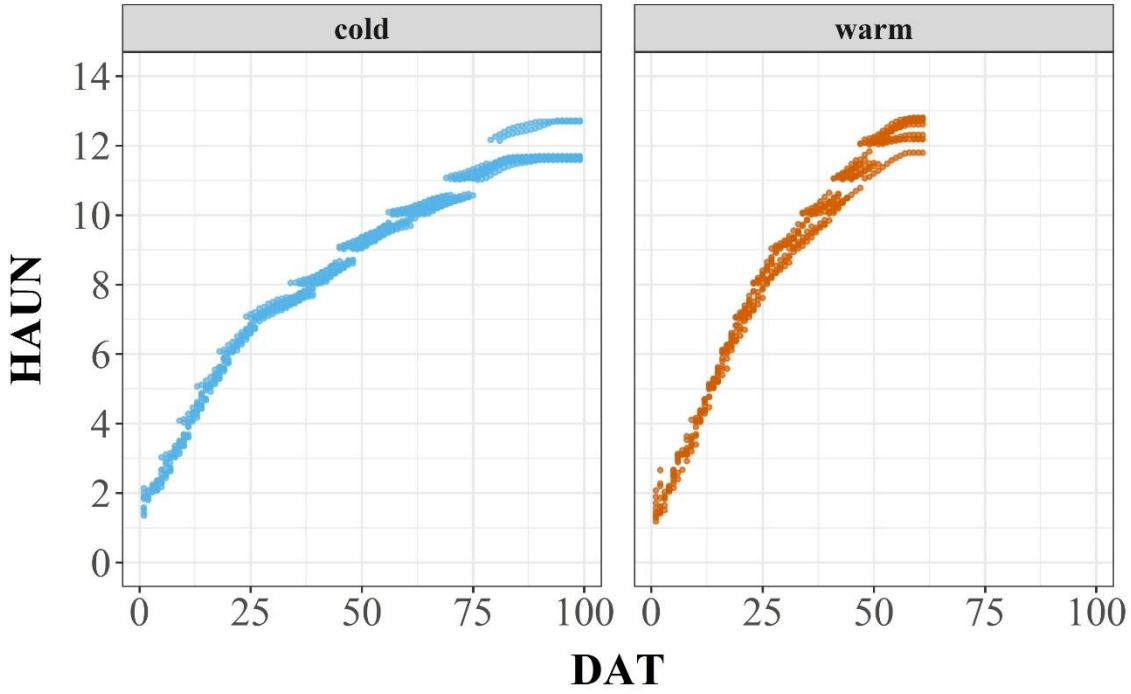


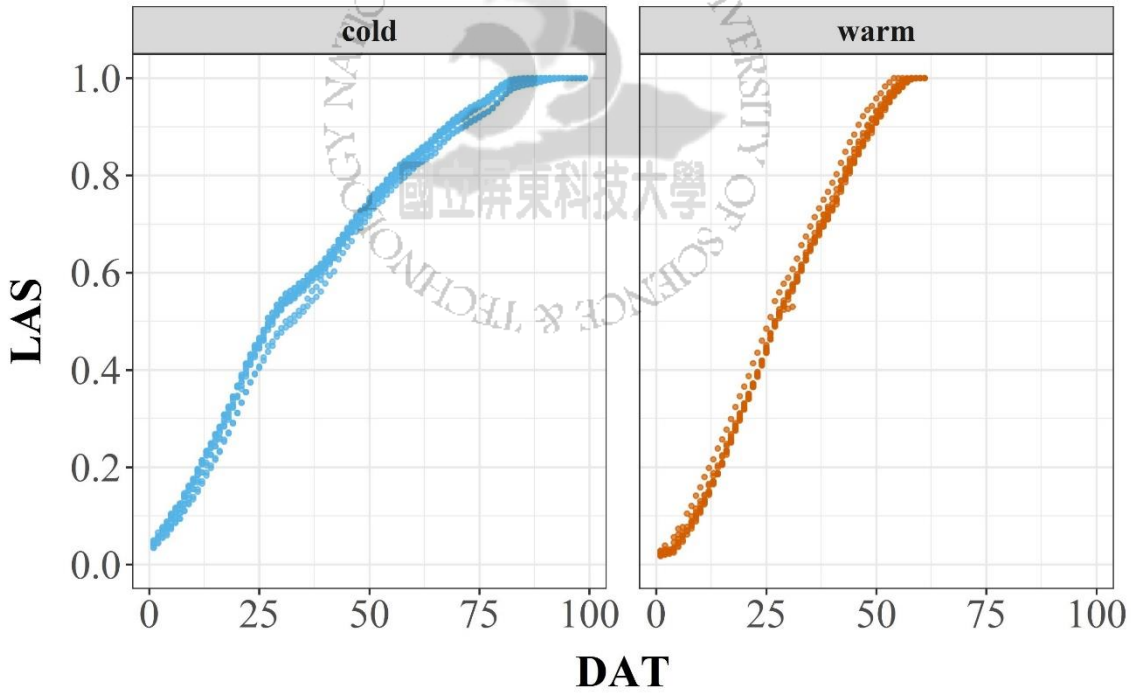
Figure 1-5. Temperature records during the experimental seasons.

1.3.2. Leaf age development and growth rate dynamic changes

The HAUN and the LAS were daily recorded from seedling stage until FFE stage in cold and warm seasons (Figure 1-6). The T_{max} and T_{min} in Pingtung during cold season are generally 25–26℃ and 14–15℃, while during warm season are generally 32–33℃ and 24–25℃. The HAUN development was stratified in both growing seasons (Figure 1-6a), while the LAS development was in continuous form (Figure 1-6b). When inferencing the dynamic changes of growth rate, the stratification of the HAUN would cause unnecessary noises (Figure 1-7), which means the increment of HAUN is caused by the leaf length variations rather than the DTU. The growth rates calculated by HAUN (d.HAUN) represent dramatic and random fluctuation along the cultivation period (Figure 1-7a). The coefficient of variation (CV) of d.HAUN in cold and warm seasons are respectively 117.10% and 88.39%. On the other hand, the growth rate calculated by the LAS (d.LAS) are more stable than the d.HAUN. The CV of d.LAS in the cold and the warm seasons are respectively 66.90% and 47.0%. Therefore, the LAS is more suitable than the HAUN in this study.



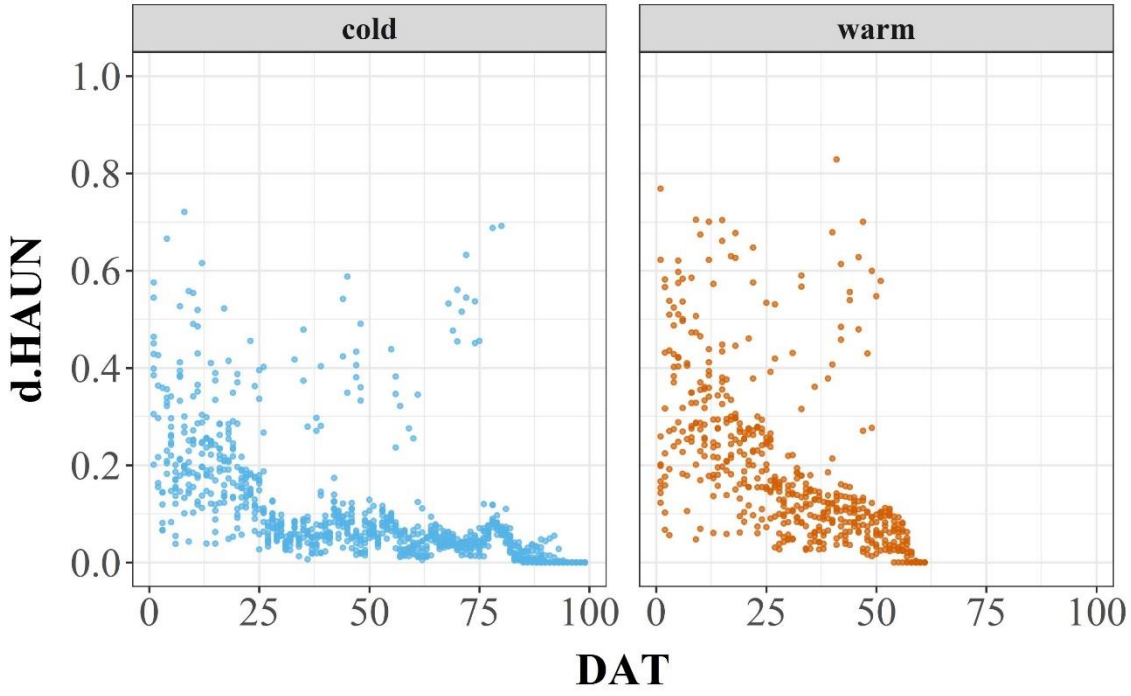
(a)



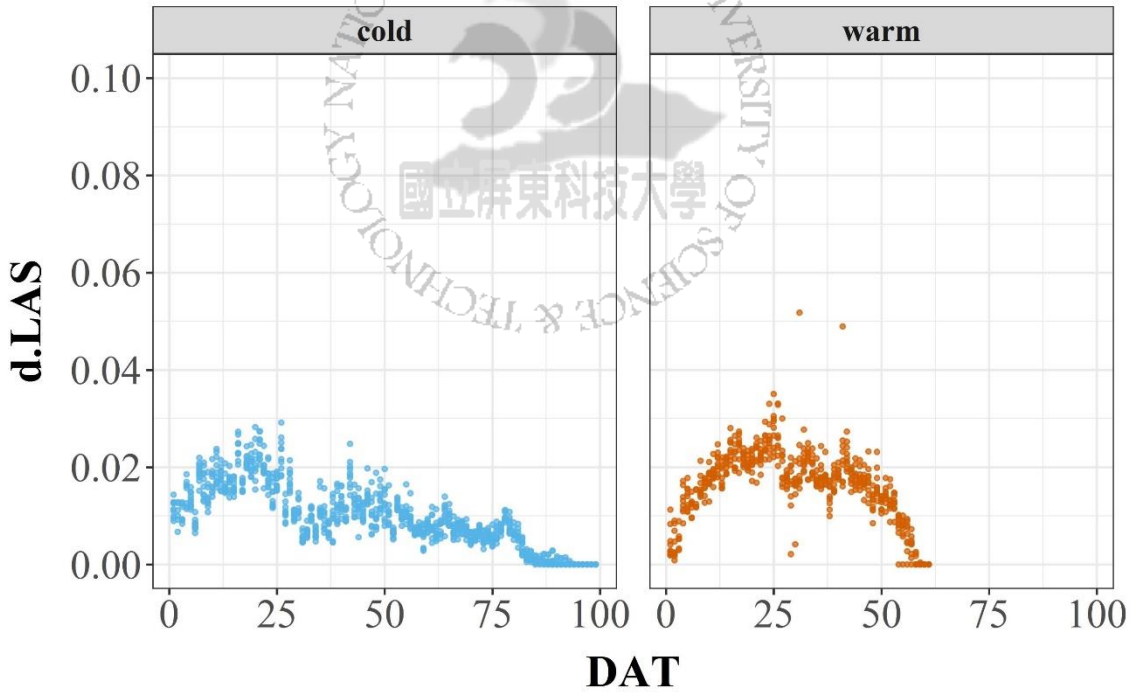
(b)

Figure 1-6. The (a) HAUN and (b) LAS records in the cold (blue) and warm (red) growing seasons.

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(a)



(b)

Figure 1-7. The (a) d.HAUN and (b) d.LAS records in the cold (blue) and warm (red) growing seasons.

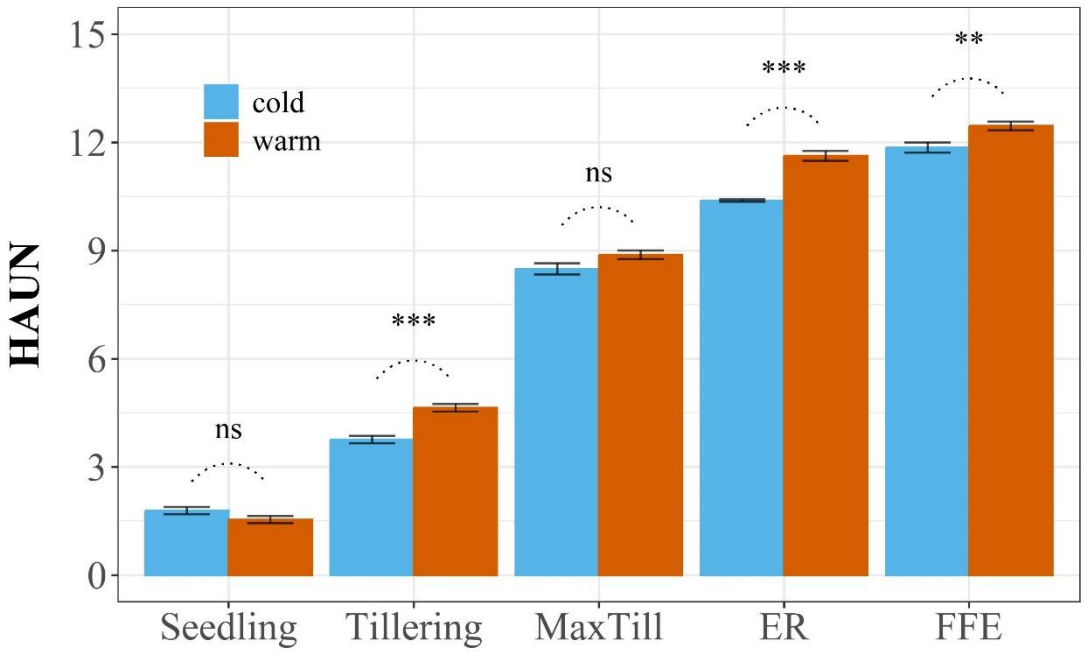
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1.3.3. Variation of leaf parameters in different seasons

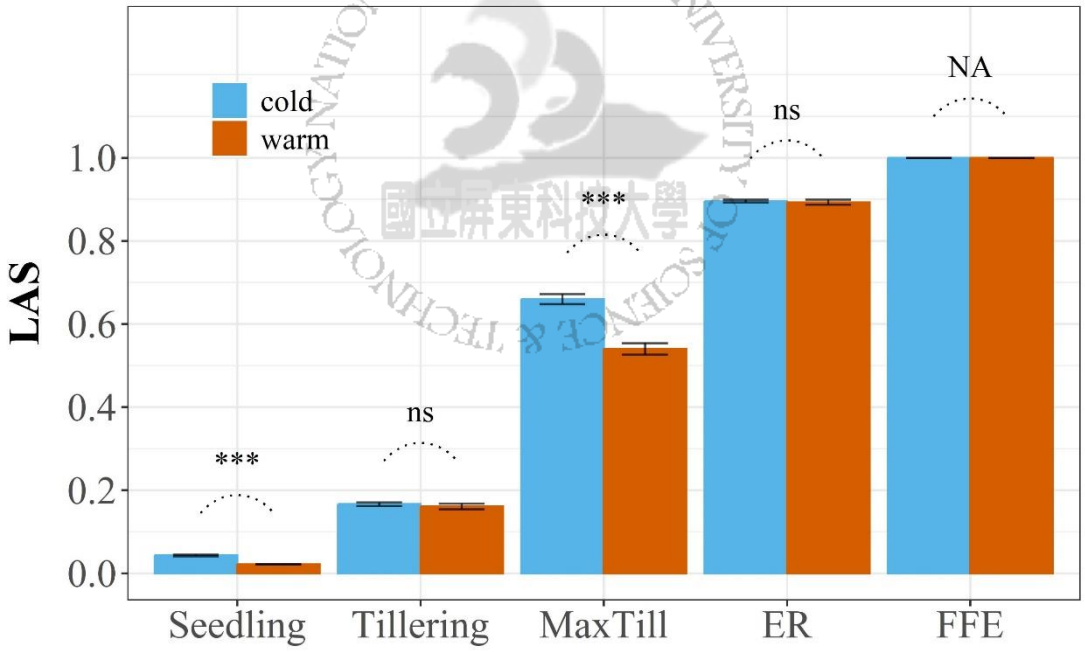
The variation of the Haun's scale (HAUN) and the leaf appearance scale (LAS) between cold and warm seasons were summarized in Figure 1-8. The HAUN is a so-called 'semi-continuous' scale that is probably ranged from 1 to 13 in this cultivar. The end number of the HAUN could be cultivar dependent. The minimum and maximum values of the HAUN across each growth stages are ranged from 1.35 to 12.72 in cold season, and from 1.18 to 12.80 in warm season (Figure 1-8a). In seedling stage, the difference between the HAUN in cold season ($HAUN_{cold}$ = 1.79 ± 0.10) and in warm season ($HAUN_{warm}$ = 1.54 ± 0.10) is insignificant. In tillering stage, the $HAUN_{cold}$ (3.76 ± 0.10) is significantly lower than the $HAUN_{warm}$ (4.65 ± 0.11). In MaxTill, the $HAUN_{cold}$ (8.49 ± 0.15) is similar with the $HAUN_{warm}$ (8.89 ± 0.12). In ER, the $HAUN_{cold}$ (10.39 ± 0.03) is significantly lower than the $HAUN_{warm}$ (11.63 ± 0.14). In FFE, the $HAUN_{cold}$ (11.86 ± 0.14) is significantly lower than the $HAUN_{warm}$ (12.46 ± 0.12).

On the other hand, the LAS is a continuous scale that is certainly ranged from 0 to 1 regardless of cultivars and growing seasons (Figure 1-8b). In seedling stage, the LAS in cold season (LAS_{cold} = 0.043 ± 0.002) is significantly higher than that in warm season (LAS_{warm} = 0.022 ± 0.001). However, this is negligible as the difference of the transplanted seedling between seasons is within one leaf. In tillering stage, the difference between the LAS_{cold} (0.167 ± 0.004) and the LAS_{warm} (0.161 ± 0.007) is insignificant. In MaxTill, the LAS_{cold} (0.660 ± 0.012) is significantly higher than the LAS_{warm} (0.540 ± 0.014). In ER, the LAS_{cold} (0.896 ± 0.003) is similar with the LAS_{warm} (0.893 ± 0.006). In FFE, there are no variations within and between seasons, therefore, two-sample test is not available (marked as 'NA', *i.e.* not available).

Due to the non-significant differences between LAS_{cold} and LAS_{warm} across each growth stages (except for the MaxTill), especially on the ER, the LAS could be a better parameter when quantifying rice growth compared to the HAUN. Therefore, the LAS was adopted as the fundamental measurement of rice growth in this study.



(a)



(b)

Figure 1-8. Comparison of the (a) HAUN and (b) LAS between cold (blue) and warm (red) growing seasons. $^{*, **, ***}$ represent significant level of $p < 0.05$, $p < 0.01$, $p < 0.001$; ^{ns} represents non-significance ($p > 0.05$). NA: two-sample test is not available.

1.3.4. d.LAS regression with DTU

Simple linear regression analysis was conducted to grid-search the optimal temperature parameters that are involved in each DTU calculation methods. The d.LAS is retrieved as a function of DTU. The models were constructed for each growth stages (seedling, tillering, MaxTill, and ER), respectively. The growth rate measurement during the FFE stage is unavailable due to its definition. Parameters combinations that yield the highest coefficient of determination (R^2) were considered as the best fit.

1.3.4.1. Seedling stage

In seedling stage, the hourly-based methods (DTU_{zhou3} and DTU_{zhou4}) performed better than the daily-based methods (Table 1-1). The highest R^2 was achieved when retrieving the d.LAS with the DTU_{zhou3} (R^2 = 0.425). The optimal T_{base}, T_{opt}, and T_{up} for calculating the DTU_{zhou3} are respectively 13.0℃, 28.5℃, and 50.0℃. Meanwhile, the performance of the DTU_{zhou4} (R^2 = 0.365) is acceptable. The overall T_{base} and T_{up} are respectively 22.7 ± 4.8℃ and 40.8 ± 10.3℃.

Among the daily-based methods, the DTU_{mc2} is outperforming (R^2 = 0.330) by defining the T_{base} and T_{up} respectively as 24.5℃ and 28.5℃. Similar with the DTU_{zhou4}, the T_{base} and T_{up} of the DTU_{mc2} are rather counter-intuitive. Further external validations are still required to ensure the reliability of these parameters.

The selected T_{base}, T_{opt}, and T_{up} were applied to reconstruct linear regression models (Figure 1-9) using the entire internal dataset. The hourly-based methods are outperforming the daily-based method. The DTU_{zhou3} remains the best retrieval performance in the final modelling process (R^2 = 0.346) (Figure 1-9e), while the DTU_{zhou4} (R^2 = 0.307) is somewhat acceptable (Figure 1-9f). The daily-based methods show poor retrieval ability on the d.LAS (Figure 1-9a-d).

Table 1-1. Best combination of parameters when retrieving the d.LAS with each DTUs during seedling stage.

Method	T_{base}	T_{opt}	T_{up}	R^2	Slope	Intercept	F
mc1	25.0	–	50.0	0.164	0.0023^{***}	0.0114^{***}	14.50^{***}
mc2	24.5	–	28.5	0.334	0.0068^{***}	0.0006^{ns}	37.19^{***}
nielsen	25.0	–	32.0	0.167	0.0029^{***}	0.0113^{***}	14.82^{***}
trial	23.5	–	50.0	0.158	0.0016^{***}	0.0114^{***}	13.89^{***}
zhou3	13.0	28.5	50.0	0.425	0.0015^{***}	0.0051^{***}	54.80^{***}
zhou4	25.0	28.5	34.0	0.365	0.0068^{***}	0.0052^{***}	42.50^{***}
mean	22.7		40.8				
sd	4.8		10.3				

*, **, *** represent significant level of $p < 0.05$, $p < 0.01$, $p < 0.001$; ^{ns} represents non-significance ($p > 0.05$); sd is the standard deviation of each column.

The reconstructed models were validated using the external dataset to test the reliability of the selected temperature parameters. During external validation, the model performance of the DTU_{zhou3} remain the best ($R^2 = 0.191$; RRMSE = 35.96%) (Figure 1-10e). The low R^2 value could be probably caused by the indirect sensing of the air temperature by the seedling organs. Most of parts of the seedling are usually located near the water surface or emerged in the water (Bonhomme, 2000). Therefore, the relation between the air temperature and the seedling growth rate is rather incompact. Although the predictive ability of the model is rather poor, several inferences could still be drawn to some extent.

Researchers found that thermal stress response of rice seedling would be induced when the air temperature is out of range of 25℃ – 28℃ (Ormrod and Bunter, 1961; Kato-Noguchi and Yasuda, 2007; Liu et al., 2018b; Wu and Yang, 2019; Zhao et al., 2020; Xu et al., 2021). The T_{opt} found with DTU_{zhou3} and DTU_{zhou4}, which both are 28.5℃, is agreeing with the preceding findings. On the aspect of cold stress, when rice seedlings were exposed to 18℃, thylakoids formed looser and grana stacking was not observed, chlorophyll fluorescence

was abnormal, ROS burst significantly increase, and cell death was significant compare to that at 28℃ (Zhao et al., 2020). When temperature dropped to 15℃, pyruvate decarboxylase (PDC) and alcohol dehydrogenase (ADH) activity were obviously increased, which is showing the induction of ethanolic fermentation process in low temperature, compare to that above 20℃ (Kato-Noguchi and Yasuda, 2007). When temperature is further decreased, dry weight and chlorophyll accumulation were impeded, and functional thylakoids and plastid protein (PSI and PSII) were not found at 12℃ (Zhao et al., 2020). According to these physiological inferences, the T_{low} of 13.0℃ found with DTU_{zhou3} could be reasonable compared to the T_{low} of 25.0℃ found with DTU_{zhou4} (Table 1-1).

On the other hand, researchers considered a transient temperature of 45℃ would result in heat stress response. The respiration rate of rice seedlings remain stable between 23 – 37℃, peaked at 43℃ and dropped at 50℃ (Ormrod and Bunter, 1961). The stomata opening and survival rate would also dramatically drop at 45℃ (Liu et al., 2018b; Wu and Yang, 2019). It is worth noting that the tested maximum value of the T_{up} in this study is 50.0℃ due to hardware limitations and computation cost. Furthermore, the samples were not encountering with such high temperature in this experiment. Therefore, the upper threshold temperature of the rice seedling is remaining uncertain.

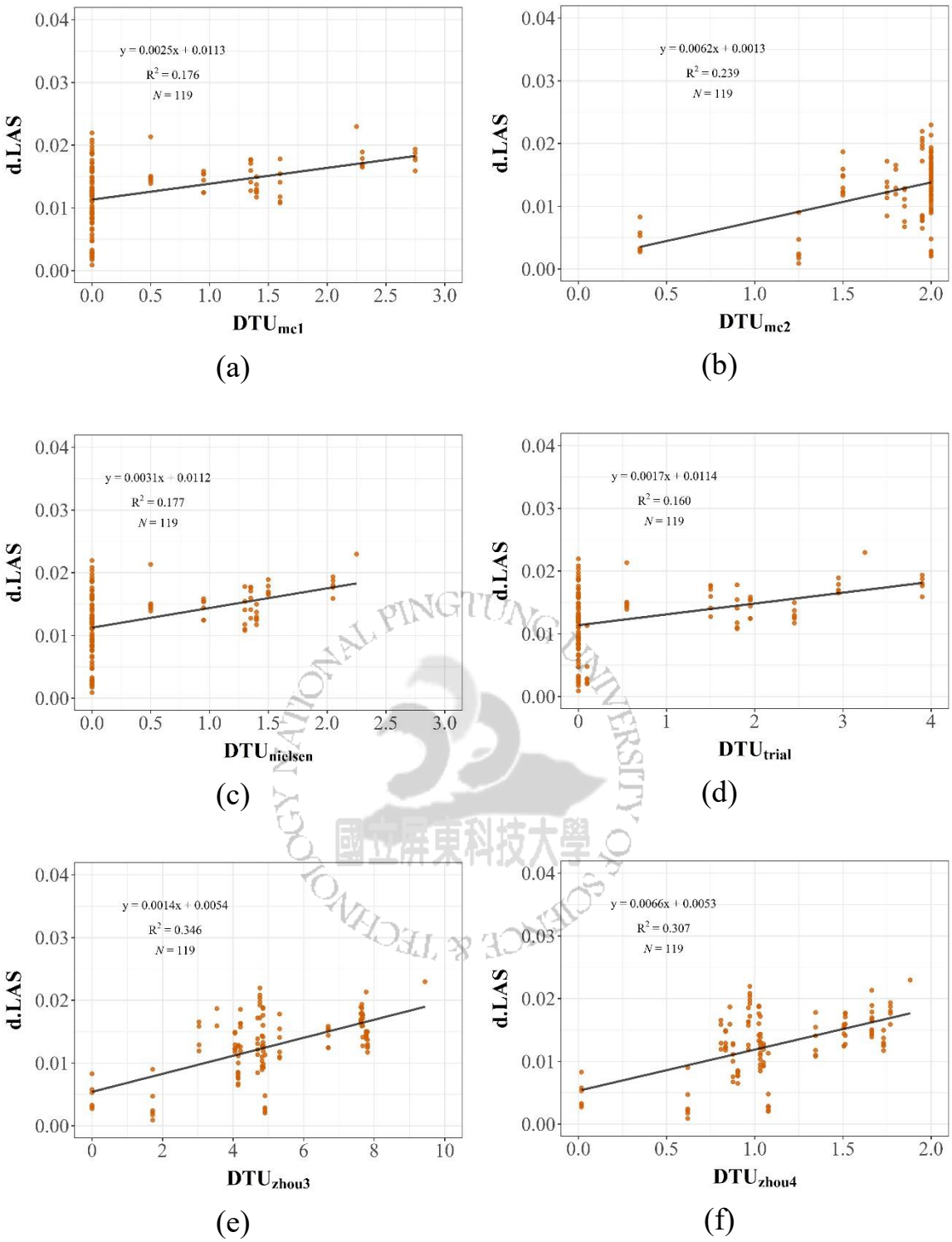


Figure 1-9. Reconstructed linear regression model in seedling stage. The d.LAS was respectively retrieved by (a) DTU_{mc1}, (b) DTU_{mc2}, (c) $DTU_{nielsen}$, (d) DTU_{trial}, (e) DTU_{zhou3}, and (f) DTU_{zhou4}.

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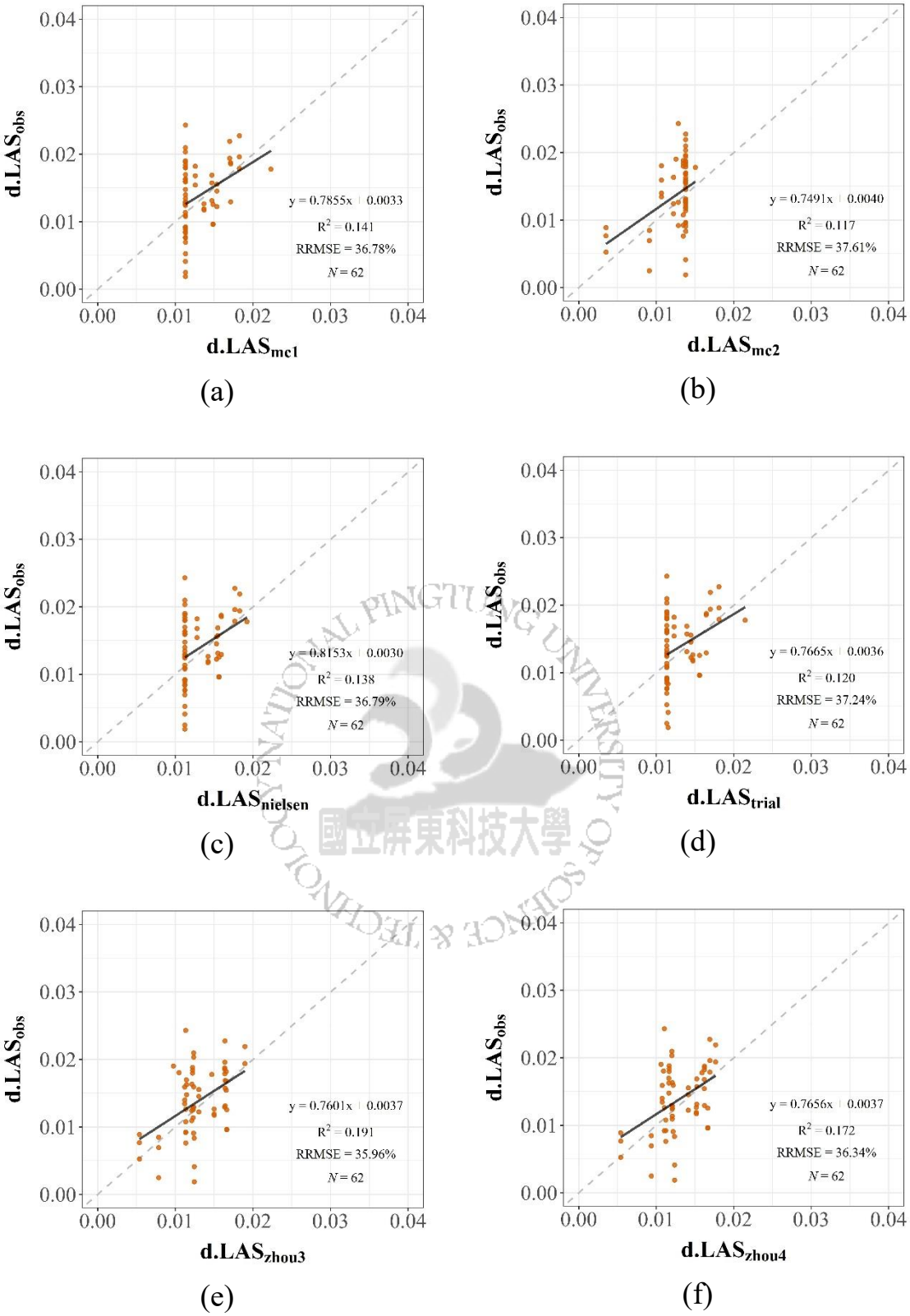


Figure 1-10. External validation result in seedling stage. The d.LAS was predicted with the retrieval models constructed with d.LAS and (a) DTU_{mc1}, (b) DTU_{mc2}, (c) $DTU_{nielsen}$, (d) DTU_{trial}, (e) DTU_{zhou3}, and (f) DTU_{zhou4}.

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1.3.4.2. Tillering stage

In tillering stage, the hourly-based methods are slightly better than daily-based methods (Table 1-2). The DTU_{zhou3} (R^2 = 0.629) achieved the highest R^2 by respectively setting the T_{base}, T_{opt}, and T_{up} as 8.5℃, 40.0℃, and 50.0℃. The DTU_{zhou4} (R^2 = 0.623) yielded comparable R^2 by respectively setting the T_{base}, T_{opt}, and T_{up} as 13.0℃, 35.5℃, and 50.0℃. On the other hand, the DTU_{mc1}, DTU_{mc2}, and $DTU_{nielsen}$ yielded comparable retrieval ability (R^2 = 0.608) by respectively setting the T_{base} and T_{up} as 10.5℃ and 50.0℃. The overall T_{base} and T_{up} are respectively 11.4 ± 2.5℃ and 50.0 ± 0.0℃.

Table 1-2. Best combination of parameters when retrieving the d.LAS with each DTUs during tillering stage.

Method	**T_{base}**	**T_{opt}**	**T_{up}**	**R^2**	**Slope**	**Intercept**	**F**
mc1	10.5	–	50.0	0.608	0.0012^{***}	0.0004^{ns}	294.78^{***}
mc2	10.5	–	50.0	0.608	0.0012^{***}	0.0004^{ns}	294.78^{***}
nielsen	10.5	–	50.0	0.608	0.0012^{***}	0.0004^{ns}	294.78^{***}
trial	15.5	–	50.0	0.612	0.0012^{***}	0.0075^{***}	300.31^{***}
zhou3	8.5	40.0	50.0	0.629	0.0010^{***}	0.0027^{**}	321.92^{***}
zhou4	13.0	35.5	50.0	0.623	0.0012^{***}	0.0001^{ns}	314.51^{***}
mean	11.4		50.0				
sd	2.5		0.0				

$^{*, **, ***}$ represent significant level of $p < 0.05$, $p < 0.01$, $p < 0.001$; ^{ns} represents non-significance ($p > 0.05$); sd is the standard deviation of each column.

After model reconstruction, the hourly-based methods performed better than daily-based methods (Figure 1-11). The DTU_{zhou3} (R^2 = 0.651) performed slightly better than the DTU_{zhou4} (R^2 = 0.648). The external validation results showed that all models constructed with each DTU methods are reliable (Figure 1-12). The DTU_{zhou3} remained outperforming predictive ability (R^2 = 0.633; RRMSE = 22.70%) among the models.

Ishii et al. (1977) reported the maximum rate of apparent photosynthesis (AP) of rice under normal CO_2 concentrations (300 ppm) during six-leaf age was obtained at 25 – 35℃. Since the highest tested temperature of that paper was 35℃, there is lack of evidence to show temperature above 35℃ leads detrimental effects to AP (Ishii et al., 1977). The paper could be summarized into three aspects to reveal the relationship between temperature and AP. Firstly, the differences of AP at 25℃ and 35℃ were not obvious, either with or without the inhibition of photorespiration. Secondly, the CO_2 compensation points at 25℃ and 35℃ are similar when inhibiting photorespiration, indicating dark respiration was not enhanced by elevating temperature from 25℃ to 35℃. Lastly the third, the activities of RuDP carboxylase increased exponentially with increasing temperature up to 40℃, showing that the optimum temperature for carbon fixation activity can be up to 40℃. The T_{opt} (40℃) found with DTU_{zhou3} in Table 1-2 is agreeing with Ishii et al. (1977). Similar with the seedling stage, the upper threshold temperature for the tillering stage is remaining uncertain.

On the other hand, temperatures between 15℃ and 20℃ usually increase rice tillers number (Dubey et al., 2018). Although temperature below 15℃ might hinder tillers development, the overall plant extension is probably ongoing. Shimono et al. (2002) also found dry matter production in rice tillering stage when the water temperature was below 15℃ (Shimono et al., 2002). Moreover, Sato (1972) reported that dry weight accumulation of rice in tillering stage could still be observed even the temperature is as low as 10℃ (Sato, 1972). According to these inferences, the T_{base} found with DTU_{zhou3} (8.5℃) seems to be more reasonable than the DTU_{zhou4} (13.0℃) (Table 1-2).

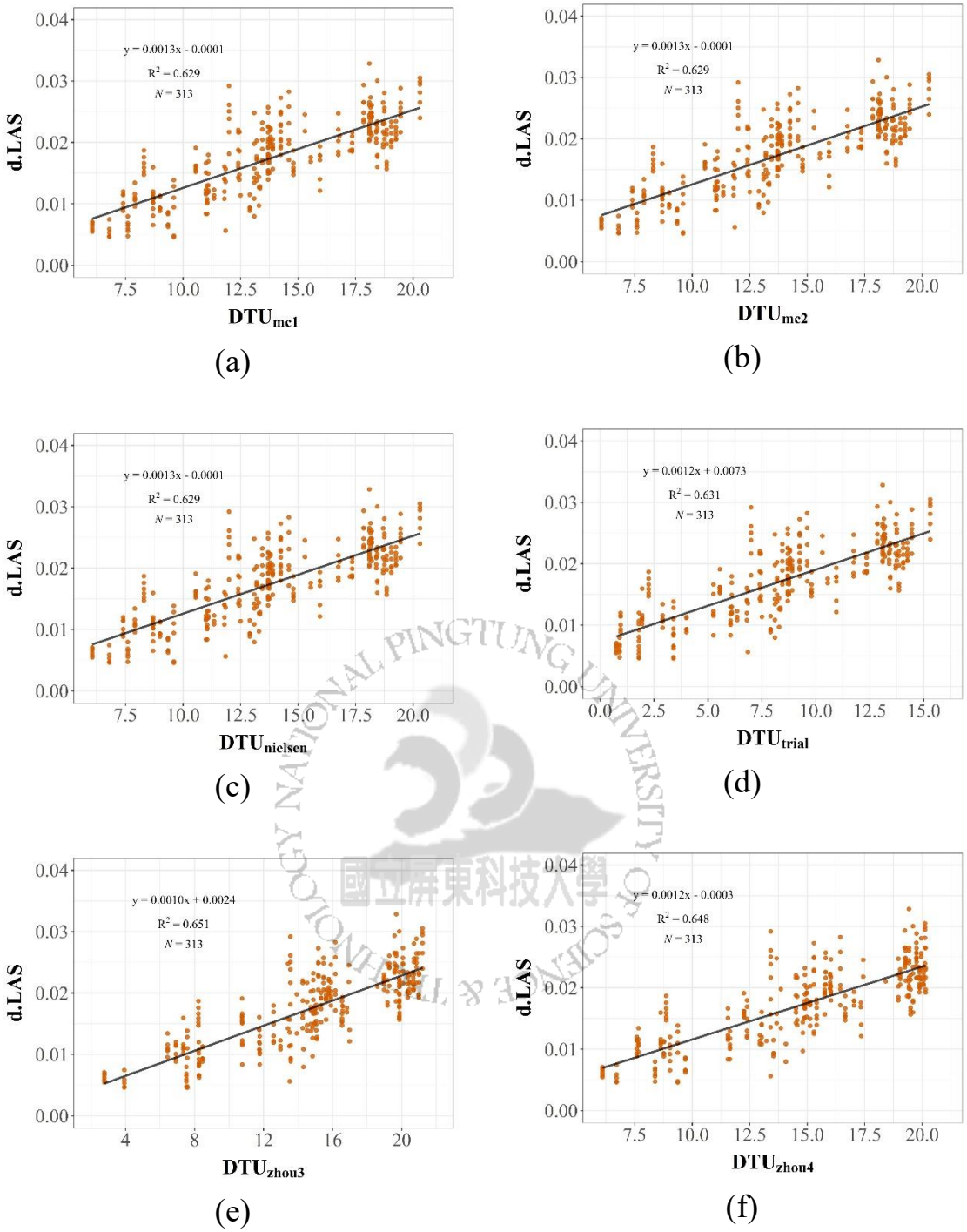


Figure 1-11. Reconstructed linear regression model in tillering stage. The d.LAS was respectively retrieved by (a) DTU_{mc1}, (b) DTU_{mc2}, (c) $DTU_{nielsen}$, (d) DTU_{trial}, (e) DTU_{zhou3}, and (f) DTU_{zhou4}.

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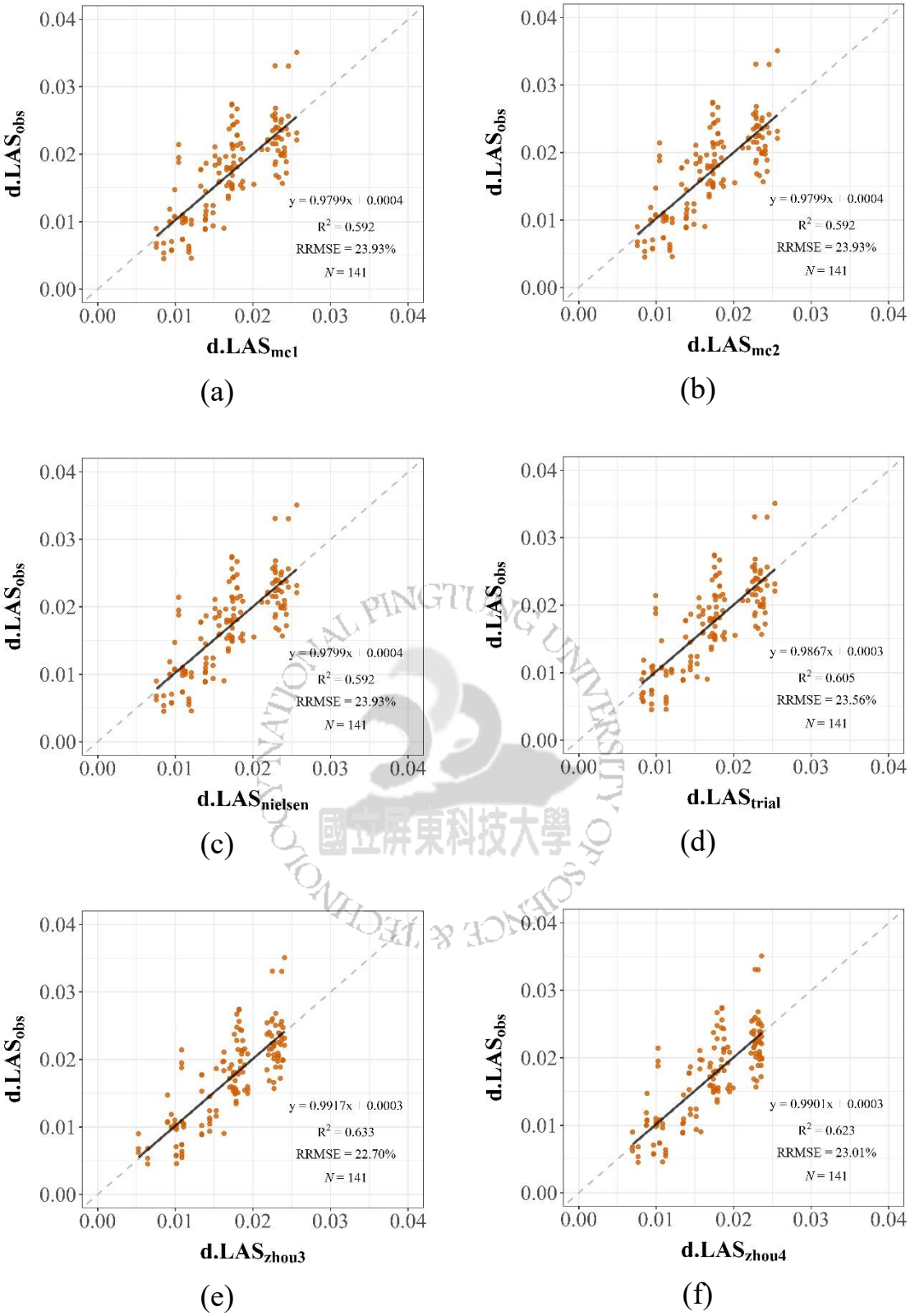


Figure 1-12. External validation result in tillering stage. The d.LAS was predicted with the retrieval models constructed with d.LAS and (a) DTU_{mc1}, (b) DTU_{mc2}, (c) $DTU_{nielsen}$, (d) DTU_{trial}, (e) DTU_{zhou3}, and (f) DTU_{zhou4}.

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1.3.4.3. Maximum tillering stage (MaxTill)

In MaxTill, the hourly-based methods performed better than the daily-based methods (Table 1-3). The DTU_{zhou4} (R^2 = 0.767) performed slightly better than DTU_{zhou3} (R^2 = 0.759). The optimal T_{base}, T_{opt}, and T_{up} for DTU_{zhou3} are respectively 10.0℃, 29.0℃, and 49.0℃. The optimal T_{base}, T_{opt}, and T_{up} for DTU_{zhou4} are respectively 17.0℃, 30.5℃, and 34.5℃. Despite the large differences of the T_{base} and T_{up} between DTU_{zhou3} and DTU_{zhou4}, T_{opt} are relatively close. This indicates that the growth rate would probably peak at 29.0 – 30.5℃ during MaxTill. The overall T_{base} and T_{up} are respectively 17.3 ± 4.1℃ and 33.3 ± 8.2℃.

Table 1-3. Best combination of parameters when retrieving the d.LAS with each DTUs during MaxTill.

Method	**T_{base}**	**T_{opt}**	**T_{up}**	**R^2**	**Slope**	**Intercept**	**F**
mc1	22.0	–	25.5	0.627	0.0031^{***}	0.0080^{***}	257.37^{***}
mc2	17.5	–	30.0	0.605	0.0020^{***}	0.0000^{ns}	234.10^{***}
nielsen	17.0	–	30.5	0.625	0.0020^{***}	0.0001^{ns}	255.09^{***}
trial	20.0	–	30.5	0.624	0.0018^{***}	0.0080^{***}	253.52^{***}
zhou3	10.0	29.0	49.0	0.759	0.0011^{***}	0.0007^{ns}	479.18^{***}
zhou4	17.0	30.5	34.5	0.767	0.0016^{***}	0.0004^{ns}	501.50^{***}
mean	17.3		33.3				
sd	4.1		8.2				

*, **, *** represent significant level of $p < 0.05$, $p < 0.01$, $p < 0.001$; ^{ns} represents non-significance ($p > 0.05$); sd is the standard deviation of each column.

After model reconstruction (Figure 1-13), the R^2 values of DTU_{zhou3} and DTU_{zhou4} declined from 0.760 to 0.639 and from 0.770 to 0.638, respectively. Meanwhile, $DTU_{nielsen}$ is relatively stable which the R^2 value changed from 0.630 to 0.640. During external validation (Figure 1-14), the $DTU_{nielsen}$ (R^2 = 0.723; RRMSE = 21.85%) remained the lowest predictive error while the DTU_{zhou4} (R^2 = 0.724; RRMSE = 21.88%) exhibited the highest R^2 value.

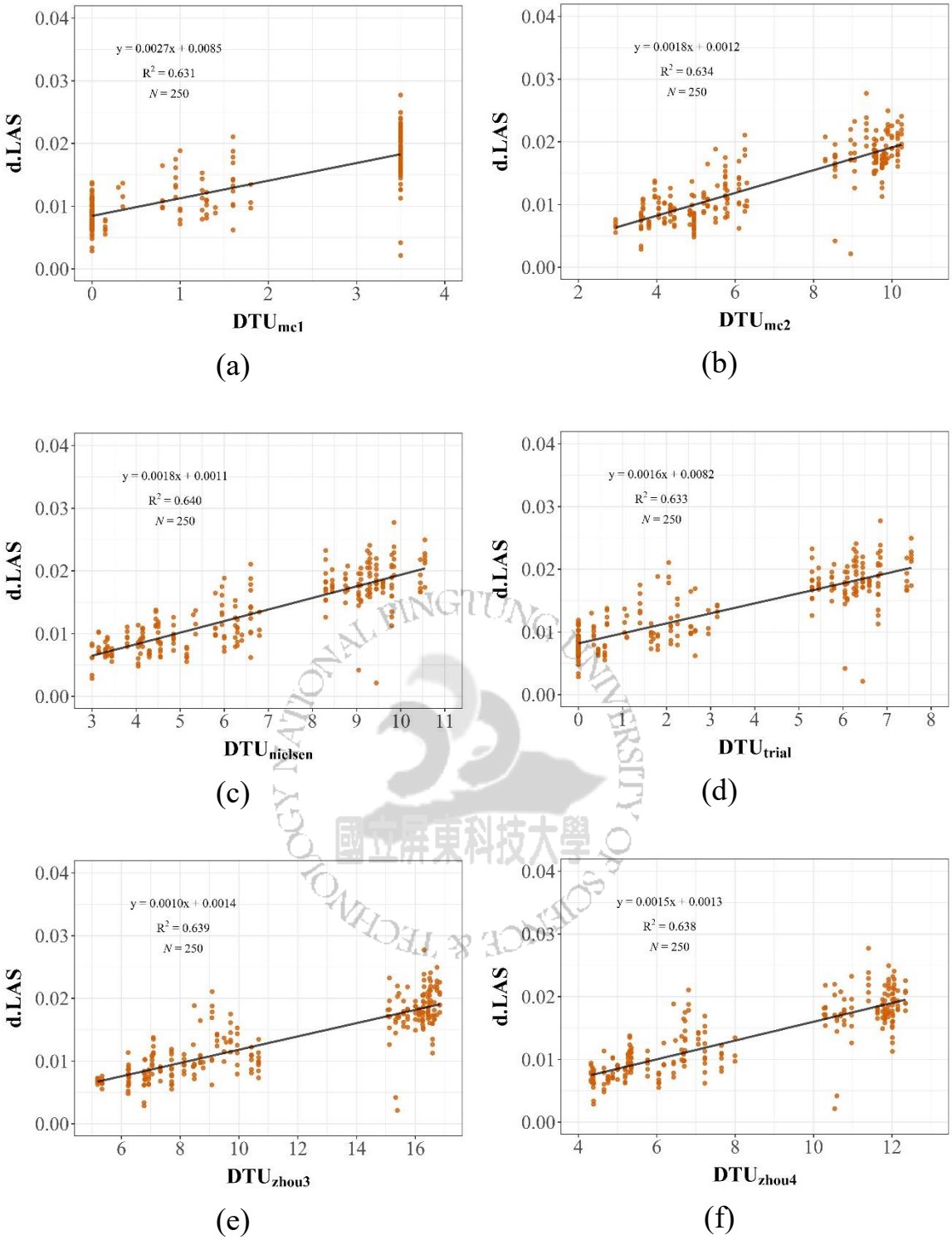


Figure 1-13. Reconstructed linear regression model in MaxTill. The d.LAS was respectively retrieved by (a) DTU_{mc1}, (b) DTU_{mc2}, (c) $DTU_{nielsen}$, (d) DTU_{trial}, (e) DTU_{zhou3}, and (f) DTU_{zhou4}.

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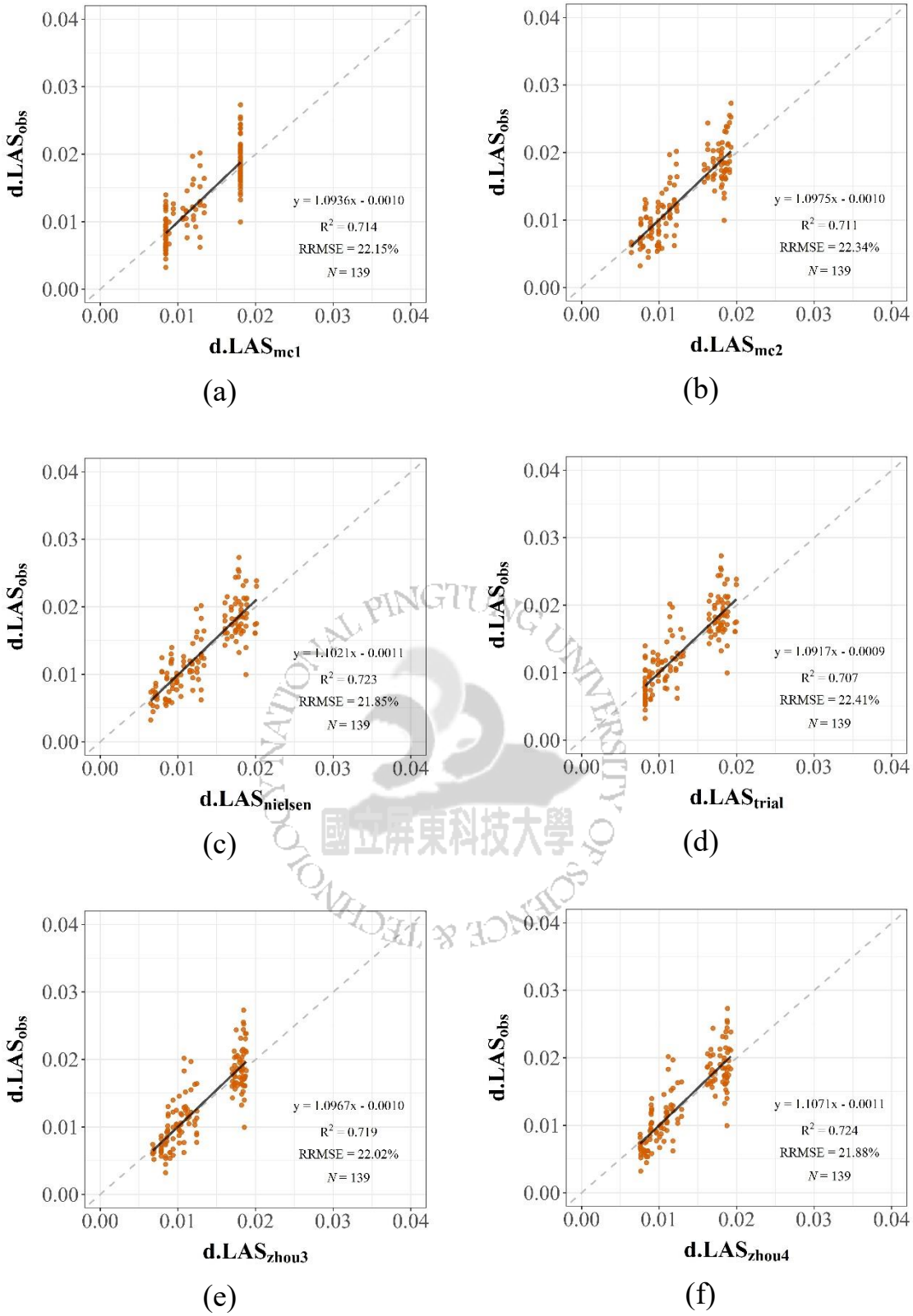


Figure 1-14. External validation result in MaxTill. The d.LAS was predicted with the retrieval models constructed with d.LAS and (a) DTU_{mc1}, (b) DTU_{mc2}, (c) $DTU_{nielsen}$, (d) DTU_{trial}, (e) DTU_{zhou3}, and (f) DTU_{zhou4}.

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1.3.4.4. Early reproductive phase (ER)

In ER, the R^2 values of the hourly-based methods are higher than the daily-based methods (Table 1-4). The DTU_{zhou3} (R^2 = 0.579) is better than DTU_{zhou4} (R^2 = 0.563) according to its greater R^2 and insignificant intercept. The optimal T_{base}, T_{opt}, and T_{up} for DTU_{zhou3} are respectively 10.0℃, 32.0℃, and 32.5℃. This indicates that a rice plant stops growing when the temperature is lower than 10.0℃ or higher than 32.5℃ during ER. Meanwhile, the growth rate probably peaked around 32.0℃. The overall T_{base} and T_{up} are respectively 14.9 ± 3.6℃ and 30.4 ± 2.9℃.

Table 1-4. Best combination of parameters when retrieving the d.LAS with each DTUs during ER.

Method	**T_{base}**	**T_{opt}**	**T_{up}**	**R^2**	**Slope**	**Intercept**	**F**
mc1	14.5	–	27.0	0.480	0.0007^{***}	0.0008^{ns}	110.89^{***}
mc2	19.5	–	26.5	0.474	0.0016^{***}	0.0003^{ns}	107.99^{***}
nielsen	14.5	–	32.0	0.476	0.0007^{***}	0.0008^{ns}	109.20^{***}
trial	12.5	–	32.5	0.483	0.0007^{***}	0.0001^{ns}	112.01^{***}
zhou3	10.0	32.0	32.5	0.579	0.0008^{***}	0.0002^{ns}	164.70^{***}
zhou4	18.5	31.5	32.0	0.563	0.0013^{***}	0.0012^{*}	154.40^{***}
mean	14.9		30.4				
sd	3.6		2.9				

*, **, *** represent significant level of $p < 0.05$, $p < 0.01$, $p < 0.001$; ^{ns} represents non-significance ($p > 0.05$); sd is the standard deviation of each column.

After model reconstruction (Figure 1-15), the R^2 values of DTU_{zhou3} (R^2 = 0.614) remains the highest among the others and no overfitting was observed (minor increase from 0.580 to 0.614). During external validation (Figure 1-16), DTU_{zhou3} (R^2 = 0.578; RRMSE = 42.70%) exhibited the highest R^2 value and the lowest predictive error.

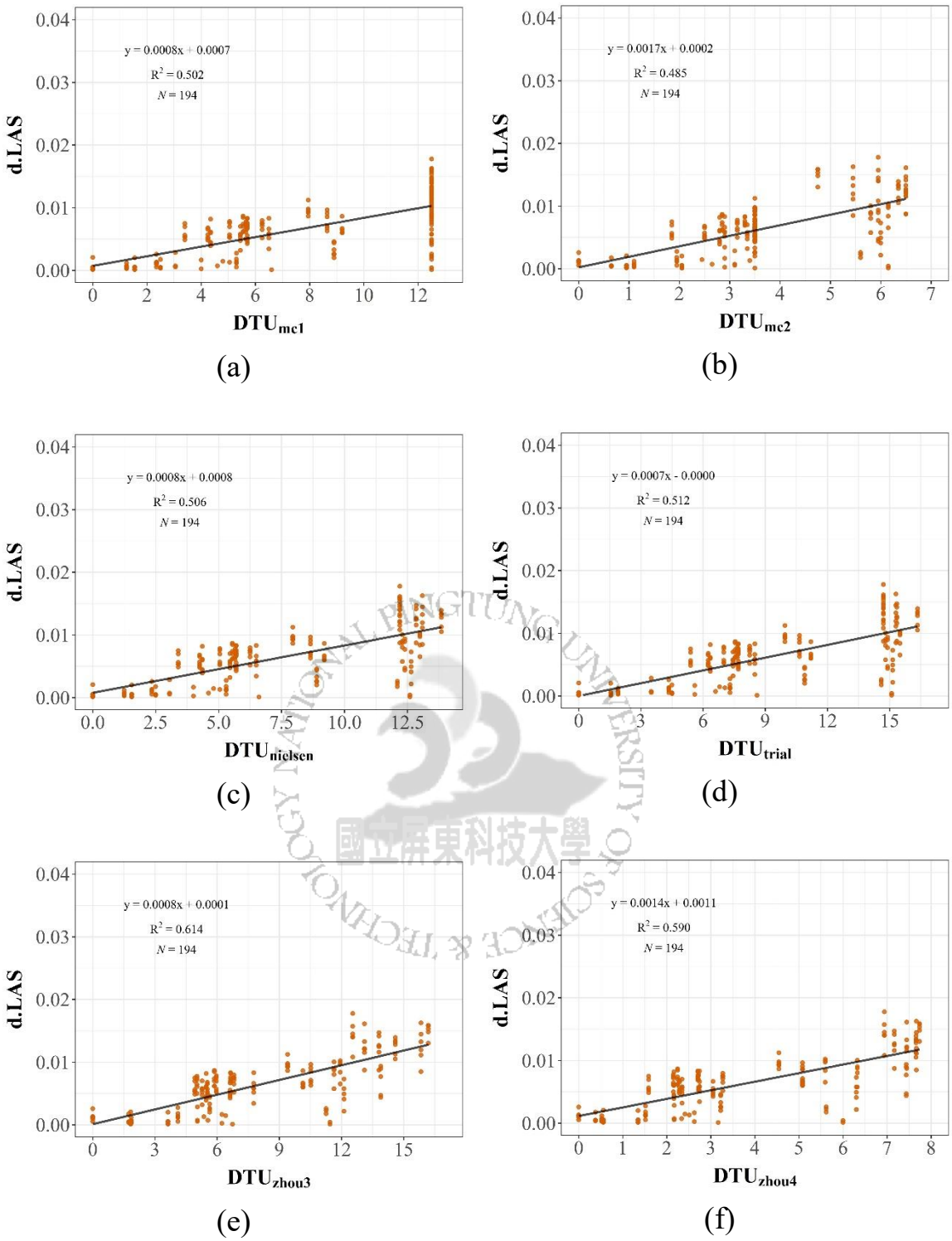


Figure 1-15. Reconstructed linear regression model in ER. The d.LAS was respectively retrieved by (a) DTU_{mc1}, (b) DTU_{mc2}, (c) $DTU_{nielsen}$, (d) DTU_{trial}, (e) DTU_{zhou3}, and (f) DTU_{zhou4}.

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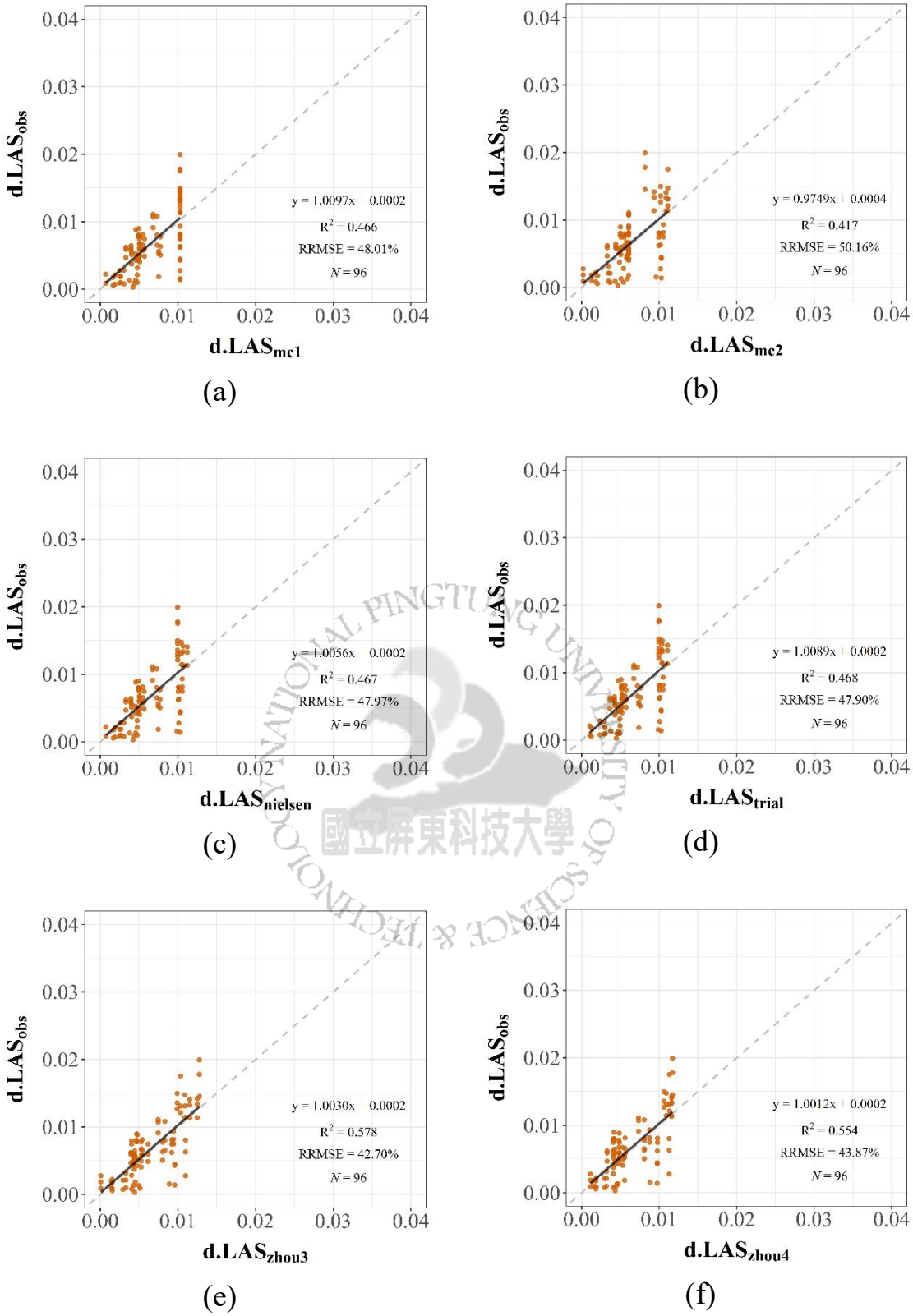


Figure 1-16. External validation result in ER. The d.LAS was predicted with the retrieval models constructed with d.LAS and (a) DTU_{mc1}, (b) DTU_{mc2}, (c) $DTU_{nielsen}$, (d) DTU_{trial}, (e) DTU_{zhou3}, and (f) DTU_{zhou4}.

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1.3.4.5. GDD variation between growing seasons

The variation of each DTU methods between cold and warm seasons at the end of each growth stages were summarized in Figure 1-17. The overall DTUs across the experiment period were summarized as the mean and standard deviation of population (mean ± sd) to figure the mean basis of daily DTU. The overall DTU_{mc1}, DTU_{mc2}, $DTU_{nielsen}$, DTU_{trial}, DTU_{zhou3}, and DTU_{zhou4} were 6.95 ± 5.84℃, 7.63 ± 5.11℃, 8.36 ± 5.22℃, 6.19 ± 4.77℃, 10.74 ± 5.29℃, and 8.33 ± 5.76℃, respectively.

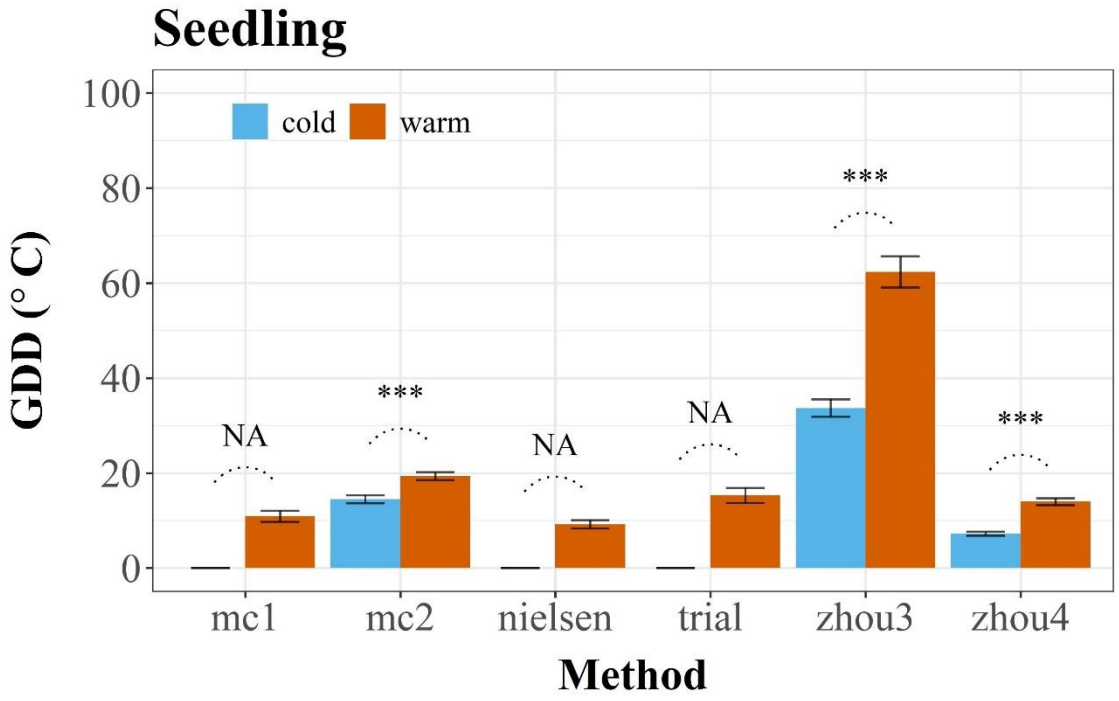
At the end of seedling stage, all GDDs between growing seasons accumulated by each DTU methods were significantly different (Figure 1-17a). The two-sample test of DTU_{mc1}, $DTU_{nielsen}$, and DTU_{trial} were unavailable as the GDD of all samples in cold season were equally zero. When applying DTU_{mc1}, the GDD_{cold} and GDD_{warm} were 0.00 ± 0.00℃ and 10.91 ± 1.14℃, which were about two days apart. When applying DTU_{mc2}, the GDD_{cold} and GDD_{warm} were 14.49 ± 0.82℃ and 19.37 ± 0.83℃, which were about one day apart. When applying $DTU_{nielsen}$, the GDD_{cold} and GDD_{warm} were 0.00 ± 0.00℃ and 9.23 ± 0.86℃, which were about one day apart. When applying DTU_{trial}, the GDD_{cold} and GDD_{warm} were 0.00 ± 0.00℃ and 15.31 ± 1.59℃, which were about two days apart. When applying DTU_{zhou3}, the GDD_{cold} and GDD_{warm} were 33.70 ± 1.84℃ and 62.35 ± 3.29℃, which were about three days apart. When applying DTU_{zhou4}, the GDD_{cold} and GDD_{warm} were 7.26 ± 0.41℃ and 13.97 ± 0.72℃, which were about one day apart.

At the end of tillering stage, all GDDs between growing seasons calculated by each DTU methods were significantly different (Figure 1-17b). When applying DTU_{mc1}, the GDD_{cold} and GDD_{warm} were 350.23 ± 8.04℃ and 273.23 ± 9.70℃, which were about 11 days apart. When applying DTU_{mc2}, the GDD_{cold} and GDD_{warm} were 366.58 ± 7.61℃ and 280.85 ± 9.87℃, which were about 11 days apart. When applying $DTU_{nielsen}$, the GDD_{cold} and GDD_{warm} were 350.23 ± 8.04℃ and 270.83 ± 9.81℃, which were about ten days apart. When applying DTU_{trial}, the GDD_{cold} and GDD_{warm} were 184.35 ± 4.76℃ and 208.75 ± 6.62℃, which were about four days apart. When applying DTU_{zhou3}, the

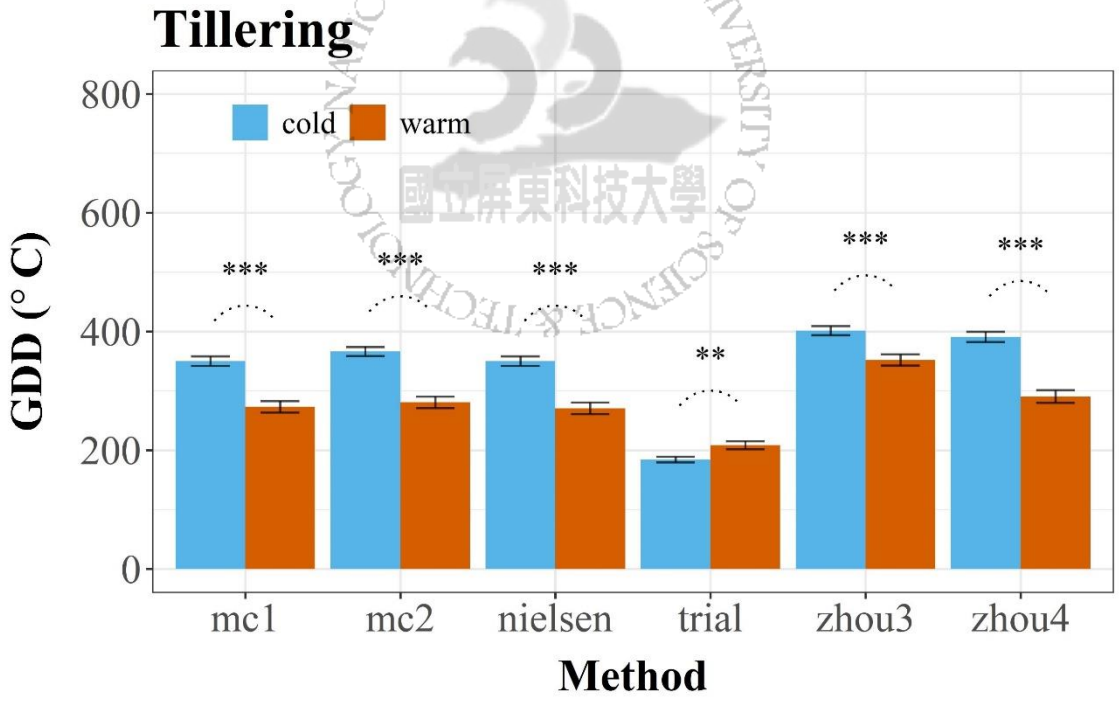
GDD_{cold} and GDD_{warm} were 401.54 ± 7.75℃ and 352.29 ± 9.47℃, which were about five days apart. When applying DTU_{zhou4}, the GDD_{cold} and GDD_{warm} were 391.19 ± 8.66℃ and 290.69 ± 10.65℃, which were about 12 days apart.

At the end of MaxTill, all GDDs between growing seasons calculated by each DTU methods are significantly different (Figure 1-17c). When applying DTU_{mc1}, the GDD_{cold} and GDD_{warm} were 375.02 ± 7.05℃ and 357.72 ± 7.02℃, which were about two days apart. When applying DTU_{mc2}, the GDD_{cold} and GDD_{warm} were 501.35 ± 5.44℃ and 479.52 ± 6.44℃, which were about three days apart. When applying $DTU_{nielsen}$, the GDD_{cold} and GDD_{warm} were 486.83 ± 5.87℃ and 466.30 ± 6.24℃, which were about two days apart. When applying DTU_{trial}, the GDD_{cold} and GDD_{warm} were 218.22 ± 4.06℃ and 342.22 ± 3.79℃, which were about 20 days apart. When applying DTU_{zhou3}, the GDD_{cold} and GDD_{warm} were 618.04 ± 5.23℃ and 677.89 ± 5.77℃, which were about six days apart. When applying DTU_{zhou4}, the GDD_{cold} and GDD_{warm} were 552.47 ± 6.38℃ and 528.15 ± 7.08℃, which were about three days apart. Although these methods yielded significant GDDs differences between growing seasons, DTU_{mc1} and $DTU_{nielsen}$ exhibited lower variation.

At the end of ER, GDDs calculated by DTU_{trial} and DTU_{zhou3} are significantly different, while DTU_{mc1}, DTU_{mc2}, $DTU_{nielsen}$, and DTU_{zhou4} had no significant differences between growing seasons (Figure 1-17d). When applying DTU_{mc1}, the GDD_{cold} and GDD_{warm} were 483.47 ± 10.78℃ and 477.88 ± 7.17℃, which were about one day apart. When applying DTU_{mc2}, the GDD_{cold} and GDD_{warm} were 562.49 ± 7.73℃ and 544.17 ± 5.51℃, which were about two days apart. When applying $DTU_{nielsen}$, the GDD_{cold} and GDD_{warm} were 599.98 ± 9.35℃ and 593.24 ± 5.85℃, which were about one day apart. When applying DTU_{trial}, the GDD_{cold} and GDD_{warm} were 363.93 ± 9.41℃ and 489.19 ± 5.22℃, which were about 20 days apart. When applying DTU_{zhou3}, the GDD_{cold} and GDD_{warm} were 744.30 ± 9.53℃ and 820.50 ± 4.27℃, which were about seven days apart. When applying DTU_{zhou4}, the GDD_{cold} and GDD_{warm} were 609.70 ± 7.97℃ and 605.16 ± 5.89℃, which were about one day apart.



(a)

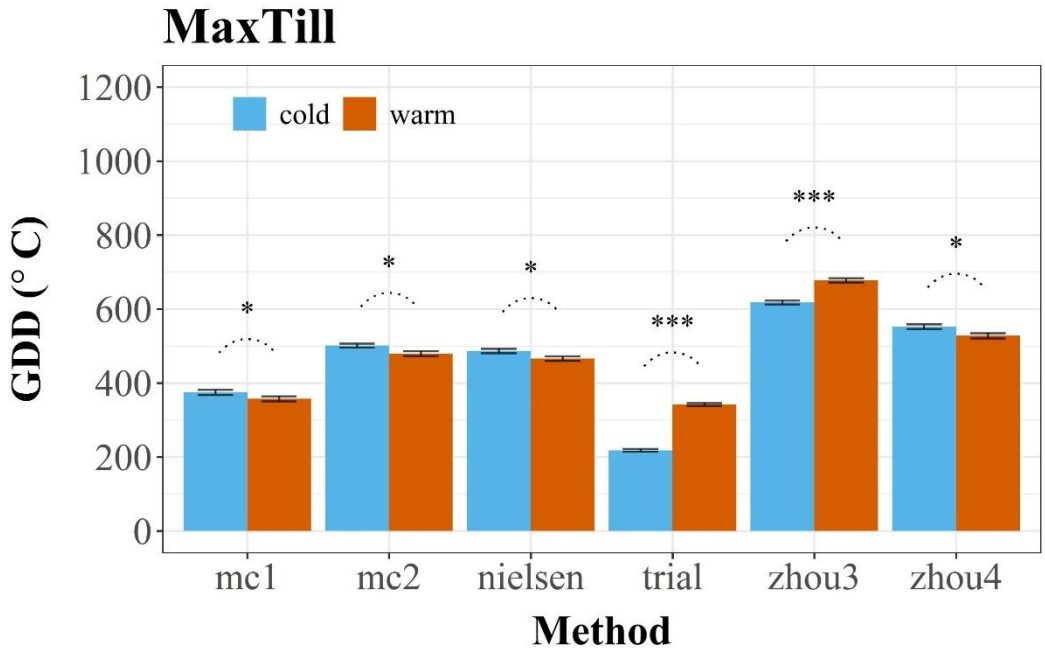


(b)

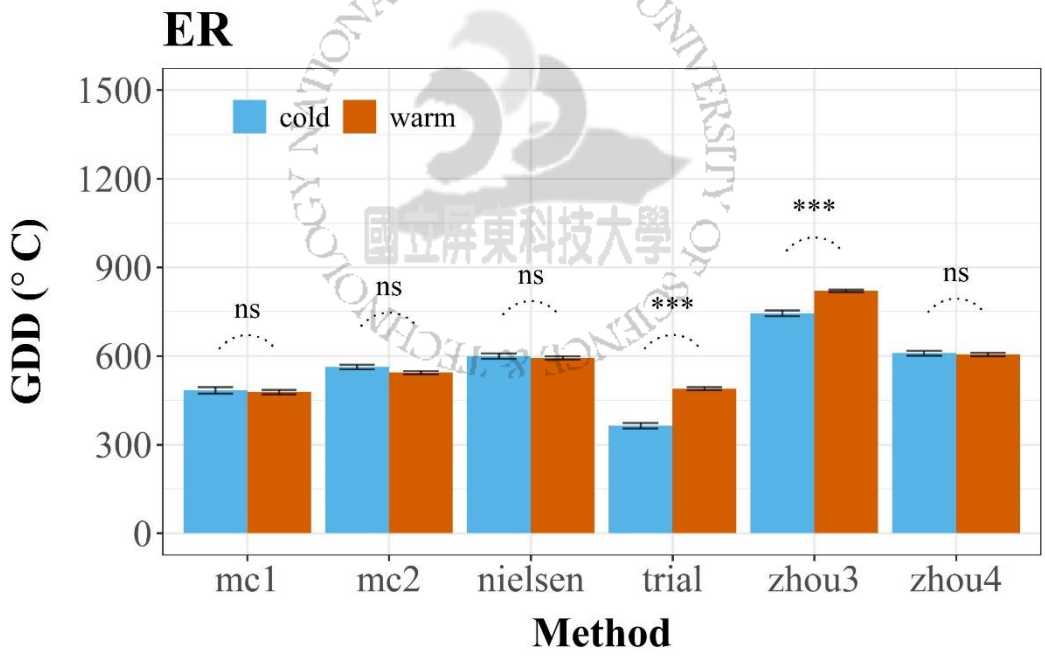
Figure 1-17. Cont.[*]

[*] *Cont. means continue, the figure caption is on backward*

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(c)



(d)

Figure 1-17. GDDs comparison between seasons at the end of (a) seedling, (b) tillering, (c) MaxTill, and (d) ER. Each GDDs were calculated with different T_{base}, T_{up}, and/or T_{opt} according to different growth stages. $^{*, **, ***}$ represent significant level of $p < 0.05$, $p < 0.01$, $p < 0.001$; ^{ns} represents non-significance ($p > 0.05$). NA: two-sample test is not available.

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1.3.5. Growth stage classification with GDD

1.3.5.1. Grid search for optimal temperature parameters

As calculating DTUs according to the growth stage are relatively complex for in-field application, determining a single set of parameters that works for all stages could be beneficial. In this section, the optimal T_{base}, T_{opt}, and T_{up} were determined by classifying growth stages with GDDs using support vector machine with radial basis kernel. Temperature parameters that exhibit the highest accuracy were considered as the optimal set.

In internal testing, each method achieved over-90% accuracy in classifying the seedling stage, tillering stage, MaxTill, ER, and FFE (Table 1-5). The daily-based methods performed slightly better than the hourly-based methods. Among the daily-based methods, the accuracy of DTU_{mc2} (accuracy = 0.9276), $DTU_{nielsen}$ (accuracy = 0.9276), and DTU_{trial} (accuracy = 0.9276) is comparable and outperforming. No overfitting was observed in the external testing for all combinations. The optimal T_{base} for DTU_{mc2}, $DTU_{nielsen}$, and DTU_{trial} are respectively 10.5℃, 16.0℃, and 10.5℃. The optimal T_{up} for DTU_{mc2}, $DTU_{nielsen}$, and DTU_{trial} were respectively 31.5℃, 28.5℃, and 32.5℃.

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Table 1-5. Grid search for optimal temperature parameters via growth stage classification with GDDs.

Method	T_{base}	T_{opt}	T_{up}	Accuracy	
				Internal	External
mc1	8.5	–	50.0	0.9220	0.9104
mc2	10.5	–	31.5	0.9276	0.9083
nielsen	16.0	–	28.5	0.9276	0.9125
trial	10.5	–	32.5	0.9276	0.9083
zhou3	7.0	36.0	50.0	0.9218	0.9104
zhou4	10.0	40.0	50.0	0.9218	0.9042

Accuracy = (TP + TN) / (TP + FP + TN + FN); TP: True positive, FP: False positive, TN: True negative, FN: False negative

1.3.5.2. GDD variation between growing seasons

The variation of each DTU methods between cold and warm seasons at the end of each growth stages (seedling, tillering, MaxTill, and ER) were summarized in Figure 1-18. The overall DTUs across the experiment period were summarized as the mean and standard deviation of population ($\mu \pm \sigma$) to figure the mean basis of daily DTU. The overall DTU_{mc1}, DTU_{mc2}, $DTU_{nielsen}$, DTU_{trial}, DTU_{zhou3}, and DTU_{zhou4} were 15.40 ± 3.96℃, 13.13 ± 3.52℃, 6.25 ± 2.29℃, 13.05 ± 3.60℃, 16.35 ± 4.26℃, and 18.14 ± 4.55℃, respectively.

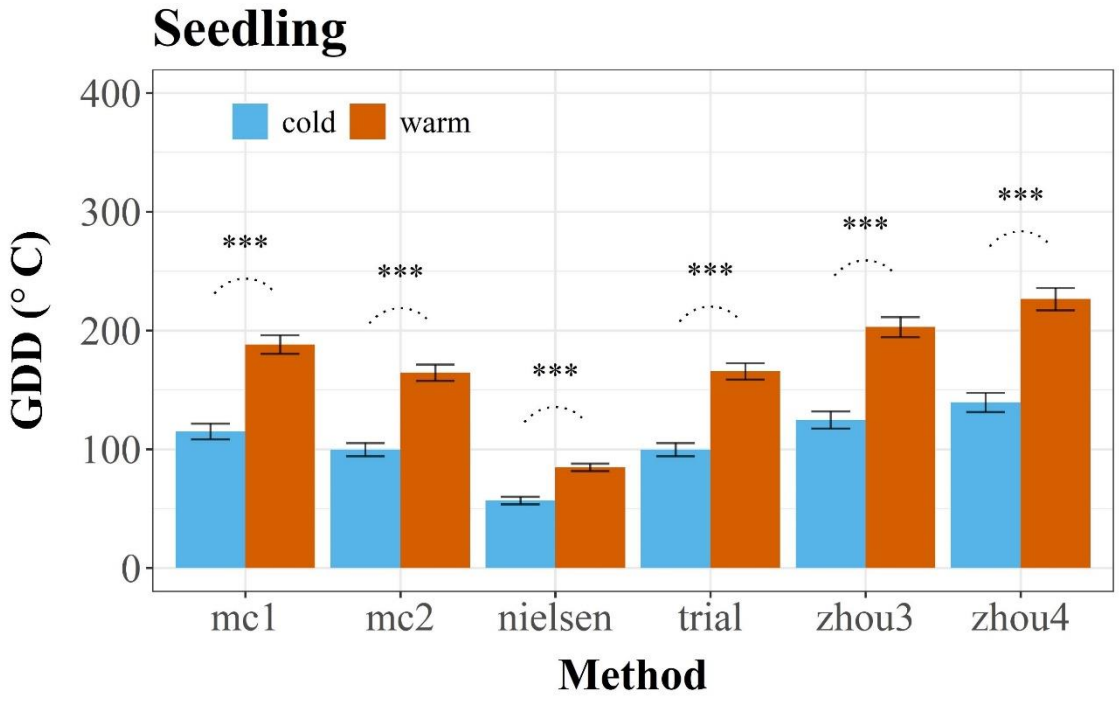
At the end of seedling stage, all GDDs between growing seasons calculated by each DTU methods were significantly different (Figure 1-18a). When applying DTU_{mc1}, the GDD_{cold} and GDD_{warm} were 115.11 ± 6.55℃ and 188.21 ± 7.87℃, which were about five days apart. When applying DTU_{mc2}, the GDD_{cold} and GDD_{warm} were 99.78 ± 5.66℃ and 164.42 ± 6.82℃, which were about five days apart. When applying $DTU_{nielsen}$, the GDD_{cold} and GDD_{warm} were 56.90 ± 3.21℃ and 84.83 ± 3.11℃, which were about four days apart. When applying DTU_{trial}, the GDD_{cold} and GDD_{warm} were 99.78 ± 5.66℃ and 165.67 ± 6.92℃, which was about five days in difference. When applying DTU_{zhou3}, the GDD_{cold} and GDD_{warm} were 124.79 ± 7.20℃ and 202.96 ± 8.52℃, which was about 5 days in difference. When applying DTU_{zhou4}, the GDD_{cold} and GDD_{warm} were 139.51 ± 8.05℃ and 226.55 ± 9.40℃, which was a five-day difference between them.

At the end of tillering stage, all GDDs between growing seasons calculated by each DTU methods were significantly different (Figure 1-18b). When applying DTU_{mc1}, the GDD_{cold} and GDD_{warm} were respectively 542.12 ± 6.23℃ and 495.03 ± 8.26℃, which were about three days apart. When applying DTU_{mc2}, the GDD_{cold} and GDD_{warm} were respectively 462.78 ± 5.35℃ and 425.77 ± 7.05℃, which were about three days apart. When applying $DTU_{nielsen}$, the GDD_{cold} and GDD_{warm} were respectively 233.29 ± 2.52℃ and 203.47 ± 3.37℃, which was about 5 days in difference. When applying DTU_{trial}, the GDD_{cold} and GDD_{warm} were respectively 462.78 ± 5.35℃ and 425.49 ± 7.01℃, which was about three days in difference. When applying DTU_{zhou3}, the

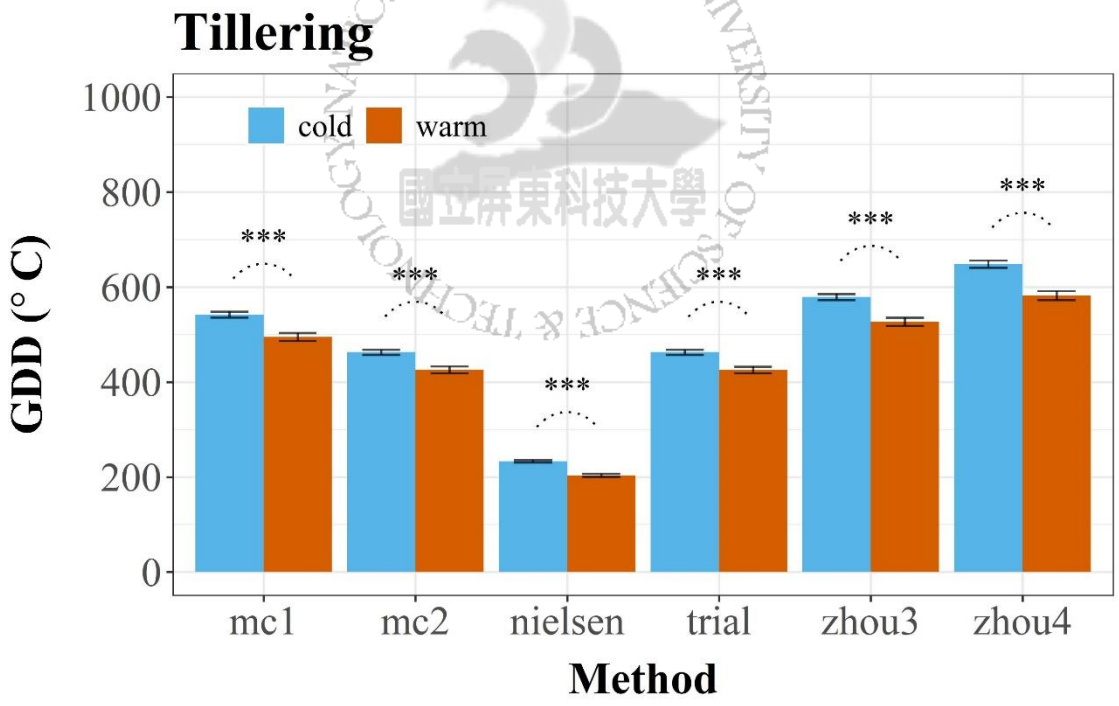
GDD_{cold} and GDD_{warm} were respectively 579.08 ± 6.59℃ and 526.77 ± 8.57℃, which were about three days apart. When DTU_{zhou4} was applied, the GDD_{cold} and GDD_{warm} values were 648.36 ± 7.25℃ and 582.06 ± 9.46℃ respectively, with a difference of about four days between them.

At the end of MaxTill, the differences of GDDs between growing seasons accumulated by DTU_{mc1}, DTU_{mc2}, $DTU_{nielsen}$, and DTU_{trial} were insignificant (Figure 1-18c). When applying DTU_{mc1}, the GDD_{cold} and GDD_{warm} were respectively 896.05 ± 3.85℃ and 886.07 ± 5.77. When applying DTU_{mc2}, the GDD_{cold} and GDD_{warm} were respectively 764.05 ± 3.28℃ and 765.87 ± 4.95℃. When applying $DTU_{nielsen}$, the GDD_{cold} and GDD_{warm} were respectively 384.25 ± 1.69℃ and 379.67 ± 2.25℃. When applying DTU_{trial}, the GDD_{cold} and GDD_{warm} were respectively 764.05 ± 3.28℃ and 765.97 ± 4.99℃. Despite the significance of DTU_{zhou3} and DTU_{zhou4}, the GDDs values between growing seasons were relatively close. When applying DTU_{zhou3}, the GDD_{cold} and GDD_{warm} were respectively 957.31 ± 4.12℃ and 939.83 ± 6.04℃, which were about one day apart. When applying DTU_{zhou4}, the GDD_{cold} and GDD_{warm} were respectively 1066.75 ± 4.54℃ and 1039.15 ± 6.65℃, with a difference of about two days between them. All DTU methods were acceptable for assessing the starting point of ER.

At the end of ER, The GDD_{mc1}, $GDD_{nielsen}$, GDD_{zhou3}, and GDD_{zhou4} exhibited significant differences between growing seasons, while the GDD_{mc2} and GDD_{trial} were in the contrary (Figure 1-18d). When applying DTU_{mc1}, the GDD_{cold} and GDD_{warm} were 1141.40 ± 11.74℃ and 1099.09 ± 6.95℃, which were about three days apart. When applying DTU_{mc2}, the GDD_{cold} and GDD_{warm} were 968.37 ± 9.63℃ and 945.60 ± 5.80℃. When applying $DTU_{nielsen}$, the GDD_{cold} and GDD_{warm} were 466.77 ± 2.18℃ and 458.94 ± 2.48℃, which were about two days apart. When applying DTU_{trial}, the GDD_{cold} and GDD_{warm} were 963.77 ± 8.55℃ and 943.11 ± 5.65℃. When applying DTU_{zhou3}, the GDD_{cold} and GDD_{warm} were 1211.80 ± 11.64℃ and 1164.36 ± 7.44℃, which were about three days apart. When applying DTU_{zhou4}, the GDD_{cold} and GDD_{warm} were 1352.57 ± 13.56℃ and 1284.68 ± 8.09℃, which were about four days apart.

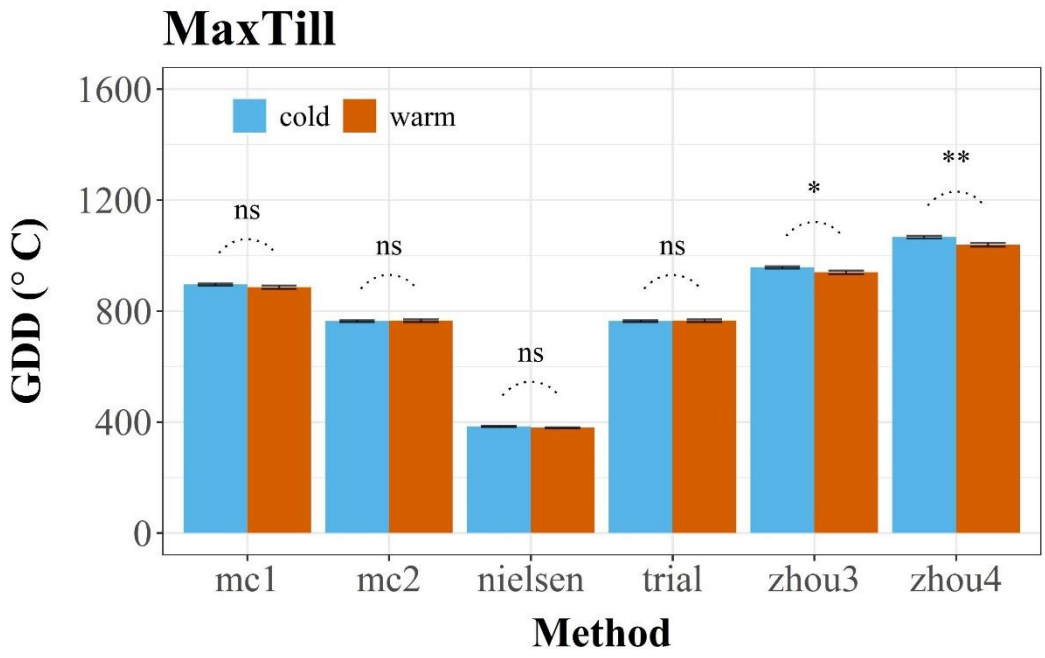


(a)

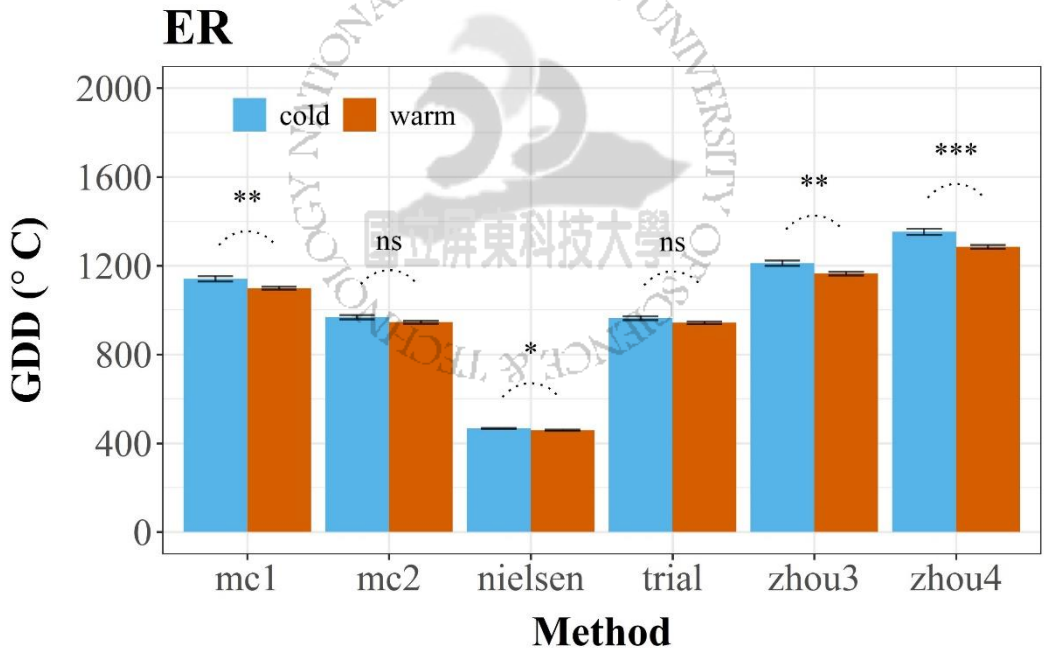


(b)

Figure 1-18. Cont.



(c)



(d)

Figure 1-18. GDDs comparison between seasons at the end of (a) seedling, (b) tillering, (c) MaxTill, and (d) ER. Each GDDs were accumulated by DTUs calculated with consistent T_{base}, T_{up}, and/or T_{opt} across different growth stages. $^{*, **, ***}$ represent significant level of $p < 0.05$, $p < 0.01$, $p < 0.001$; ^{ns} represents non-significance ($p > 0.05$).

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1.3.5.3. Field application

The GDD_{mc2} and GDD_{trial} were additionally tested with a two-years (four crop seasons) field experiment. The threshold values for both methods are 765℃, therefore, the evaluation results for them are identical (Figure 1-19). The accuracy, precision, sensitivity, and specificity are respectively 0.6579, 0.6484, 0.8939, and 0.3333. When either the GDD_{mc2} or GDD_{trial} reached 765℃, there are probably 64.84% of plants in the field were transited to reproductive phase (precision = 0.6484). Besides, by setting the threshold of GDD at 765℃, the positive classification covered 89.39% of the in-field occurrences of phase transition of the plants (sensitivity = 0.8939).

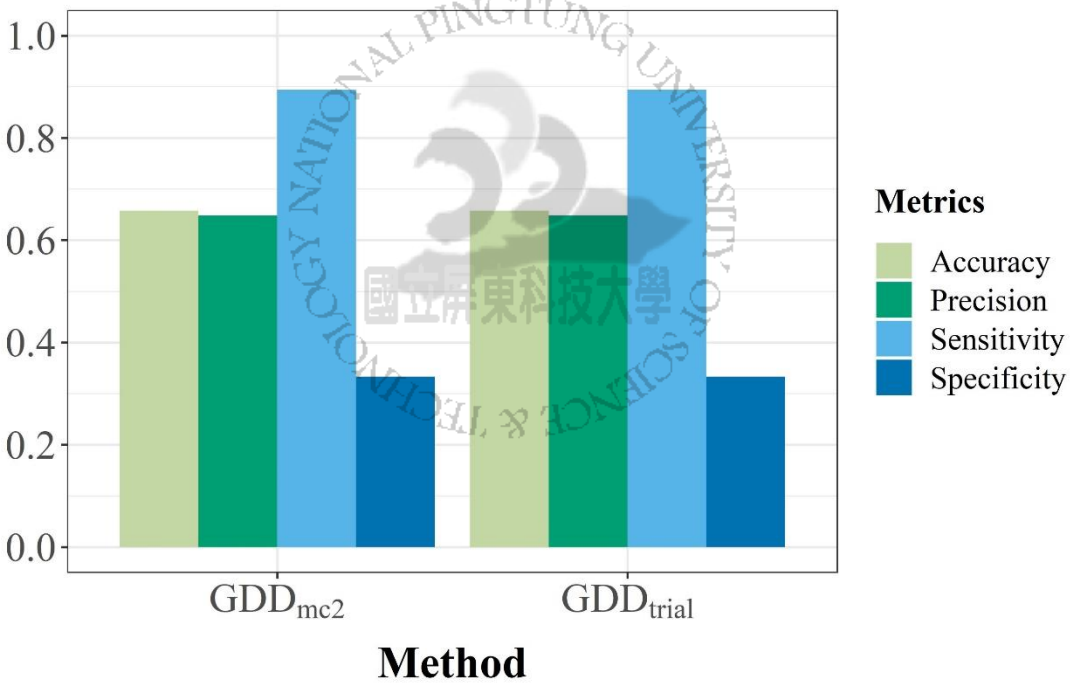


Figure 1-19. Field application by using the threshold values of GDD_{mc2} and GDD_{trial}. The threshold values for both methods are 765℃. The T_{base} for both methods is 10.5℃. The T_{up} for both methods are respectively 31.5℃ and 32.5℃.

1.4. Conclusion

This work applied an uncommon scaling system (*i.e.* leaf appearance scale, LAS), which is probably a novel approach, to quantify the growth of rice plant. The advantages for using LAS compared to HAUN can be summarized to three points. The first, LAS is a continuous scale, which is critically important to investigate the relationship between temperature and growth rate. The second, the LAS is necessarily ranged from zero to one regardless of cultivars and weather conditions. The third, LAS values during ER in different growing seasons have no significant differences, which could improve the integrity of GDD system for rice cultivation. By retrieving growth rate and DTU calculated by different combinations of temperature parameters (T_{base}, T_{opt}, and T_{up}) in different growth stages, DTU_{zhou3} generally exhibits the most precise and stable d.LAS estimation. Unfortunately, the GDD values accumulated with dynamic DTU calculation during each growth stages yield significant differences between growing seasons. On the other hand, by classifying growth stages using GDDs accumulated with fixed DTU calculation, it is also possible to figure out the optimal temperature parameters. On the other hand, the optimal T_{base}, T_{opt}, and T_{up} could also be obtained by direct classification of each growth stages with GDDs. The accuracy of each method is comparable and satisfying. Among the methods, the GDD values between growing seasons of DTU_{mc2} and DTU_{trial} have no significant differences after the end of MaxTill. The T_{base} for DTU_{mc2} and DTU_{trial} are both 10.5℃, while the T_{up} are respectively 31.5℃ and 32.5℃. This result revealed the possibility of monitoring rice growth stages with a single set of GDD values regardless of different growing seasons. Further efforts are still required to evaluate the feasibility of this methodology by testing in different sites, incorporating with different cultivation practices, and involving a variety of rice cultivars.

1.5. Appendix

Appendix 1-A. Temperature records

DAT	Cold season				Warm season			
	Date (y/m/d)	T_{min}	T_{mean}	T_{max}	Date (y/m/d)	T_{min}	T_{mean}	T_{max}
1	2019/11/6	20.5	23.9	28.9	2020/4/22	20.7	24.4	29.3
2	2019/11/7	19.1	23.3	28.2	2020/4/23	21.1	23.9	27.0
3	2019/11/8	19.5	23.8	28.7	2020/4/24	20.6	22.5	25.2
4	2019/11/9	17.8	23.1	28.5	2020/4/25	19.7	23.9	28.0
5	2019/11/10	18.1	22.7	27.5	2020/4/26	20.9	26.2	32.3
6	2019/11/11	18.0	22.9	28.4	2020/4/27	22.6	26.3	30.2
7	2019/11/12	17.3	23.1	28.4	2020/4/28	22.5	25.6	29.4
8	2019/11/13	18.3	23.5	28.7	2020/4/29	20.6	25.6	30.4
9	2019/11/14	19.6	23.6	28.1	2020/4/30	20.8	26.3	31.9
10	2019/11/15	18.4	23.0	27.5	2020/5/1	21.8	27.1	32.8
11	2019/11/16	18.7	23.4	29.2	2020/5/2	22.8	28.2	32.7
12	2019/11/17	17.3	23.8	31.1	2020/5/3	22.5	27.7	32.0
13	2019/11/18	20.6	25.0	30.0	2020/5/4	24.9	28.4	33.8
14	2019/11/19	21.6	23.2	25.8	2020/5/5	23.1	28.4	32.6
15	2019/11/20	21.3	23.6	27.1	2020/5/6	24.1	29.2	34.4
16	2019/11/21	20.3	23.7	29.2	2020/5/7	24.5	29.5	33.9
17	2019/11/22	18.7	23.5	29.8	2020/5/8	24.6	28.6	32.7
18	2019/11/23	20.2	24.3	28.4	2020/5/9	24.4	28.9	33.4
19	2019/11/24	20.2	24.4	29.3	2020/5/10	25.2	28.9	33.9
20	2019/11/25	20.9	24.7	29.3	2020/5/11	25.8	29.2	33.7
21	2019/11/26	21.2	24.0	27.2	2020/5/12	25.2	27.8	32.2
22	2019/11/27	19.2	24.1	29.6	2020/5/13	25.0	27.7	32.2
23	2019/11/28	19.5	22.8	26.3	2020/5/14	23.6	28.1	33.1
24	2019/11/29	17.9	21.1	25.3	2020/5/15	23.5	29.4	36.4
25	2019/11/30	15.6	21.2	26.5	2020/5/16	25.3	29.7	36.3

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26	2019/12/1	17.6	22.1	27.4	2020/5/17	23.7	28.2	33.5
27	2019/12/2	16.9	20.6	26.2	2020/5/18	25.1	28.4	32.8
28	2019/12/3	13.4	18.7	24.2	2020/5/19	23.8	25.1	29.1
29	2019/12/4	16.8	18.4	22.2	2020/5/20	23.2	25.5	29.7
30	2019/12/5	15.8	17.6	21.0	2020/5/21	24.2	25.3	27.9
31	2019/12/6	15.8	17.0	18.8	2020/5/22	24.2	25.4	27.4
32	2019/12/7	14.9	16.6	18.2	2020/5/23	24.8	26.3	28.9
33	2019/12/8	12.6	17.8	23.2	2020/5/24	24.8	27.3	30.7
34	2019/12/9	12.1	18.3	24.1	2020/5/25	25.0	28.4	32.4
35	2019/12/10	12.0	18.7	26.4	2020/5/26	24.7	27.1	30.8
36	2019/12/11	11.7	18.9	26.7	2020/5/27	23.6	25.6	28.5
37	2019/12/12	13.6	19.4	26.1	2020/5/28	24.6	27.3	32.4
38	2019/12/13	13.1	19.3	27.1	2020/5/29	24.5	27.8	31.9
39	2019/12/14	14.0	21.0	29.0	2020/5/30	24.1	28.7	32.9
40	2019/12/15	16.0	21.9	28.6	2020/5/31	25.3	29.4	33.4
41	2019/12/16	16.6	22.4	30.2	2020/6/1	25.3	29.8	33.8
42	2019/12/17	16.9	22.6	30.3	2020/6/2	25.5	29.5	33.9
43	2019/12/18	18.3	23.3	29.3	2020/6/3	24.1	29.2	33.6
44	2019/12/19	17.4	22.0	27.3	2020/6/4	24.2	29.6	34.6
45	2019/12/20	18.5	21.5	25.2	2020/6/5	24.2	29.0	34.3
46	2019/12/21	19.1	22.4	28.1	2020/6/6	24.5	28.0	32.9
47	2019/12/22	18.1	22.8	29.1	2020/6/7	23.1	27.9	33.5
48	2019/12/23	17.6	22.0	28.4	2020/6/8	23.4	26.2	32.2
49	2019/12/24	16.9	22.3	29.0	2020/6/9	22.0	27.5	32.6
50	2019/12/25	18.0	22.6	28.5	2020/6/10	25.2	28.5	32.5
51	2019/12/26	17.6	23.2	29.0	2020/6/11	24.4	29.5	35.0
52	2019/12/27	18.8	20.0	22.1	2020/6/12	25.5	30.2	36.1
53	2019/12/28	18.4	20.8	24.5	2020/6/13	25.5	29.4	34.8
54	2019/12/29	18.4	20.7	23.9	2020/6/14	24.8	29.1	33.8
55	2019/12/30	19.5	21.9	26.1	2020/6/15	24.1	29.4	34.5
56	2019/12/31	19.6	21.1	24.1	2020/6/16	24.4	29.2	34.6

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57	2020/1/1	18.3	20.4	24.3	2020/6/17	24.5	29.4	34.2
58	2020/1/2	15.9	20.0	24.7	2020/6/18	23.7	29.4	34.5
59	2020/1/3	15.3	19.9	24.7	2020/6/19	24.8	29.7	34.6
60	2020/1/4	14.3	19.7	26.4	2020/6/20	25.4	30.2	35.2
61	2020/1/5	14.4	20.1	25.9	2020/6/21	25.0	28.7	33.5
62	2020/1/6	14.7	20.3	27.4	2020/6/22	24.4	29.4	33.7
63	2020/1/7	15.0	20.5	27.2				
64	2020/1/8	15.1	20.9	27.9				
65	2020/1/9	16.0	20.8	25.6				
66	2020/1/10	16.9	21.6	27.4				
67	2020/1/11	15.7	20.9	27.4				
68	2020/1/12	14.9	19.4	25.2				
69	2020/1/13	14.5	19.3	25.8				
70	2020/1/14	12.9	19.8	27.5				
71	2020/1/15	13.0	19.0	24.5				
72	2020/1/16	13.7	20.0	27.8				
73	2020/1/17	13.7	18.9	26.2				
74	2020/1/18	12.6	17.8	23.2				
75	2020/1/19	12.6	18.4	25.1				
76	2020/1/20	12.4	19.2	26.7				
77	2020/1/21	14.3	19.9	26.1				
78	2020/1/22	15.9	22.2	29.0				
79	2020/1/23	17.9	23.1	28.4				
80	2020/1/24	18.1	23.1	29.3				
81	2020/1/25	16.7	20.5	25.3				
82	2020/1/26	17.8	21.4	29.0				
83	2020/1/27	16.2	19.0	23.4				
84	2020/1/28	15.2	16.5	18.5				
85	2020/1/29	13.2	16.2	20.8				
86	2020/1/30	9.8	14.6	21.7				
87	2020/1/31	6.7	14.1	21.4				

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88	2020/2/1	8.5	15.6	23.6
89	2020/2/2	9.4	17.3	25.7
90	2020/2/3	11.7	18.6	25.3
91	2020/2/4	12.8	19.0	26.3
92	2020/2/5	14.2	19.6	24.4
93	2020/2/6	13.4	19.2	24.8
94	2020/2/7	14.7	20.3	27.5
95	2020/2/8	14.5	19.2	25.3
96	2020/2/9	15.9	19.4	24.0
97	2020/2/10	13.2	19.1	25.1
98	2020/2/11	15.0	19.4	23.1
99	2020/2/12	15.3	22.8	29.7
100	2020/2/13	18.6	23.5	28.0

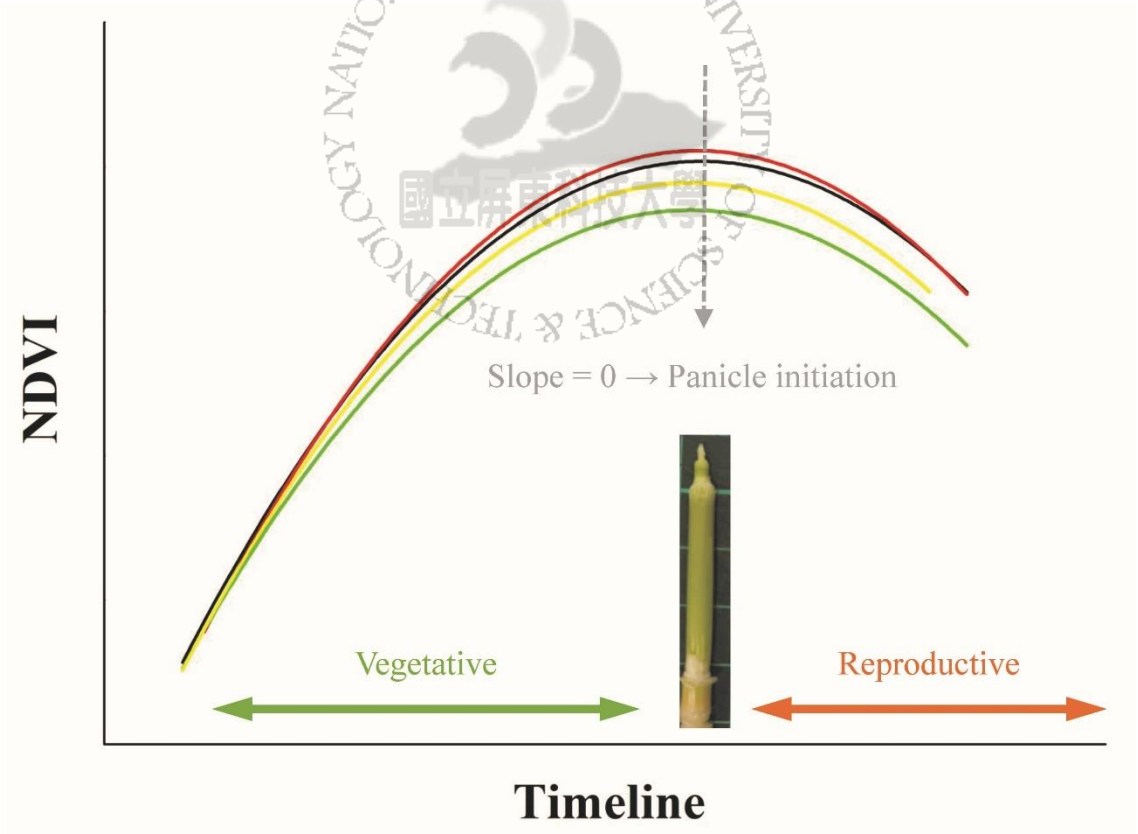
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Chapter 2. Assessment of the rice panicle initiation by using NDVI-based vegetation indices

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Highlights

- The turning point of NDVI (quadratic polynomial regression) represent the transition from vegetative phase to reproductive phase.
- NDVI (700, 720), NDVI (700, 730), NDVI (708, 730), NDVI (700, 760), NDVI (708, 760), and NDVI (708, 800) are potential indices to determine panicle initiation stage.



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2.1. Introduction

Rice (*Oryza sativa* L.) is one of the most important staple foods for more than half of the world population. Due to the rapid growth of food demand and limited arable land, improving yield potential to boost up future rice production is an urgent need. Rice yield is known to be increased by the nitrogen topdressing at the panicle initiation (PI) stage that is the beginning of the reproductive stage (Turner and Jund, 1991; Inamura et al., 2003; Ghaley, 2012). According to the recommendation of the Agricultural Improvement Committee in Taiwan, when the length of the panicle is found to be 2 mm, the nitrogen fertilizer should be applied within two days. Applying a large amount of nitrogen fertilizer before PI would easily cause excessive stem elongation and thus tend to increase lodging risk. On the contrary, applying nitrogen fertilizer after PI is less effective on rice yield improvement. Therefore, accurate determination of the PI stage is crucial for rice production.

The PI is generally considered as the turning point between the vegetative phase and the reproductive phase. When a rice plant has reached maximum tillering, the internodes of the rice stem start elongating and subsequently panicle initiating. The overlapping period between maximum tillering and PI is also termed as vegetative-lag phase. The differences of the rice appearance between the vegetative-lag and PI are obscure to the naked eye. Therefore, they are difficult to be distinguished directly by human observation.

The general methods to assess PI in the farm are identifying the internode-elongation, green-ring (Counce et al., 2000; Moldenhauer and Slaton, 2001), the length of the young panicle (1–2 mm) in the cross section of dissected stem (Moldenhauer and Slaton, 2001), and leaf number index/leaf appearance (Ellis et al., 1993). When 30% of the main culms have panicles 2 mm or longer, it is considered as the panicle differentiation (PD) stage (Turner and Jund, 1991) that is late for nitrogen topdressing. Those methods are inconvenient and inefficient for large scale estimation. Some other convenient, large-scale, and non-destructive approaches that help to monitor the plant growth stages, such as modified-calendar days and heat units, are potential candidates to be used

for the precise estimation of PI. The calendar days method is the easiest approach but is not reliable enough as it is largely affected by the weather variability during the cultivation period (Gilmore and Rogers, 1958; Nielsen and Hinkle, 1996; Snyder et al., 1999). Growing degree day (GDD) is an excellent heat unit that has been widely applied in corn production (Angel et al., 2017) since it was first proposed to describe the timeline of biological development (Gilmore and Rogers, 1958). However, these methods are rudimentary and are not able to distinguish the variability between fields. Nowadays, quantitative assessment, real-time and site-specific management of precision agriculture are the objectives of the community.

Spectral remote sensing is another potential approach for the estimation of various variables that are correlated to plant architecture and physiology, providing high-throughput information non-destructively and rapidly for precision agriculture. The arithmetic combinations of vegetation spectral reflectance, which are usually termed as vegetation indices (VIs), became useful indicators for studying plant health and status. The normalized difference vegetation index (NDVI), which is calculated through a normalization procedure (Rouse et al., 1974), seems to be the most popular and long-established VI. The NDVI is sensitive to responses on the green vegetation (Tucker, 1979) as it correlates with the biophysical and physiological changes in plants (Gamon et al., 1992; Myneni and Williams, 1994; Carlson and Ripley, 1997; Sims and Gamon, 2002; Zhou et al., 2017b; Tan et al., 2018). Moreover, the NDVI has been subsequently developed to the other NDVI-based VIs, such as normalized differential red-edge (NDRE), green NDVI (GNDVI), and normalized difference spectral index (NDSI) (Gitelson et al., 1996; Tanaka et al., 2015; Yang et al., 2017). Recent studies put efforts on the quantification of nitrogen contents in plants through remote sensing to optimize the fertilization efficiency and increased the yield (Peng et al., 2010; Xue et al., 2014; Ali et al., 2015; Onoyama et al., 2015; Cao et al., 2016). In the field of rice research, NDVI is one of the most frequently used vegetation indexes. For example, preceding studies revealed that the NDVI is

useful in the research of rice breeding, nitrogen use efficiency monitoring, and rice yield prediction (Phyu et al., 2020; Son et al., 2020; Jiang et al., 2021). Furthermore, NDVI has also been used to monitor the rice growth stages, including the panicle development stage (Wang et al., 2014). The preceding studies revealed the trend of NDVI changes during the rice growth cycle; however, the evaluation of PI prediction is currently unavailable.

The main objective of this study, therefore, is to investigate the relationship between NDVI and PI occurrence. Meanwhile, a non-destructive and high temporal resolution approach was also expected to be established for PI assessment in this study.

2.2. Materials and methods

2.2.1. Field experiment design and management

Pot experiments were conducted in 2019 at the experimental field (22◦38′ N, 120◦36′ E) of National Pingtung University of Science and Technology, Pingtung, Taiwan, to examine the canopy reflectance behavior as a function of panicle initiation on rice (*Oryza sativa* L.). A japonica cultivar, 'Kaohsiung147' (KH147), was planted into 4 groups with 5 replications. Ammonium sulfate was applied at a rate of 150 kg/ha N in each group (20% for basal, 20% for 1st tillering topdressing, 30% for 2nd tillering topdressing, and 30% for panicle initiation topdressing).

2.2.2. Determination of PI through dissection

The PI stage was determined through manual dissection and feature observation. The entrance of the PI stage was verified via either the green ring (Figure 2-1a), internode elongation, or when a 1–2 mm panicle was observed (Figure 2-1b). The ideal timing for nitrogen topdressing is the interval between green ring formation and 2 mm panicle appearance. When the length of the panicle was over 2 mm, it was considered as the panicle differentiation (PD) stage (Figure 2-1c) that was not included in PI determination. The PI

occurrence of each group was determined if more than 50% of dissected samples were verified to PI.

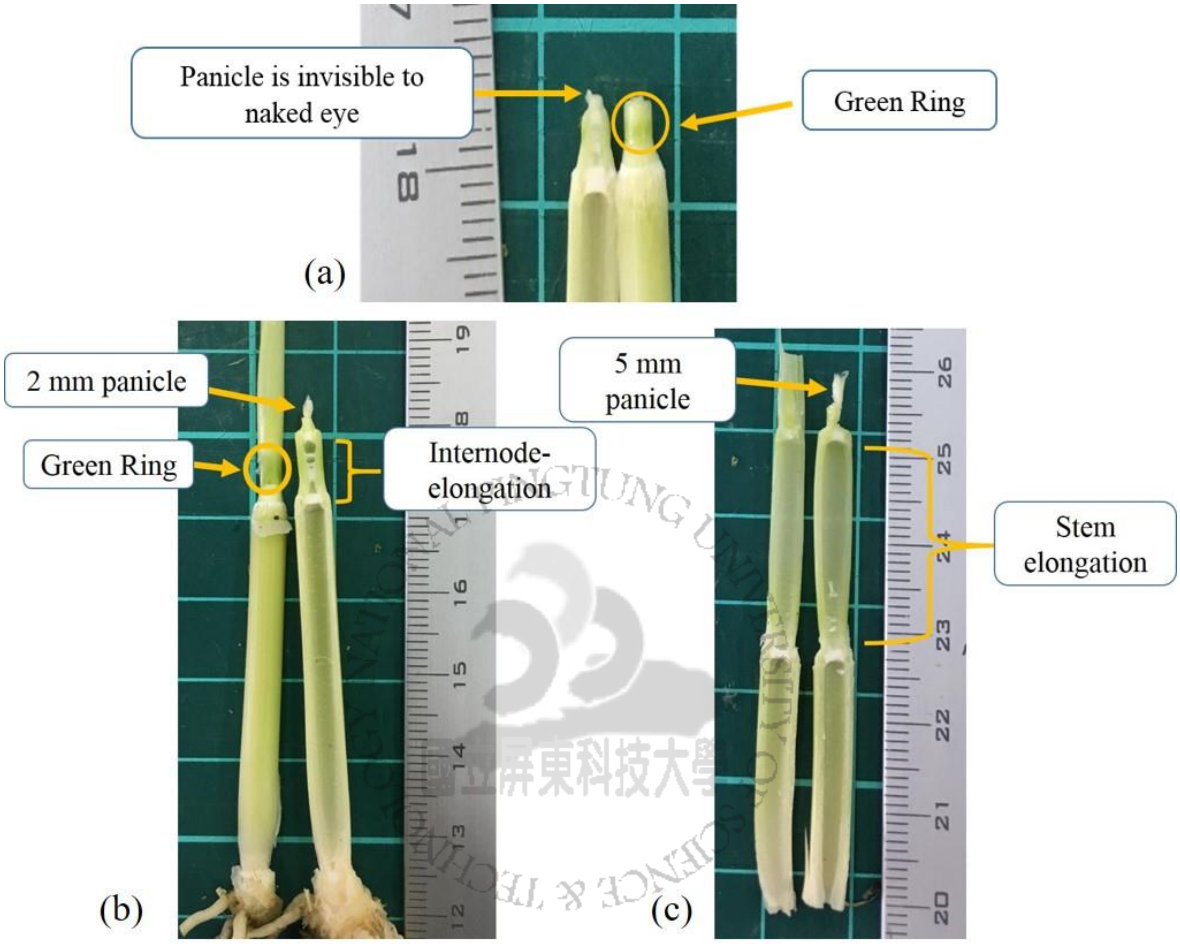


Figure 2-1. The early stage of panicle initiation (a), and the optimum stage for topdressing (b). The green ring disappeared, significant stem elongation, panicle above 2 mm, is considered as panicle differentiation stage (c).

2.2.3. Spectral measurement and NDVI calculation

The spectral data of the rice canopy was collected by using SpectraPen SP-100 (range 640-1050 nm, 2 nm scan-range, Photon Systems Instruments, Czech), which is a non-imaging, handheld hyperspectral sensor. The data collection period was between 10 days after transplanting (DAT) and 80 DAT. The acquisition time was between 10:00 a.m. and 2:00 p.m., while the sensor was held horizontally at nadir view in the position about 30-50 mm above the highest leaf of the plants. The integration time of spectrum collection was set to auto-sensitivity to minimize the interference of sunlight intensity variability. The collected spectrum data were used to calculate NDVI-based VIs (Equation 2-1), which can be expressed as:

$$\text{NDVI-based VIs} = \frac{\lambda_a - \lambda_b}{\lambda_a + \lambda_b} \qquad \text{(Equation 2-1)}$$

where the λ_a and λ_b respectively denote the reflectance of near-infrared (NIR) and red wavelengths.

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2.2.4. Reference wavelengths and recombined NDVI-based VIs

The key wavelengths were selected by some physiologically related reference wavelengths with the equation of NDVI that have been used in previous studies (Table 2-1). Those reference wavelengths were recombined and re-calculated as the normalized difference procedure with all samples.

2.2.5. Reference wavelengths and recombined NDVI-based VIs

The trends of the NDVI-based VIs were changed along the rice plant development, and the quadratic slope always gradually decreased after reaching the reproductive phase. We assumed that the NDVI might have reached maxima during PI, and therefore, the slope would be zero (Figure 2-2).

Table 2-1. Reference wavelengths correlating with physiological traits

Reference wavelengths	Related components	References
NDVI (720, 800)	LAI, yield	(Zhou et al., 2017c)
NDVI (708, 760)	Photosynthesis (FPAR)	(Tan et al., 2018)
NDVI (660, 740)	$NDVI_{leaf}$ and $NDVI_{canopy}$	(Yu et al., 2016)
NDVI (680, 800)	Plant N concentration. Chlorophyll content	(Sims and Gamon, 2002; Zheng et al., 2018)
NDVI (670, 780)	Plant N concentration. Plant N uptake	(Lu et al., 2017)
670, 700, 730 nm	N, P, S-related	(Mahajan et al., 2017)

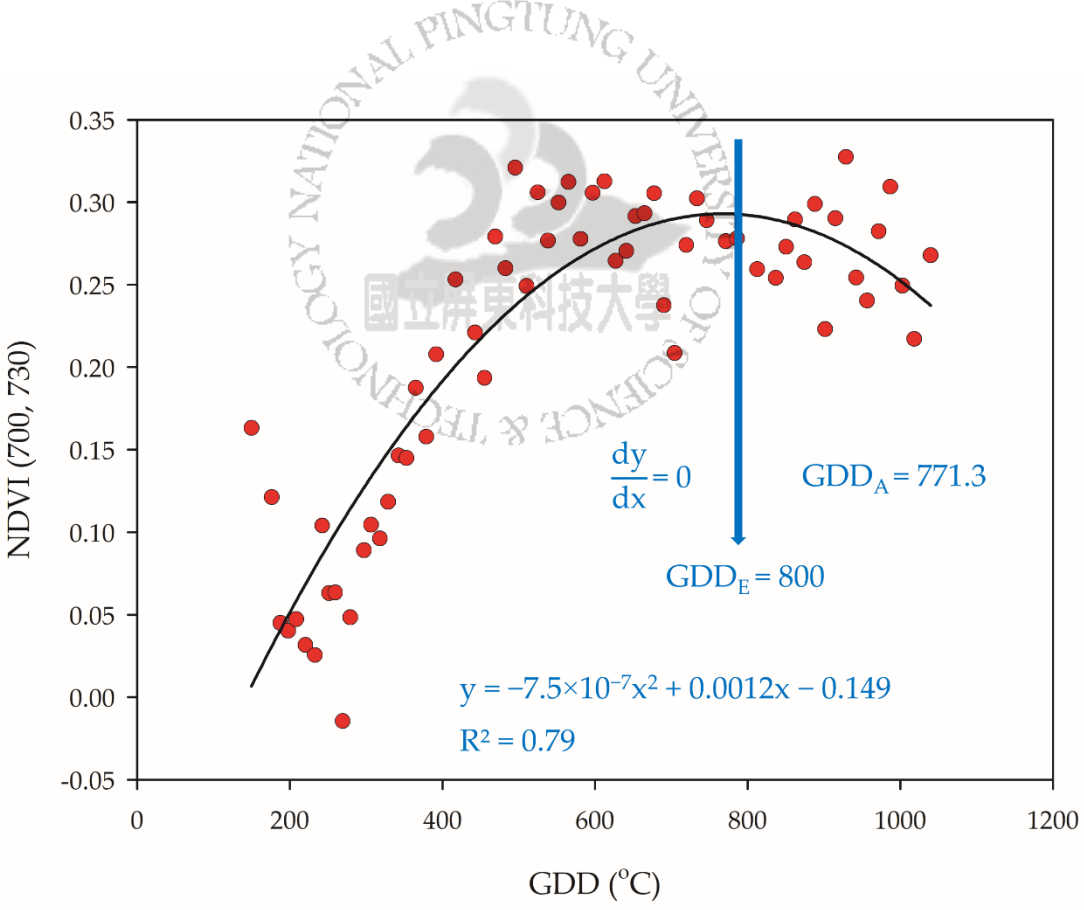


Figure 2-2. Illustration for the estimation of PI occurrence. When the quadratic slope is equal to zero, the actual determination of PI was $GDD_A = 771.3$ through dissection, and the estimation of PI was $GDD_E = 800$.

2.2.6. Reference wavelengths and recombined NDVI-based VIs

Temperature data were obtained from the 'Chishan' meteorological station (22°35′ N, 120°36′ E), which is located approximately seven kilometers away from the experimental site. The GDD of "Method 1" as stated by preceding study was selected as a substitution to represent the cultivation timing (Equation 2-2), according to which (McMaster and Wilhelm, 1997):

$$\mathrm{GDD} = \sum[\frac{\mathrm{T_{max}} + \mathrm{T_{min}}}{2}] - \mathrm{T_{base}} \qquad \text{(Equation 2-2)}$$

where the T_{max} and T_{min} are the daily maximum and minimum air temperature, and the T_{base} is the base temperature that was set to 10 °C (Nielsen and Hinkle, 1996).

2.2.7. Statistical analysis

The statistical analysis of the data was done by using Microsoft Excel 2013 (Microsoft Corporation, Redmond, Washington, USA). The quadratic polynomial regression analysis was performed in SigmaPlot 10.0 (Systat Software Inc., San Jose, California, USA).

The model was validated by leave-one-out cross validation (LOOCV). During the LOOCV process, each group was progressively and alternately held out for model validation, while the remaining groups were used for model construction. The appropriateness of the estimation of PI occurrence and prediction of the NDVI-based VIs was tested based on relative error (RE) (Equation 2-3):

$$\mathrm{RE} = \sqrt{\frac{\sum_{i=1}^{n}[(A_i - P_i)/A_i]^2}{n}} \times 100\% \qquad \text{(Equation 2-3)}$$

in which the A_i and P_i are the actual and predicted value of the ith data point and n is the number of data points. Estimations were considered excellent if RE is <10%, good between 10% and 20%, fair between 20% and 30%, and poor if it is >30% (Jamieson et al., 1991; Mahajan et al., 2017).

2.3. Results and Discussion

2.3.1. PI determination through manual dissection

The PI of each group was observed through manual dissection. The actual determination of PI for group 1 was 60 DAT (GDD = 771.3 °C), group 2 was 56 DAT (GDD = 713.9 °C), group 3 was 57 DAT (GDD = 783.4 °C), and group 4 was 55 DAT (GDD = 791.1 °C) (Table 2-2). The group 1 encountered with low temperature during the tillering stage. As a result, the required DAT of group 1 for PI was five days longer than group 4, which was thoroughly grown under warm conditions. Besides this, group 2 encountered low temperature before the tillering stage. Consequently, the required GDD of group 2 for PI was approximately 80°C lower than group 4. These biases indicated that the DAT and GDD are both largely affected by weather variability.

Table 2-2. Observation of PI in each group.

Group	Date	DOY	DAT	GDD (°C)
1	2019-03-04	63	60	771.3
2	2019-03-14	73	56	713.9
3	2019-03-29	88	57	783.4
4	2019-04-10	100	55	791.1

DOY: day of year; DAT: days after transplanting; GDD: Growing degree day.

2.3.2. NDVI-based VIs selection

All samples were regressed with quadratic polynomial models, as we assumed that the trend of this model (when the slope is equal to zero) might be correlated with the entrance of the PI stage. Considering the ability of explanation, NDVI (700, 720), NDVI (700, 730), NDVI (708, 730), NDVI (660, 760), NDVI (700, 760), NDVI (708, 760), and NDVI (708, 800) were selected by the R^2 value that are above 0.7 of quadratic polynomial regression model (Table 2-3). On the other hand, all the NDVI-based VIs of groups was stratified, where the distribution of groups 1 and 2 were lower than groups 3 and 4 (Figure 2-3). The reason could be that group 1 encountered low temperature during the tillering stage, while group 2 encountered low temperature before the tillering stage. A preceding study indicated that more uniquely expressed proteins were found at 20/12 °C (day/night) and frequently alternating stress/non-stress temperature changes, leading the rice plant to complex stress conditions (Gammulla et al., 2011). This also indicated that cold stress at an early stage would have an irreversible impact on rice plants, since the NDVI-based VIs of groups 1 and 2 remained lower than that of groups 3 and 4.

2.3.3. Estimation of PI occurrence through first-order differentiation

The seven NDVI-based VIs were then tested with first-order differentiation in each group to compare the estimated GDD with the actual GDD of PI occurrence (Table 2-4). Due to the excellent estimation of GDDs for PI occurrence (RE < 10%), NDVI (700, 720), NDVI (700, 730), NDVI (708, 730), NDVI (700, 760), NDVI (708, 760), and NDVI (708, 800) were considered as the suitable NDVI-based VIs for PI determination. Among these combinations, NDVI (708, 760) that was proposed by Tan et al. (2018) coincidentally had the lowest RE value (RE = 5.63%) (Tan et al., 2018). This indicated that the entrance of the reproductive phase might have influenced the photosynthetic capacity of the rice plant, such as a lower rate of crop growth and absorption of photosynthetically active radiation. The other suitable combinations of NDVI-based VIs have not been researched in previous studies.

Table 2-3. Recombination and selection of NDVI-based VIs from reference wavelengths. The calculation was $[(\lambda_a - \lambda_b)/(\lambda_a + \lambda_b)]$, and the R^2 values of quadratic polynomial regression models were recorded.

λ_a \ λ_b	**660**	**670**	**680**	**700**	**708**	**720**	**730**	**740**	**760**	**780**	**800**
670	0.24										
680	0.31	0.47									
700	0.37	0.33	0.38								
708	0.51	0.45	0.47	0.57							
720	0.62	0.58	0.60	0.71	0.58						
730	0.68	0.64	0.65	0.73	0.74	0.55					
740	0.67	0.64	0.64	0.68	0.69	0.43	0.29				
760	0.71	0.69	0.69	0.73	0.75	0.64	0.68	0.26			
780	0.66	0.64	0.64	0.66	0.66	0.43	0.34	0.41	0.00		
800	0.67	0.65	0.65	0.68	0.70	0.49	0.43	0.37	0.10	0.10	

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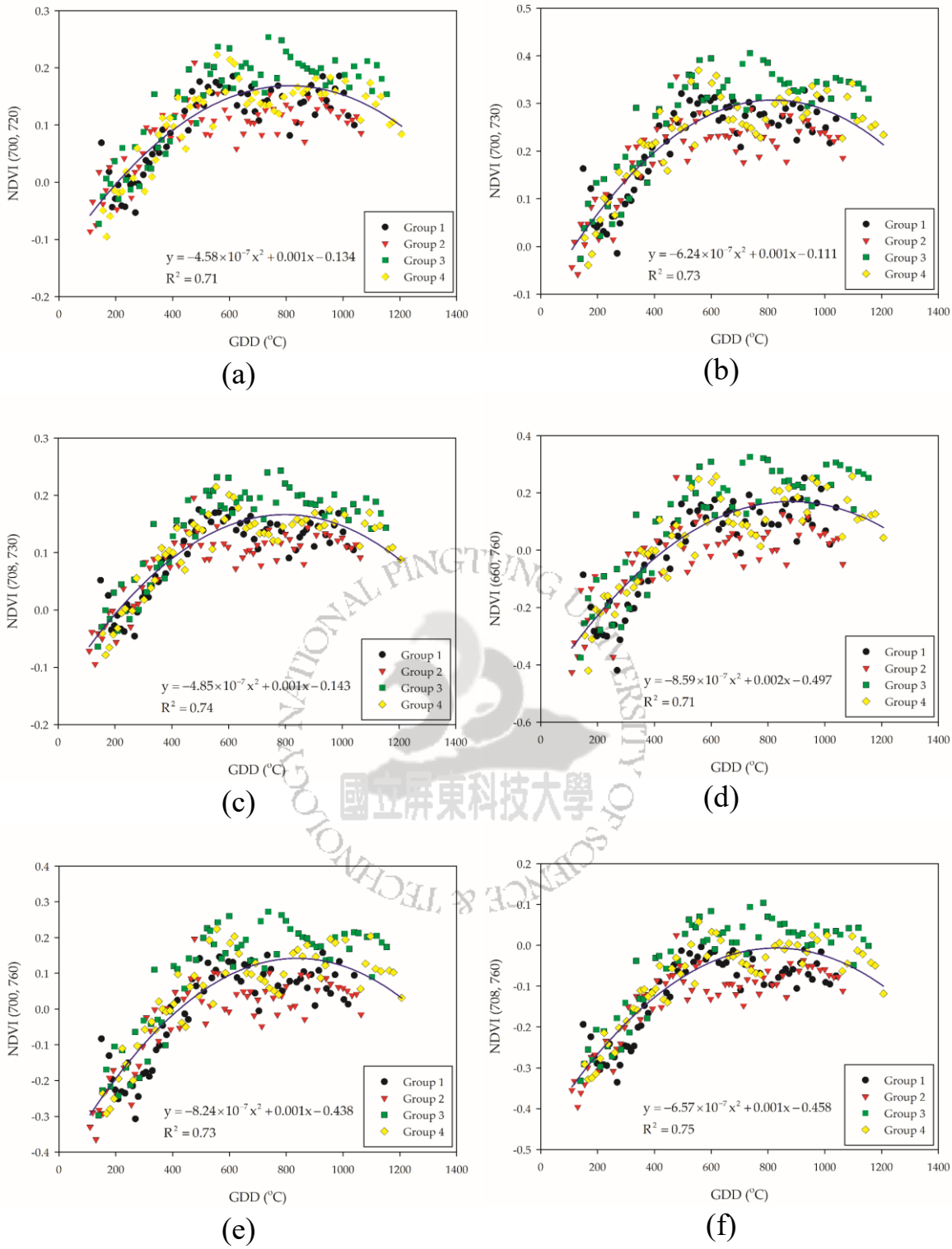
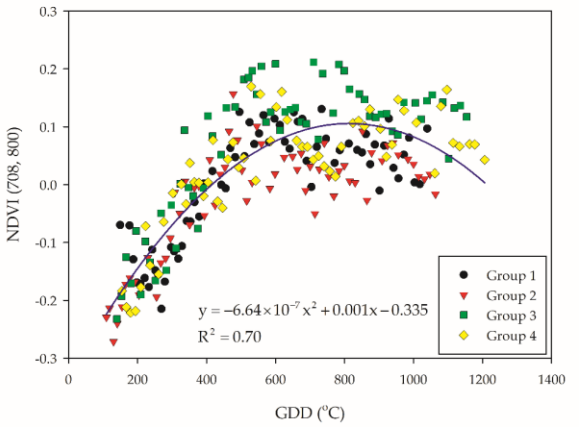


Figure 2-3. Cont.

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(g)

Figure 2-3. Illustration for the selected NDVI-based VIs combination: (a) NDVI (700, 720); (b) NDVI (700, 730); (c) NDVI (708, 730); (d) NDVI (660, 760); (e) NDVI (700, 760); (f) NDVI (708, 760); and (g) NDVI (708, 800). The black solid lines indicate the quadratic polynomial regression lines. The red, green, blue, and yellow symbols indicate the NDVI-based VIs NDVI-based VIs data of each group.

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Table 2-4. Seven selected NDVI-based VIs were tested by the first-order differentiation of quadratic polynomial regression models in each group.

NDVI-based VIs	Equation	Estimated GDDs	Actual GDDs	R^2	RE (%)
700, 720	$y = -5.52\times10^{-7}x^2 + 0.0008x - 0.161$	724.6	771.3	0.77	5.82
	$y = -4.15\times10^{-7}x^2 + 0.0006x - 0.099$	722.9	713.9	0.71	
	$y = -5.40\times10^{-7}x^2 + 0.0009x - 0.160$	833.3	783.4	0.85	
	$y = -5.29\times10^{-7}x^2 + 0.0009x - 0.171$	850.7	791.1	0.82	
700, 730	$y = -7.50\times10^{-7}x^2 + 0.0012x - 0.149$	800	771.3	0.79	6.29
	$y = -6.65\times10^{-7}x^2 + 0.0010x - 0.100$	751.9	713.9	0.77	
	$y = -7.44\times10^{-7}x^2 + 0.0012x - 0.139$	806.5	783.4	0.84	
	$y = -6.30\times10^{-7}x^2 + 0.0011x - 0.125$	873	791.1	0.8	
708, 730	$y = -5.74\times10^{-7}x^2 + 0.0009x - 0.162$	784	771.3	0.79	6.23
	$y = -4.45\times10^{-7}x^2 + 0.0007x - 0.123$	786.5	713.9	0.78	
	$y = -5.98\times10^{-7}x^2 + 0.0010x - 0.171$	836.1	783.4	0.88	
	$y = -5.16\times10^{-7}x^2 + 0.0008x - 0.165$	775.2	791.1	0.84	
660,760	$y = -1.11\times10^{-6}x^2 + 0.0018x - 0.608$	810.8	771.3	0.8	10.38
	$y = -9.10\times10^{-7}x^2 + 0.0014x - 0.449$	769.2	713.9	0.69	
	$y = -9.93\times10\text{-}^{-7}x^2 + 0.0018x - 0.530$	906.3	783.4	0.84	
	$y = -9.20\times10^{-7}x^2 + 0.0016x - 0.522$	869.6	791.1	0.8	

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700, 760	$y = -1.03\times10^{-6}x^2 + 0.0016x - 0.505$	776.7	771.3	0.79	6.36
	$y = -8.88\times10^{-7}x^2 + 0.0013x - 0.435$	732	713.9	0.79	
	$y = -1.00\times10^{-6}x^2 + 0.0017x - 0.472$	850	783.4	0.86	
	$y = -8.11\times10^{-7}x^2 + 0.0014x - 0.435$	863.1	791.1	0.82	
708, 760	$y = -8.20\times10^{-7}x^2 + 0.0012x - 0.504$	731.7	771.3	0.8	5.63
	$y = -6.41\times10^{-7}x^2 + 0.0010x - 0.447$	780	713.9	0.82	
	$y = -8.20\times10^{-7}x^2 + 0.0013x - 0.491$	792.7	783.4	0.89	
	$y = -6.71\times10^{-7}x^2 + 0.0011x - 0.463$	819.7	791.1	0.86	
708, 800	$y = -8.24\times10^{-7}x^2 + 0.0012x - 0.373$	728.2	771.3	0.77	5.80
	$y = -7.27\times10^{-7}x^2 + 0.0011x - 0.341$	756.5	713.9	0.78	
	$y = -8.48\times10^{-7}x^2 + 0.0014x - 0.375$	825.5	783.4	0.82	
	$y = -5.95\times10^{-7}x^2 + 0.0010x - 0.320$	840.3	791.1	0.75	

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2.3.4. Leave-one-out cross-validation (LOOCV)

The LOOCV was subsequently performed to test the accuracy of the quadratic polynomial regression of NDVI (700, 720), NDVI (700, 730), NDVI (708, 730), NDVI (700, 760), NDVI (708, 760), and NDVI (708, 800). The LOO regression line of groups 1, 3, and 4 showed the highest plateau, while groups 1, 2, and 4 the lowest (Figure 2-4). Due to the great variabilities of the plateau distribution, neither setting up an absolute threshold value nor extended the range for the PI occurrence determination seems to be an ideal method. Although the plateau of regression curve between groups was different, the trend of the slopes was relatively similar. This agreed with preceding research (Liu et al., 2017); however, the plateau of the curve in this study was different, which was at the PI stage rather than the booting-heading stage. The stability between the trend of the slopes and PI occurrence were tested through first-order differentiation in the leave-one-out cross-validation model. The results showed that NDVI (700, 720) was the most reliable NDVI-based VIs for PI determination since it had the lowest RE value (RE = 4.90%). A previous study represented NDVI application on wheat phenology monitoring and obtained satisfying results that achieved the lowest error of 4.61 days during the jointing stage (Zhou et al., 2020). Our study performed a slightly better approach with the application of NDVI (700, 720) with a quadratic polynomial regression model, which represented a maximum error of 68.9 GDDs (approximately three days) (Table 2-5). The others, NDVI (700, 730), NDVI (708, 730), and NDVI (708, 800), are potential candidates that are competent for PI assessment. At this point, the relationship between NDVI-based VIs and PI occurrence could be confirmed through the first-order differentiation of quadratic polynomial regression models.

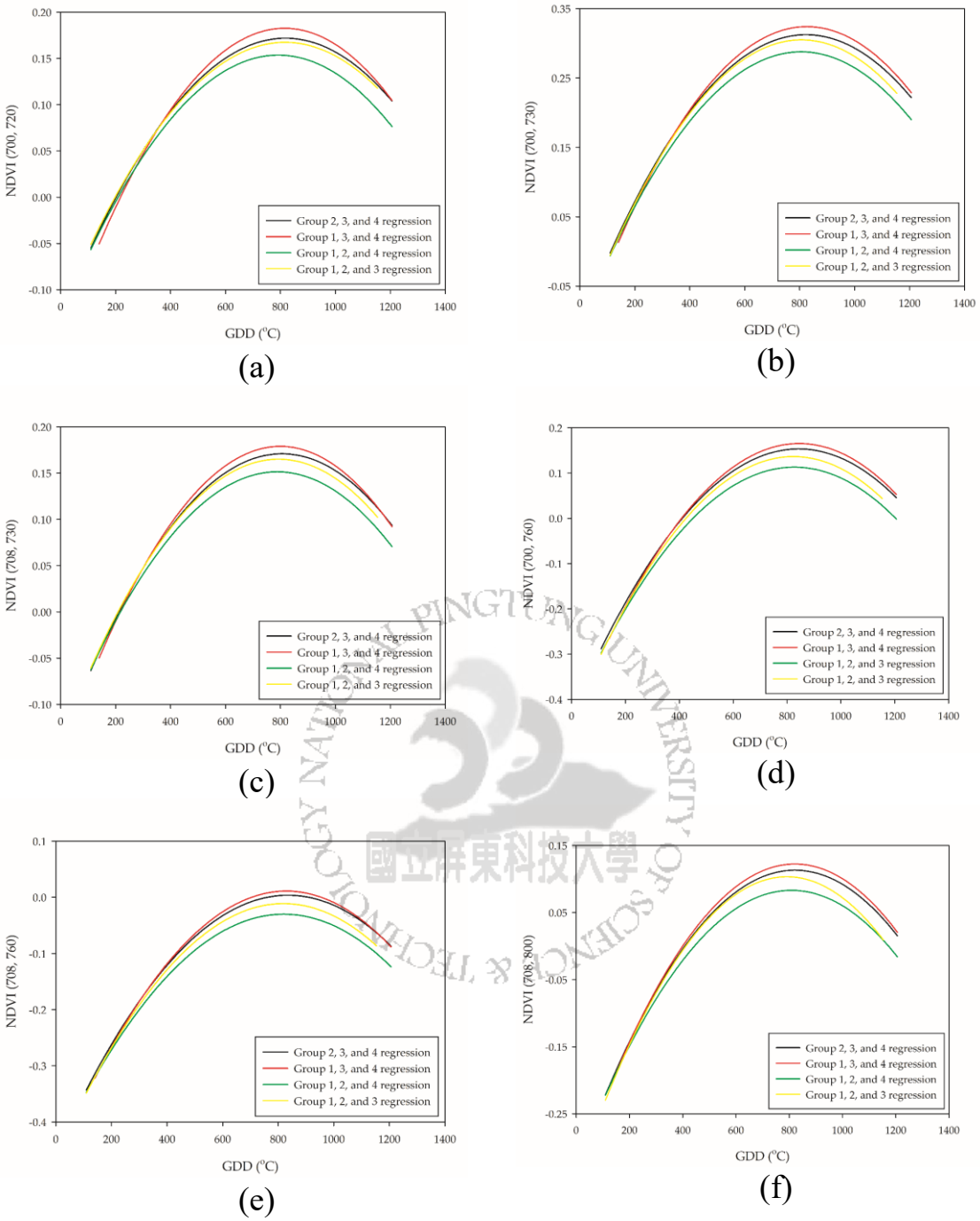


Figure 2-4. LOOCV of the six selected PI-related NDVI-based VIs: (a) NDVI (700, 720); (b) NDVI (700, 730); (c) NDVI (708, 730); (d) NDVI (700, 760); (e) NDVI (708, 760); and (f) NDVI (708, 800). The black line is the validation of group 1, the red line is the validation of group 2, the green line is the validation of group 3, and the yellow line is the validation of group 4.

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Table 2-5. LOOCV of the six selected PI-related NDVI-based VIs by first-order differentiation.

NDVI-based VIs	Equation	Estimated GDDs	Actual GDDs	R^2	RE (%)
700, 720	$y = -4.51\times10^{-7}x^2 + 0.0007x - 0.131$	776.1	771.3	0.70	4.90
	$y = -5.11\times10^{-7}x^2 + 0.0008x - 0.157$	782.8	713.9	0.77	
	$y = -4.51\times10^{-7}x^2 + 0.0007x - 0.130$	776.1	783.4	0.73	
	$y = -4.37\times10^{-7}x^2 + 0.0007x - 0.124$	800.9	791.1	0.68	
700, 730	$y = -6.18\times10^{-7}x^2 + 0.001x - 0.107$	809.1	771.3	0.72	9.08
	$y = -6.60\times10^{-7}x^2 + 0.0011x - 0.127$	833.3	713.9	0.77	
	$y = -6.08\times10^{-7}x^2 + 0.001x - 0.107$	822.4	783.4	0.75	
	$y = -6.40\times10^{-7}x^2 + 0.001x - 0.111$	781.3	791.1	0.71	
708, 730	$y = -4.83\times10^{-7}x^2 + 0.0008x - 0.143$	828.2	771.3	0.74	5.76
	$y = -5.26\times10^{-7}x^2 + 0.0008x - 0.157$	760.5	713.9	0.78	
	$y = -4.64\times10^{-7}x^2 + 0.0007x - 0.137$	754.3	783.4	0.77	
	$y = -4.83\times10^{-7}x^2 + 0.0008x - 0.139$	828.2	791.1	0.72	
700,760	$y = -8.19\times10^{-7}x^2 + 0.0014x - 0.430$	854.7	771.3	0.73	12.30
	$y = -8.65\times10^{-7}x^2 + 0.0015x - 0.453$	867.1	713.9	0.76	
	$y = -7.98\times10^{-7}x^2 + 0.0013x - 0.433$	814.5	783.4	0.76	
	$y = -8.55\times10^{-7}x^2 + 0.0014x - 0.445$	818.7	791.1	0.71	

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708, 760	$y = -6.59\times10^{-7}x^2 + 0.0011x - 0.456$	834.6	771.3	0.76	11.03
	$y = -7.00\times10^{-7}x^2 + 0.0012x - 0.471$	857.1	713.9	0.77	
	$y = -6.28\times10^{-7}x^2 + 0.001x - 0.453$	796.2	783.4	0.78	
	$y = -6.7\times10^{-7}x^2 + 0.0011x - 0.461$	820.9	791.1	0.72	
708, 800	$y = -6.61\times10^{-7}x^2 + 0.0011x - 0.333$	832.1	771.3	0.7	7.46
	$y = -6.87\times10^{-7}x^2 + 0.0011x - 0.342$	800.6	713.9	0.72	
	$y = -6.26\times10^{-7}x^2 + 0.0010x - 0.326$	798.7	783.4	0.73	
	$y = -7.17\times10^{-7}x^2 + 0.0011x - 0.345$	767.1	791.1	0.69	

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2.4. Conclusions

This study revealed the relationship between rice panicle initiation and NDVI-based VIs by the first-order differentiation of quadratic polynomial regression. The results showed that NDVI (700, 720), NDVI (700, 730), NDVI (708, 730), NDVI (700, 760), NDVI (708, 760), and NDVI (708, 800) are potential candidates to determine PI stage. Although the observed values had great variabilities between groups, the trend is stable—when the population of the target reached their plateau of NDVI-based VIs distribution, PI occurred. Due to the synchronistic correlation between PI and plateau of the trend, the PI estimation might be achieved through deter-mining the increasing proportion of saturated values of NDVI-based VIs. Cooperating with the UAV-mounted multispectral or hyperspectral camera and pixel-based analysis of the region of interest, NDVI-based VIs could be a useful indicator to assess rice plant PI stage, thus optimizing the management of fertilization practice.

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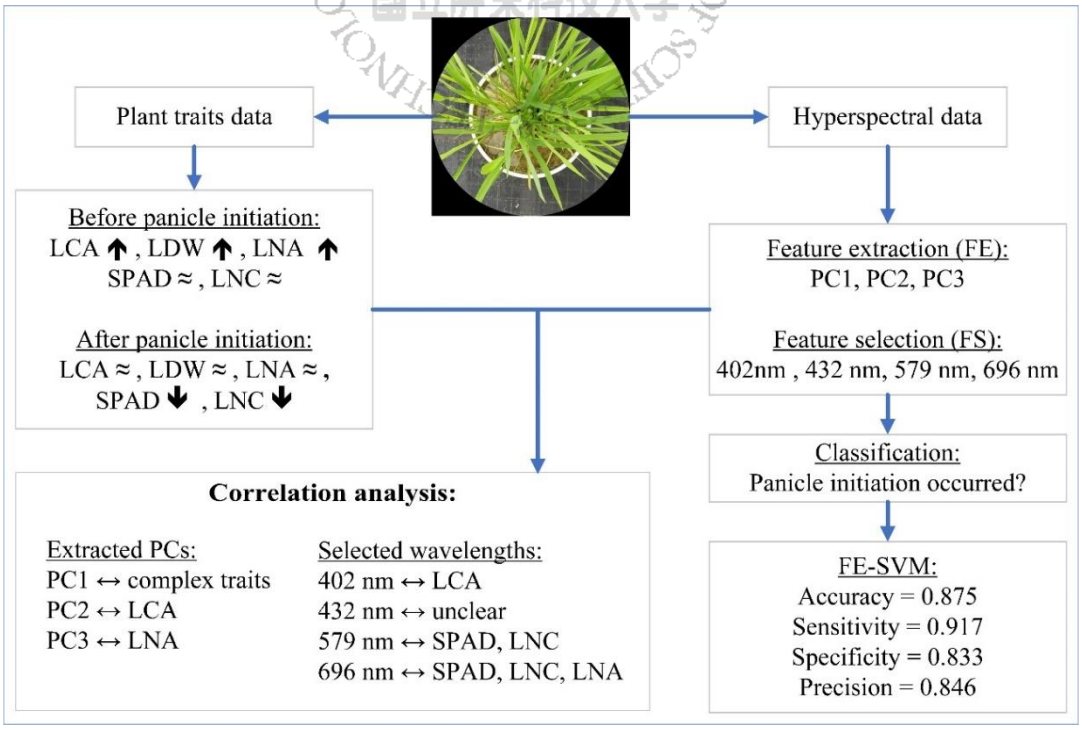
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Chapter 3. Real-time detection of rice growth phase transition for panicle nitrogen application timing assessment

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Highlights

- Leaf cover area (LCA) and leaf nitrogen accumulation (LNA) are the possible primary extracted features from principal component analysis.
- 402 nm, 432 nm, 579 nm, and 696 nm are the selected features from random forest algorithm. 402 nm is correlating with LCA, 579 nm is correlating with leaf chlorophyll readings (SPAD) and leaf nitrogen content (LNC), while 696 nm is correlating with SPAD, LNC, and LNA.
- Feature extraction-based support vector machine with radial basis kernel (FE-SVM) is more accurate and precise.



“

3.1. Introduction

Rice (*Oryza sativa* L.) is an important staple food for more than half of the global population. The worldwide milled rice requirement was predicted to reach 555 million tons in 2035, from 439 million tons in 2010 (Fahad et al., 2019). However, rice yield growth is decreasing because of inappropriate farming methods. One of the major constraints for rice production is the inefficient use of nitrogen fertilizer at the wrong time (Fageria and Baligar, 1999; Fahad et al., 2019). Therefore, improving nitrogen use efficiency, i.e., maximizing the grain yield while minimizing the nitrogen input, in terms of proper timing is an important task to facilitate future rice production.

A number of studies indicated that nitrogen topdressing at the early reproductive phase (ER) could enhance nitrogen use efficiency and is critical to rice yield (Kamiji et al., 2011; Zhou et al., 2017a; Ju et al., 2021a; b). The ER of rice consisted of the panicle initiation stage, panicle differentiation stage, and flag leaf collar formation stage (Counce et al., 2000). Furthermore, applying nitrogen at the panicle initiation stage or panicle differentiation stage could improve nitrogen absorption and utilization (Muhammad et al., 1974; Zhou et al., 2019; Ju et al., 2021a), leaf nitrogen conservation (Peng and Cassman, 1998; Wang et al., 2018), spikelet production (Kamiji et al., 2011; Zhou et al., 2017a), and lodging resistance (Quang Duy et al., 2004; Pan et al., 2019). Moreover, applying nitrogen at the flag leaf collar formation stage helps to reduce spikelet degeneration and to increase leaf photosynthetic efficiency and functional period (Zhou et al., 2017a; Liu et al., 2020). On the other hand, applying large amounts of nitrogen at the late vegetative phase (LV) tends to increase lodging risk (Pan et al., 2019).

For field management, the general method to assess whether a rice plant has entered ER is inefficient by directly observing either internode elongation, green-ring formation, or 1–2 mm panicle appearance through stem dissection (Counce et al., 2000; Moldenhauer and Slaton, 2001; Dunand and Saichuk, 2014). On the contrary, some other indirect approaches such as calendar day and thermal units are likely more effective for field management. Calendar day

is a long-established, simple, and convenient approach for farming guides; however, it is rather unreliable resulting from contrasting environments and weather variability (Gilmore and Rogers, 1958; Aslam et al., 2017). Subsequently, the thermal units appear to overcome the shortcoming of calendar day. However, there are still several limitations on practical usage, such as the distance between the measurement site and practical site, the susceptibility of maximum and base temperature estimation, and the dynamic changes of sensitive organs (Bonhomme, 2000).

Moreover, remote sensing either with satellite-based (Zeng et al., 2020), aerial-based (Yang et al., 2020), or ground-based (Zheng et al., 2016; Hufkens et al., 2019) platforms appears to be an effective tool for plant phenology monitoring. However, despite satellite-based observation offering large-scale screening, it often encounters limitations such as coarse resolution, low revisit frequency, susceptibility to the atmospheric condition (Zheng et al., 2016), and lack of biological meanings (Guo et al., 2016). In recent years, due to high spatial and temporal resolutions, unmanned aerial vehicles (UAV) have emerged as a robust monitoring platform. Nevertheless, UAV flight missions are often disrupted by weather conditions, no-fly zones, or regulations, particularly in Taiwan. In addition, UAV image processing requires considerable computational cost. According to the above limitations, ground-based remote sensing seems to be a competent approach for plant growth observation.

Previous studies performed promising methods, such as threshold determination and shape model fitting, using ground-based sensors to monitor crop phenology (Zheng et al., 2016; Hufkens et al., 2019; Zhou et al., 2020). Nevertheless, these methods require time-series data, and thus are rather difficult for real-time assessment. Meanwhile, a number of studies had successfully accomplished real-time detection of several important stages, such as heading and flowering, by adopting computer vision techniques based on significant phenotypic changes (Guo et al., 2015; Zhu et al., 2016; Bai et al., 2018). Unfortunately, the study about the growth phase transition of rice plants

from the late vegetative to early reproductive phase has not been found yet. There are several hints to discover the transition, such as leaf area and leaf length reduction (Tivet et al., 2001) and leaf nitrogen translocation (Peng and Cassman, 1998). However, these changes are rather obscure to human observation. Therefore, a real-time detection method with a high temporal resolution is still needed for the assessment of panicle-fertilizer topdressing timing.

Since hyperspectral sensing could provide detailed information about a target with numerous, continuous, and meticulous spectral reflectance data, it has been a potential tool for growth phase discrimination. Nonetheless, these advantages often raise problems that impede the performance of hyperspectral sensing, such as data collinearity and issues with dimensionality. In consequence, hyperspectral data are often pre-treated with dimensionality reduction techniques to reduce the data dimensionality. There are two major approaches for data dimensionality reduction: feature extraction (FE) and feature selection (FS) (Xu et al., 2019).

Principal component analysis (PCA) is one of the most common FE methods in agricultural/vegetation studies for reducing data dimensionality and removing wavelength correlations (Asner, 1998; Song et al., 2011; Liu et al., 2018a). By combining and transforming all of the original features with PCA, the numerous hyperspectral data could be substituted by a few components while still holding most of the variation within the original data (Almeida and Filho, 2004). Moreover, the factor loadings (or eigenvectors) resulting from PCA could also be used for wavelength selection (Song et al., 2011). On the other hand, random forest (RF) is a useful FS method for data dimensionality reduction; at the same time, it is also a robust learning algorithm to solve both classification and regression problems (Breiman, 2001; Liaw and Wiener, 2002; Abdel-Rahman et al., 2013). The RF estimates the importance of a variable via measuring either Gini impurity or permutation accuracy. Subsequently, the dimensionality-reduced data could be accompanied with commonly used

classifiers such as logistic regression, random forest (RF), and support vector machine (SVM).

The main goal of this study is to establish an unprecedented method for rice late vegetative phase (LV) and early reproductive phase (ER) discrimination to improve the precision of N application timing. Both FE and FS methods were evaluated for dimensionality reduction of hyperspectral data. Furthermore, the specific objectives in this manuscript were that the realistic implication of the modeling inputs was expected to be inferenced for the specific plant traits changes. Moreover, the physical and physiological changes of the rice plant were also illustrated for better understanding the rice growth phase transition. Finally, a real-time and high-temporal-resolution approach would be constructed for precise nitrogen application.

3.2. Materials and methods

3.2.1. Study site and materials

An outdoor pot experiment was conducted in the experimental field of National Pingtung University of Science and Technology in southern Taiwan in 2019–2020. A semi-dwarf japonica rice cultivar, Kaohsiung 147 (KH147), was transplanted when the seedlings had reached the two-leaf age. Each pot had five seedlings. Ten pots were used for daily hyperspectral data acquisition, while the remaining 20 pots were prepared for destructive sampling. The nitrogen fertilizers were applied in five splits (20% at basal, 10% at 50%-tillering stage, 30% at active tillering stage, 30% at panicle initiation stage, and 10% at booting stage) with a total amount of 150 kg/ha. The rice plants were submerged from the transplanting stage until the flag leaf collar formation stage to ensure that the spectral changes were not influenced by the variation of water reflectance.

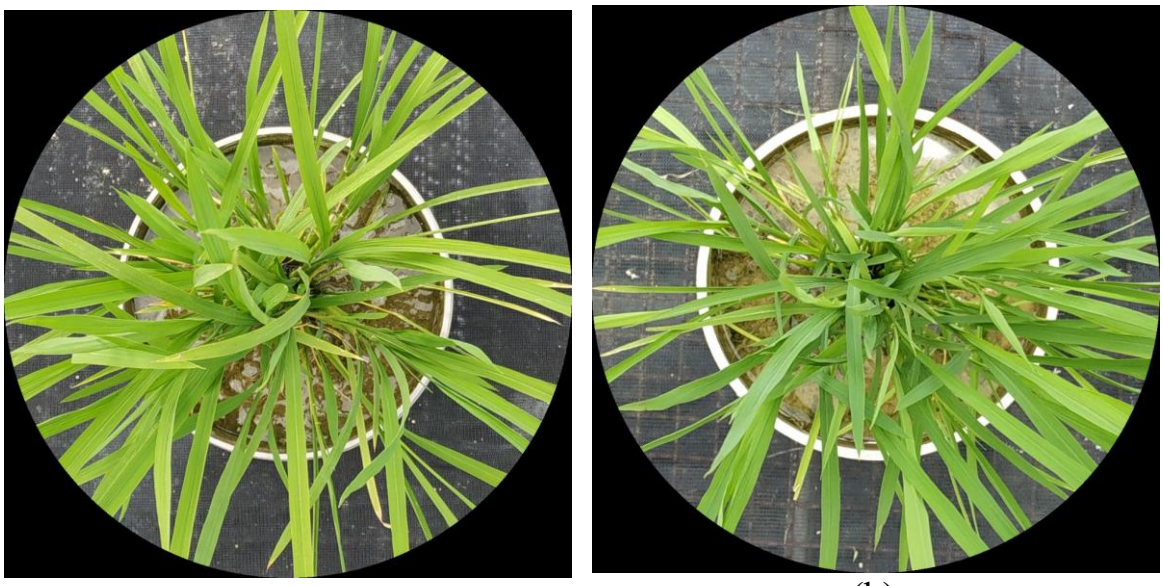
3.2.2. Hyperspectral data acquisition

Hyperspectral data were daily measured from the maximum tillering stage until the flag leaf collar formation stage by using the ASD FieldSpec HandHeld 2 Portable Spectroradiometer (Analytical Spectral Devices, Inc., Colorado, CO, USA). The spectroradiometer, which is a non-imaging hyperspectral instrument, collects reflectance data in a 25° field of view with a wavelength range of 325–1075 nm (1.5 nm sampling interval). Before measurement, the spectroradiometer was calibrated with the Spectralon white reference panel (Labsphere Ltd., North Sutton, NH, USA) to standardize different light conditions. The instrument was held vertically at around one meter above the soil surface, providing a detection spot approximately 0.45 m in diameter, covering the pots (0.25 m in diameter) in the center (Figure 3-1). Since the measurements were conducted during 10:00 a.m. to 2:00 p.m., when the sun is found at nearly zenith position, the sun-spot on the water surface was carefully considered to avoid abrupt reflectance changes. The collected hyperspectral data were averaged over the ten pots.

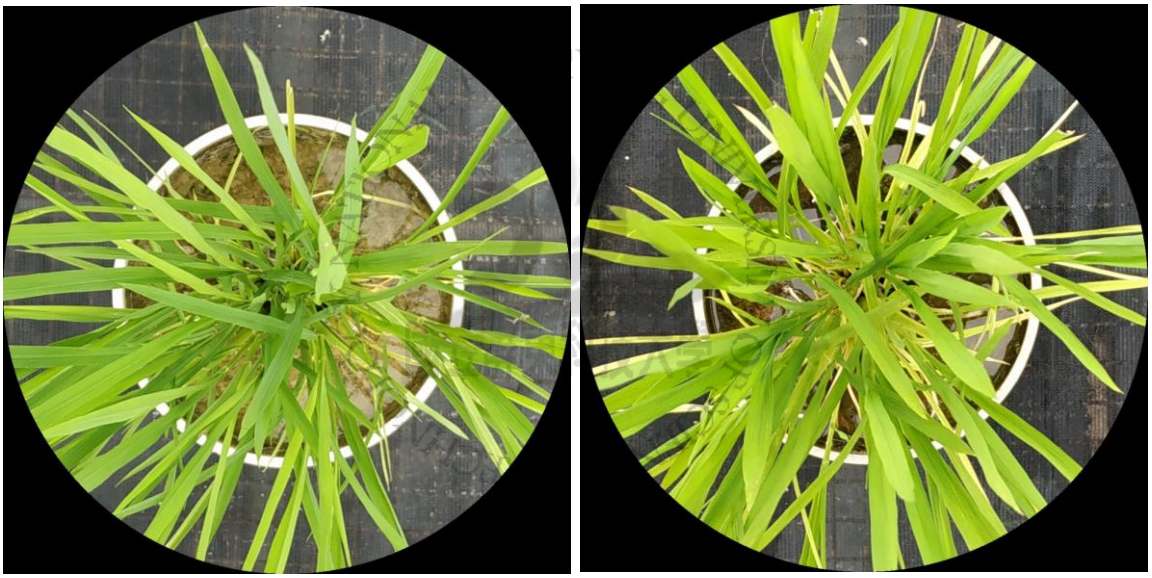
3.2.3. Anatomical observation

Destructive sampling was conducted weekly from the maximum tillering stage until the flag leaf collar formation stage (Figure 3-1). Three pots of plants were randomly selected and sent to the laboratory for dissection. All culms of the plants were dissected for anatomical observation. A culm having either green-ring or 1–3 mm panicle is considered as a reproductive culm (Figure 3-2). The beginning of the reproductive phase is defined when the proportion of reproductive culm has reach 50%.

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(a) (b)



(c) (d)

Figure 3-1. Illustration of late vegetative phase (LV), which includes (a) maximum tillering stage and (b) vegetative-lag stage; and early reproductive phase (ER), which consists of (c) panicle initiation stage and (d) flag leaf collar formation stage. This figure is a rough simulation of the spectroradiometer's detection spot instead of the real instrumental view.



Figure 3-2. Anatomical observation of first internode-elongation during LV (left); and green-ring with a 1-3 mm panicle during ER (right).

3.2.4. Plant traits examination

3.2.4.1. Leaf cover area (LCA)

Before destructive sampling, LCA and leaf chlorophyll content were measured. The LCA measurement, which is an uncommon method for leaf area-related ground-truth observation, was done with a smartphone-RGB camera and ImageJ software (version 1.53) (National Institutes of Health, Bethesda, MD, USA). The procedure was graphically illustrated in Figure 3-3. A white panel was placed beside the target plant as a reference length for next image analysis. Then, an RGB-image was taken with a smartphone-RGB camera that was held vertically and approximately one meter above the soil surface. Subsequently, the captured image was loaded into the ImageJ software for LCA estimation. Firstly, a straight line was drawn according to the border of the white panel within the raw image, and a known scale was set to define the pixel unit to distance unit (Figure 3-3a). Secondly, the image was converted to 8-bit RGB color channels (Figure 3-3b–d). Then, the 8-bit images were transformed by the equation (G − R) + (G − B) to extract the plant pixels (Figure 3-3e–g). It is worth noting that, since the default image data type of the software is uint8, rearranging the formula might be unwise. For example, (G + G) − (R + B) could yield different results, although it is mathematically equivalent to the (G − R) + (G − B) equation. Lastly, the plant pixels were masked out by Otsu thresholding, and finally the LCA was estimated (Figure 3-3h).

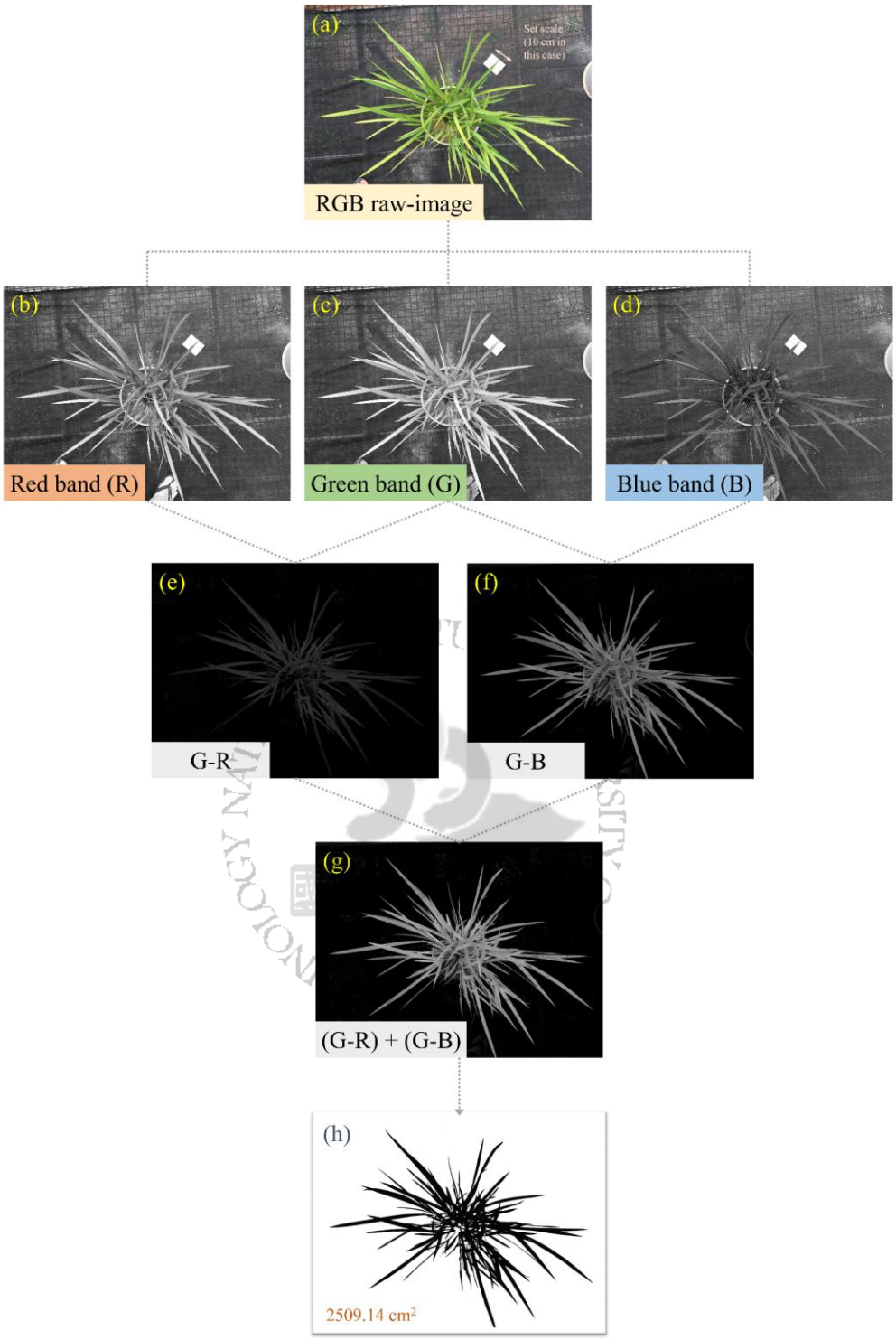


Figure 3-3. Illustration of leaf cover area (LCA) estimation procedure. (a) Define the drawn pixels unit; (b–d) split channel; (e–g) transformation with (G − R) + (G − B) formula; (h) estimating the threshold area.

3.2.4.2. Plant traits measurements

The leaves were oven-dried until constant weight at 80 °C to obtain the leaf dry weight (LDW). The leaf N content (LNC) was determined with Kjeldahl method. The leaf nitrogen accumulation (LNA) was calculated by multiplying LDW with LNC (Equation 3-1).

$$LNA = LDW \times LNC \quad \text{(Equation 3-1)}$$

The leaf chlorophyll content (SPAD) was investigated with a chlorophyll meter (Chlorophyll meter SPAD-502Plus, Konica Minolta, Tokyo, Japan). The SPAD measurements were made at the 2/3 position on the third fully expanded leaf from the top (Yuan et al., 2016).

3.2.5. Data analysis

The data analysis was implemented in R software (version 4.1.0) (R Core Team, 2022). The procedure is illustrated in Figure 3-4.

3.2.5.1. Data pretreatment and partition

The raw spectral data were smoothed before subsequent analysis (Liu et al., 2018a). As a result, the smoothed spectral data for subsequent analysis ranged from 327 nm to 1073 nm. Next, the smoothed data were randomly partitioned into two parts: 70% as the training dataset, and the remaining were held out for validation.

3.2.5.2. Dimensionality reduction

Two dimensionality-reduction techniques, which are the feature extraction (FE) and the feature selection (FS), were applied on the smoothed hyperspectral data. The FE approach was performed with PCA. The eigenvalues of each component were calculated with a correlation matrix. Subsequently, a suitable number of principal components (PCs) were

determined according to the proportion of variance explained (PVE) in the scree-plot. The extracted features were then used to establish logistic regression (FE-logistic), RF (FE-RF), and SVM (FE-SVM) models.

On the other hand, the FS approach was performed with RF. Several wavelengths were selected according to the variable importance that was measured with the permutation accuracy. The selected wavelengths were then used to construct logistic regression (FS-logistic), RF (FS-RF), and SVM (FS-SVM) models.

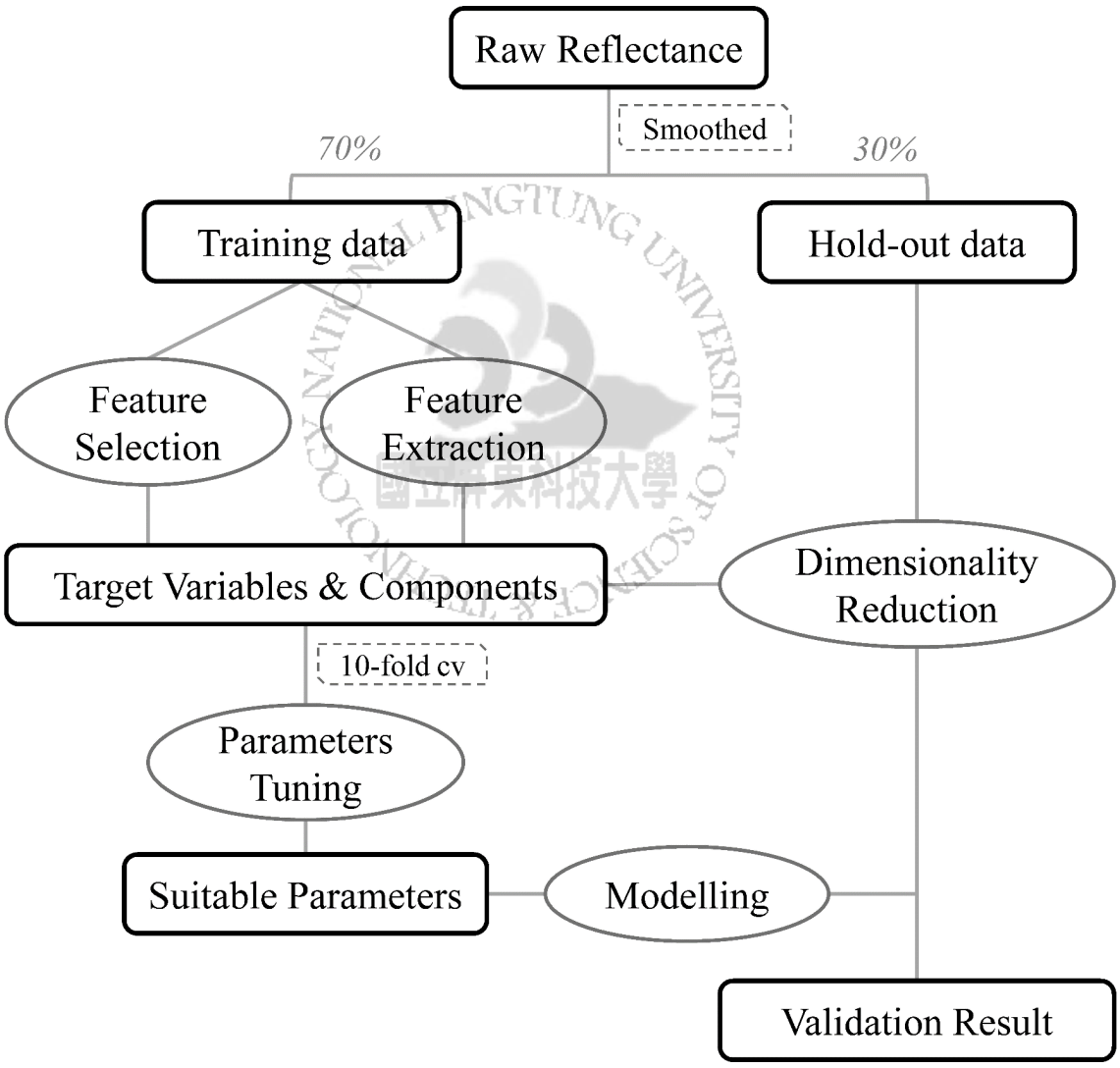


Figure 3-4. Illustration of the data analysis procedure.

3.2.5.3. Parameters tuning

The parameters of each algorithm were tuned in a 10-fold cross-validation procedure. The optimal cutoff threshold of logistic regression was determined as the cross-point of sensitivity and specificity (Peng and So, 2002). The number of trees (ntree), the number of variables (mtry) of RF, and the cost and gamma of SVM (RBF kernel) were decided based on the highest accuracy obtained in the grid search method. The fine-tuned parameters were then applied to internal data modelling (Table 3-1) and so validated by the hold-out dataset.

Table 3-1. Fine-tuned parameters for each algorithm.

Models	**Parameters**
FE-logistic	threshold = 0.50
FS-logistic	threshold = 0.60
FE-RF	mtry = 1, ntree = 160
FS-RF	mtry = 1, ntree = 300
FE-SVM	cost = 100, gamma = 0.01
FS-SVM	cost = 10, gamma = 0.01

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3.2.5.4. Model performance evaluation

A confusion matrix was adopted to deal with binary classification purposes (Table 3-2). The true positive (TP) is correct prediction of reproductive phase. The true negative (TN) is accurate prediction of vegetative phase. The false positive (FP) is incorrect classification of vegetative to reproductive phase. The false negative (FN) represents incorrect classification of reproductive phase to vegetative phase. Subsequently, four metrics (accuracy, sensitivity, specificity, precision) derived from the confusion matrix were used to evaluate the performance of each classifier. The derivatives are expressed as follows:

- Accuracy: This is an overall proportion of accurate classification (correctly classified both vegetative phase and reproductive phase, respectively) (Equation 3-2).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad \text{(Equation 3-2)}$$

- Sensitivity: This shows how well the classifiers can identify the reproductive phase when the rice plant enters the reproductive phase (Equation 3-3).

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad \text{(Equation 3-3)}$$

- Specificity: This shows how well the classifiers can identify vegetative phase when the rice plant has not entered reproductive phase (Equation 3-4).

$$\text{Specificity} = \frac{TN}{TN + FP} \quad \text{(Equation 3-4)}$$

- Precision: This represents the probability that a positive prediction is accurate (Equation 3-5).

$$\text{Precision} = \frac{TP}{TP + FP} \quad \text{(Equation 3-5)}$$

Table 3-2. Confusion matrix.

Predicted	**Observed**	
	Vegetative	Reproductive
Vegetative	TN	FN
Reproductive	FP	TP*

* TN: True negative; FN: False negative; FP: False positive; TP: True positive.

3.3. Results and Discussion

3.3.1. Plant trait changes during phase transition

The LCA progressively expanded during LV and reached maxima before reaching ER (Figure 3-5a). This might have resulted from the decrease in leaf area and leaf length after panicle initiation (Tivet et al., 2001). At the same time, the LDW gradually increased before ER and started to decline after LV (Figure 3-5b). This revealed that the leaf biomass accumulation was retarded after the transition to ER.

On the other hand, the dynamic changes of SPAD values represented the color changes of the leaves during phase transition (Figure 3-5c). The SPAD values in LV were higher than those in ER, depicting the color of leaves turned from green to yellowish-green during phase transition. The reason is generally related to the degradation of chlorophyll that is a common phenomenon during leaf senescence. Besides, LNC declined during ER regardless of the decrement in LDW (Figure 3-5d). The result indicated that the process of nitrogen translocation from leaf (source) to stem (sink) had synchronously occurred along with chlorophyll degradation. The nitrogen requirement of the panicle is continuously elevated after the transition to ER, while the plant total nitrogen absorption declines after the vegetative phase's end (Sheehy et al., 2004). Consequently, nitrogen is translocated from leaves to panicles to compensate for the demand of panicle development. The nitrogen translocation process is especially obvious when nitrogen supply is inefficient during ER (Peng and Cassman, 1998). The reduction in LNA during ER further revealed the nitrogen translocation process (Figure 3-5e).

In summary, there are three symptoms showing the transition from LV to ER. The first is that the coverage is maximally extended before ER; the second one, the leaf growth is retarded after LV; and lastly, the leaves appear from green to yellowish-green that could be caused by the nitrogen translocation process.

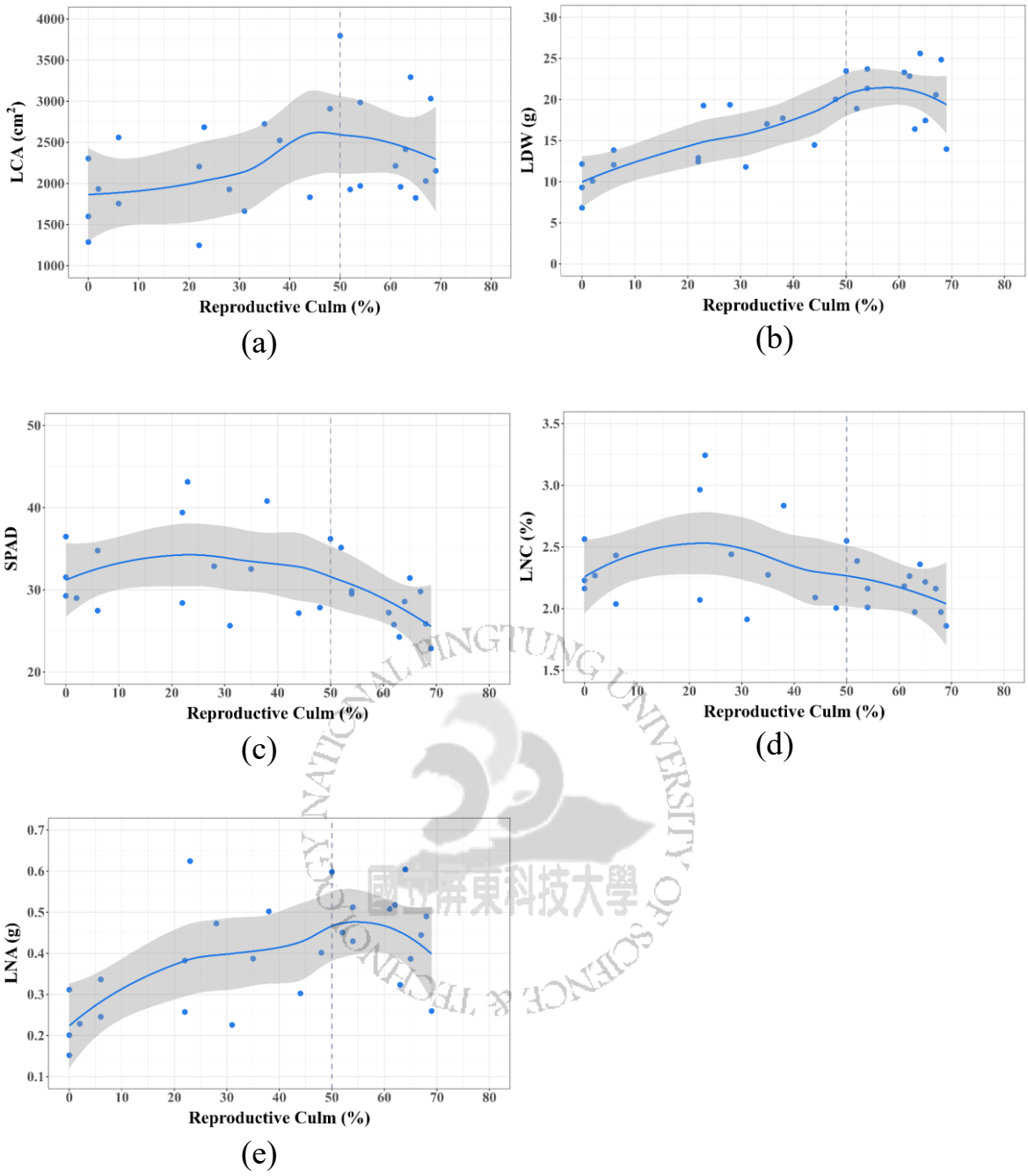


Figure 3-5. Illustration of (a) leaf cover area (LCA), (b) leaf dry weight (LDW), (c) chlorophyll meter readings (SPAD), (d) leaf nitrogen content (LNC), and (e) leaf nitrogen accumulation (LNA). The grey-dashed vertical line denotes the beginning of the reproductive phase.

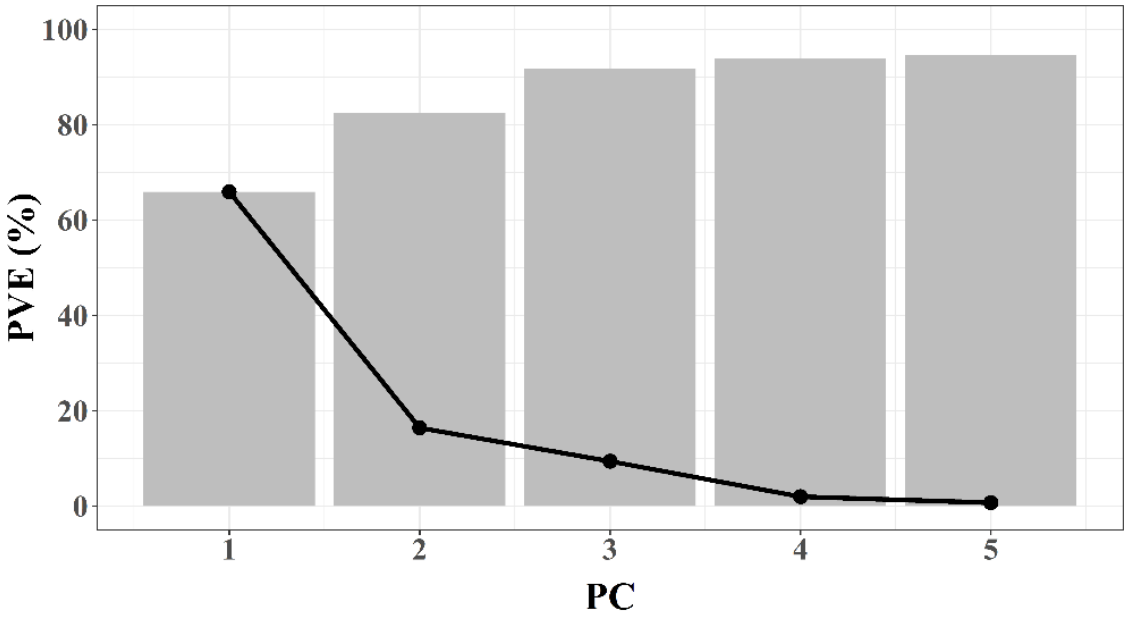
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3.3.2. Dimensionality reduction

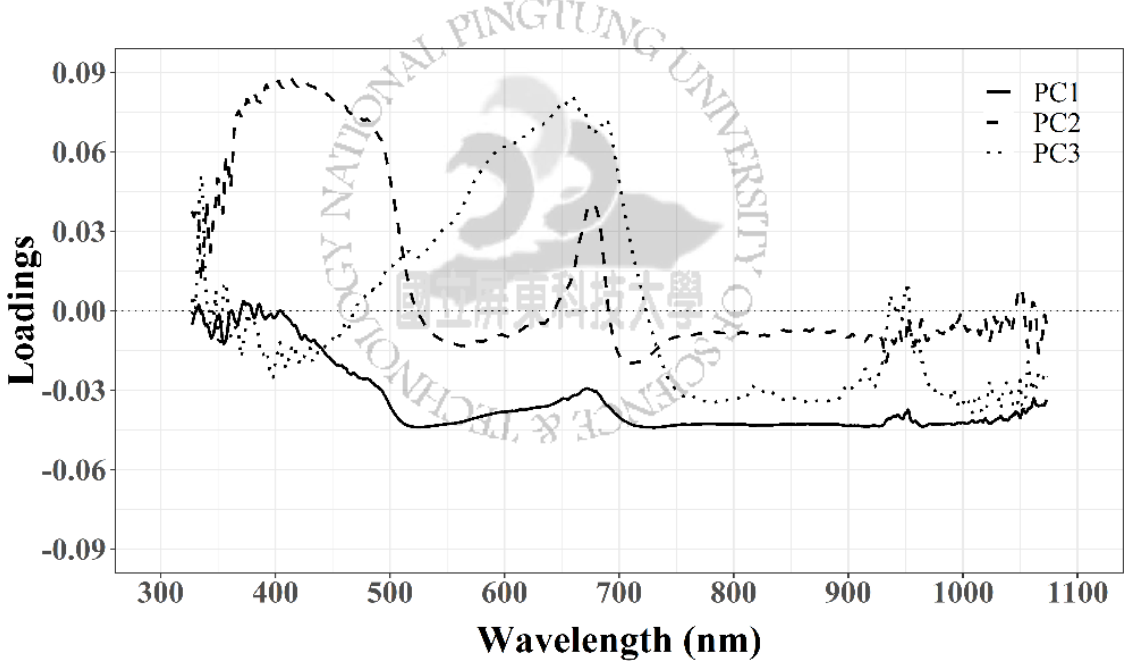
3.3.2.1. Feature extraction (FE)

The FE-based dimensionality reduction was done with the PCA algorithm according to the PVE of each PCs. The result showed that the former five PCs explained 65.93%, 16.45%, 9.43%, 2.00%, and 0.74% of the data variances, respectively (Figure 3-6a). The scree-plot's elbow bent on PC2; however, PC3 was also taken into consideration to achieve an over 90% cumulative PVE (Liu et al., 2018a). The contributions of each wavelength to the extracted PCs were further represented with a loadings plot (Figure 3-6b).

The major contents of PC1 could be roughly separated into three parts according to the peaks of eigenvectors and are the green region (500 – 600 nm), the red-edge region (680 – 750 nm), and the NIR region (>750 nm) (Figure 3-6b). The trough around 510–540 nm involved pigment variation (chlorophyll, carotenoids, and anthocyanin) (Zur et al., 2000; Gitelson et al., 2002b; Merzlyak et al., 2003) and the xanthophyll signal (Gamon et al., 1992). The result indicates PC1 might be a physiological indicator. However, the PC1 subsequently inflected on the red-edge region (around 675 nm) and appeared as a basin throughout the red-edge and NIR regions. The red-edge region is comprehensive in plant monitoring and is related to the plant's structural and physiological status, such as leaf area index and chlorophyll content (Horler et al., 1983). Besides, the NIR region correlates with biophysical changes such as intra- and inter-cellular properties, leaf orientation, and plant coverage (Knipling, 1970; Woolley, 1971). Due to the confounding effects of PC1, it is rather difficult, if not impossible, to be interpreted as a single or several plant traits.



(a)



(b)

Figure 3-6. Feature extraction is based on (a) the proportion of variance explained (PVE) (black dots and line) and cumulative PVE (grey bars) of each principal component (PC). The variable importance was depicted by (b) the loadings plot that consists of the eigenvectors of the PC1 (solid line), PC2 (dashed line), and PC3 (dotted line).

The PC2 provoked a major peak in the blue region (350 – 450 nm) and a short peak in the red region (670 – 680 nm) (Figure 3-6b). The blue region is related to plant pigment absorption, which provides much information on plant physiological status and indicates the phenological stage (Blackburn, 2007). Meanwhile, chlorophyll absorbs strongly in the red region, with maximum absorbance between 660 and 680 nm. These led the PC2 to be assumed as a physiological feature. However, the correlation of PC2 with physiological measurements (SPAD, LNC, and LNA) in this work was insignificant. The reason might be due to the overlapping absorption features of major pigments and other constituents (Gitelson and Merzlyak, 1996; Sims and Gamon, 2002; Blackburn, 2007). Furthermore, the wavelengths around 675 nm were suggested to be insensitive to pigment variation unless the chlorophyll content is very low (Gitelson and Merzlyak, 1994). On the other hand, this work suggested that PC2 was negatively correlated with LCA to some extent (r = −0.373; $p < 0.09$) (Table 3-3). The negative correlation between PC2 and LCA was reasonable, as the extension of LCA would probably increase the area of blue light absorption, thus reducing blue light reflectance. Hence, PC2 could be partly considered as a physical feature rather than a physiological feature.

PC3 mainly constituted the red and red-edge regions (Figure 3-6b). Preceding studies showed that several wavelengths within the red and red-edge region correlate with nitrogen content (Song et al., 2011; Mahajan et al., 2017). However, the correlation between PC3 and LNC was rather insignificant in this work (Table 3-3). The reason might be the impact of considerable loadings from the NIR region that is insensitive to chlorophyll and thus nitrogen content (Knipling, 1970; Wada et al., 2015). Nonetheless, PC3 negatively correlated with LNA ($r = -0.440$; $p < 0.05$). The LNA depicts the growth amount of plants, particularly the enrichment of nitrogen and carbon. Besides, the in-significant correlation between PC3 and LCA is reasonable, as previous research proved that the red-edge region is unaffected by ground cover (Horler et al., 1983). In overall, PC3 tended to be a physiological feature due to the stronger correlation with the LNA.

Table 3-3. Pearson's correlation between extracted features and plant traits.

Traits	PC1	PC2	PC3
LCA	−0.277	−0.373	−0.329
LDW	−0.294	−0.289	−0.358
SPAD	0.049	−0.013	−0.258
LNC	−0.035	−0.062	−0.333
LNA	−0.258	−0.262	−0.440 *

LCA: leaf cover area; LDW: leaf dry weight; SPAD: chlorophyll meter readings of Soil Plant Analysis Development; LNC: leaf nitrogen content; LNA: leaf nitrogen accumulation. The symbol * denotes significant correlation at $p < 0.05$.

3.3.2.2. Feature selection (FS)

The FS-based dimensionality reduction was performed with RF algorithm. The importance of each wavelength was measured with permutation accuracy. Figure 3-7 represents four groups of peaks that are located within blue (350 – 450 nm), blue-green edge (450 – 499 nm), green (500 – 600 nm), and red-edge (680 – 750 nm) regions. Considering the problem of data collinearity, only a single representative wavelength would be selected from each region. Consequently, 402, 432, 579, and 696 nm were selected for subsequent modeling (Figure 3-7).

Although major plant pigments absorb blue wavelengths, their strong absorbance peak is above 430 nm, which is quite far from 402 nm (Blackburn, 2007). Meanwhile, the wavelengths around 402 nm were of lesser concern. Interestingly, 402 nm was also one of the highest loadings of PC2 in the previous section. The result showed that 402 nm negatively correlated with LCA to some extent ($r = −0.346$; $p < 0.09$) (Table 3-4). The weak correlation between LCA and either PC2 or 402 nm might be caused by the insufficient area of the detection spot. The detection spot could be enlarged by elevating the detector's height; however, it is rather difficult to achieve due to the observer's height limitation. On the contrary, 432 nm is very near 430 nm, one of the main absorption peaks of carotenoids (Gitelson et al., 2002b). The 432

nm wavelength probably correlated with carotenoid content and could be an indicator of leaf senescence.

On the other hand, 579 nm is intuitively a yellow wavelength representing the plant's yellowness. Although the carotenoids are the principal pigments that correspond to the yellowish appearance of leaves, the correlation between 579 nm and carotenoid content might be indirect. Preceding research found that those wavelengths above 550 nm within the green region are only governed by chlorophyll absorption (Gitelson et al., 2002b). Therefore, the yellowness of leaf appearance during phase transition was probably caused by the degradation of chlorophyll. This statement was agreed by the results of this work. Table 3-4 represented the significant correlation between 579 nm and SPAD ($r = -0.758$; $p < 0.001$) and LNC ($r = -0.712$; $p < 0.001$).

Besides, 696 nm within the range of 690 – 705 nm could be a sensitive indicator of chlorophyll content (Gitelson and Merzlyak, 1996). Furthermore, several studies also stated that 696 nm correlates with chlorophyll content and N content (Tkachuk et al., 1988; Dedeoglu, 2020). The situations were also similar, as shown in the results, that 696 nm had significant correlation with SPAD ($r = -0.777$; $p < 0.001$), LNC ($r = -0.741$; $p < 0.001$), and LNA ($r = -0.427$; $p < 0.05$).

3.3.3. Model evaluation

To construct the predicted model, the extracted features (PC1, PC2, and PC3) and selected features (402, 432, 579, and 696 nm) were applied for modelling and comparing the performances. Four metrics evaluated the constructed models: accuracy, sensitivity, specificity, and precision, as shown in Figure 3-8. The FE-logistic (accuracy = 0.833) and FE-SVM (accuracy = 0.875) represented a higher overall proportion of correct classification. Besides, the FE-logistic (sensitivity = 0.917) and FE-SVM (sensitivity = 0.917) exhibited greater ability for recognizing the ER. On the contrary, the FS-RF (specificity = 0.833) and FE-SVM (specificity = 0.833) showed better performance in identifying the LV. Nonetheless, the FE-SVM (precision =

0.846) and FS-RF (precision = 0.818) manifested higher reliability of positive estimation.

The performance of the FE-logistic was comparable to the FE-SVM to some extent if considering the computational cost. However, the lower precision indicated the FE-logistic could be somewhat over-sensitive. An over-sensitive model would easily misclassify the LV as ER and consequently mislead N application at LV. When a cultivar with high lodging resistance is planted, the FE-logistic could be a competent model. On the contrary, the FE-SVM is more appropriate for general application.

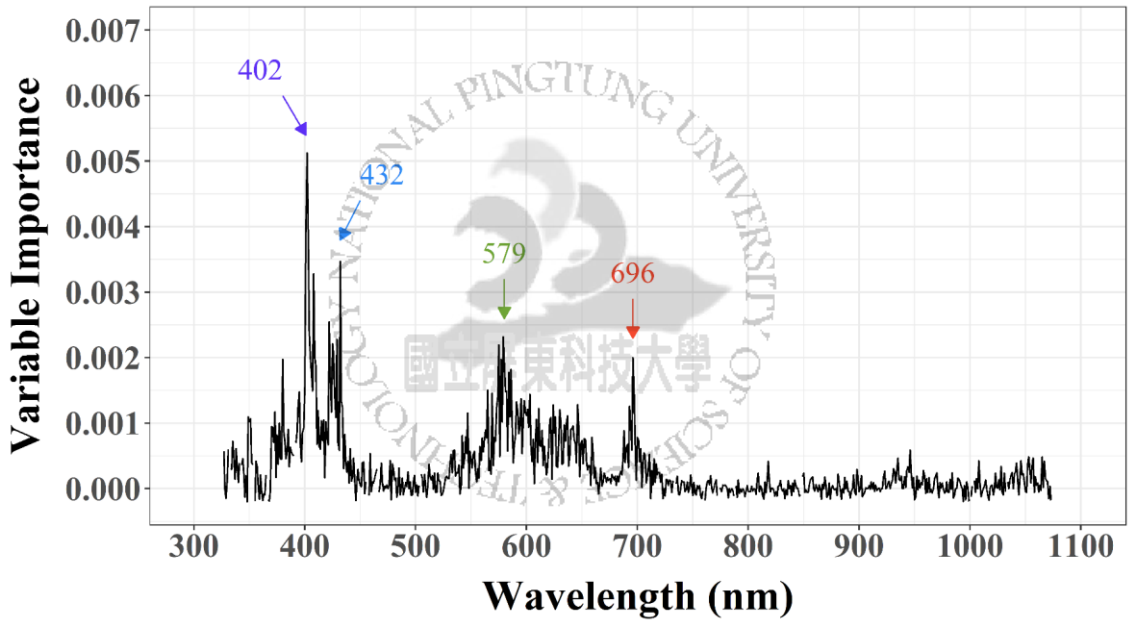


Figure 3-7. Feature selection with the random forest algorithm. The arrows point out the selected wavelengths.

Table 3-4. Pearson's correlation between selected features and plant traits.

Traits	402 nm	432 nm	579 nm	696 nm
LCA	−0.346	−0.274	−0.019	−0.044
LDW	−0.067	−0.016	−0.028	−0.124
SPAD	0.068	−0.080	−0.758 ***	−0.777 ***
LNC	0.113	−0.023	−0.712 ***	−0.741 ***
LNA	0.000	−0.014	−0.332	−0.427 *

LCA: leaf cover area; LDW: leaf dry weight; SPAD: chlorophyll meter readings of Soil Plant Analysis Development; LNC: leaf nitrogen content; LNA: leaf nitrogen accumulation. The symbols * and *** respectively denote significant correlation at $p < 0.05$ and $p < 0.001$.

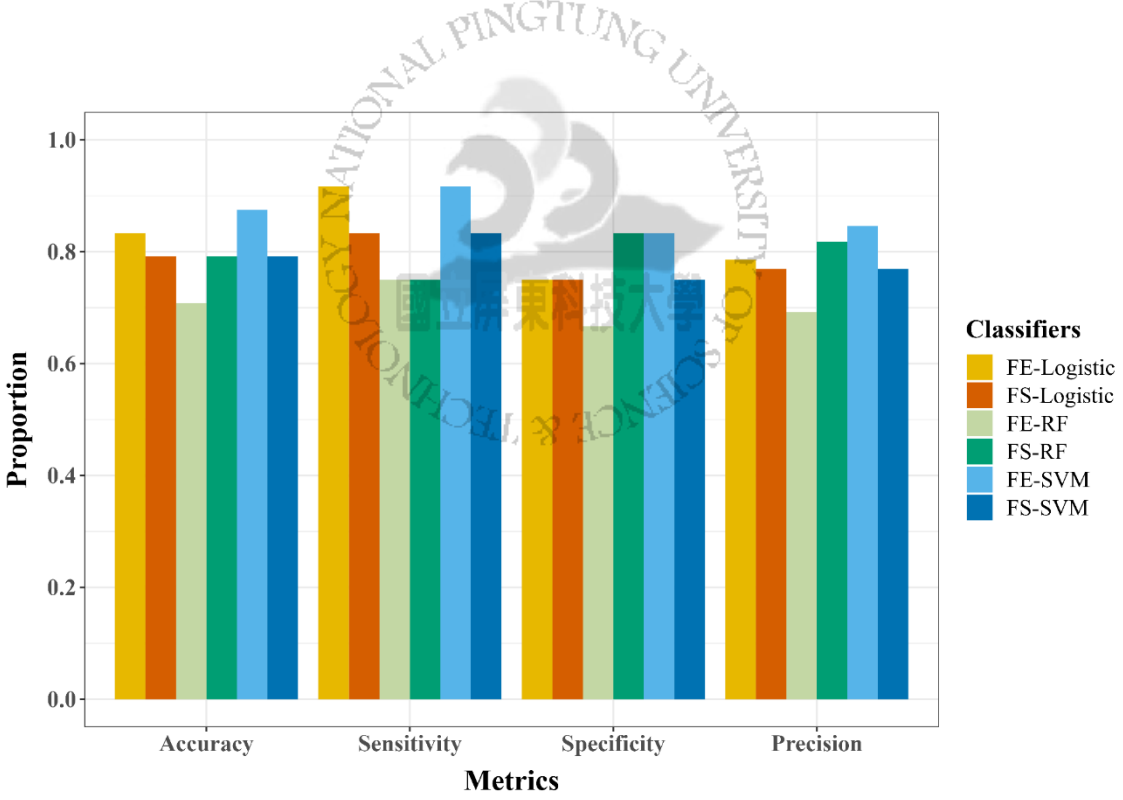


Figure 3-8. Validation result for the constructed models.

3.4. Conclusions

This study represented the workflow of hyperspectral sensing for the discrimination of rice late vegetative phase (LV) and early reproductive phase (ER). Moreover, the physical and physiological changes of the rice plant during phase transition were also illustrated. Furthermore, the realistic implications of the spectral features were inferenced. As shown in the experimental results, the leaf cover area (LCA), leaf dry weight (LDW), and leaf nitrogen accumulation (LNA) of rice plants progressively increased, while the chlorophyll content (SPAD) and leaf nitrogen content (LNC) remained stable during LV. After reaching ER, the nitrogen translocation from leaves to emerging panicles led to chlorophyll degradation. Although these differences were obscure to human observation, hyperspectral sensing could still distinguish the differences in a non-destructive and high-temporal-resolution manner. Considering the detrimental effect of applying nitrogen at LV, the feature extraction-based support vector machine (FE-SVM) was suggested for the timing assessment of panicle fertilizer application. Further effort is still required to integrate this timing assessment approach with extensively published nitrogen status monitoring techniques for precise nitrogen management.

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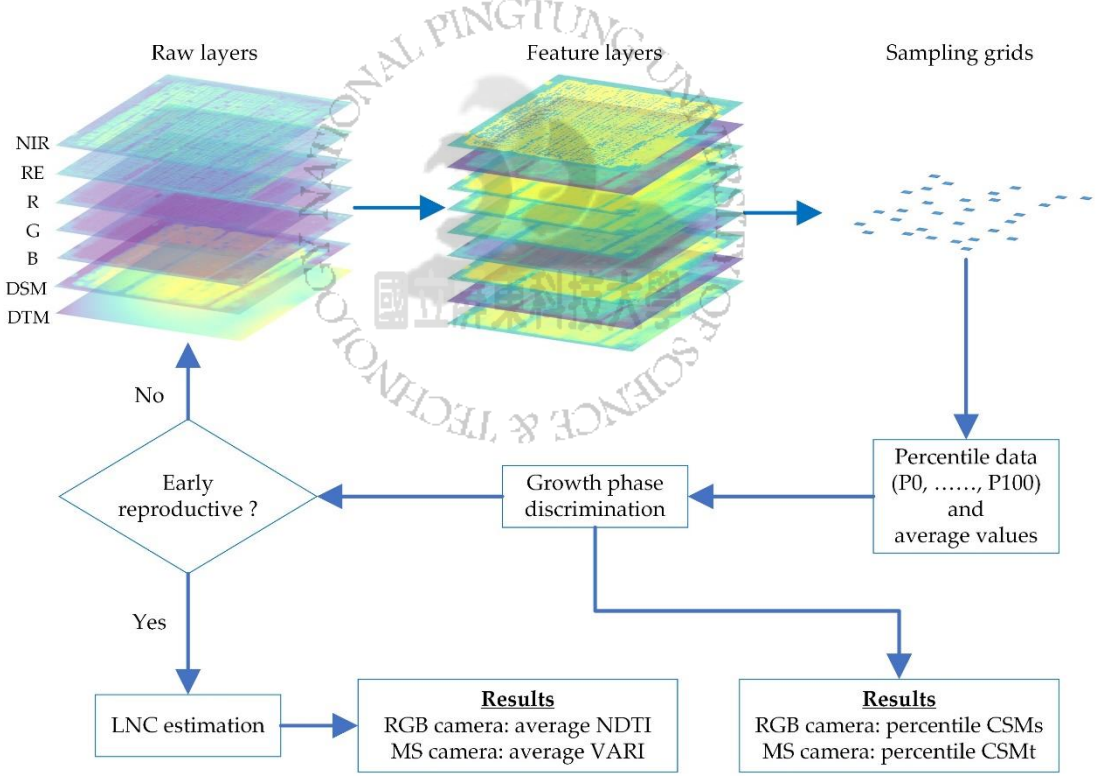
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Chapter 4. A preliminary study on timely and precise rice panicle nitrogen topdressing via UAV-mounted RGB and multispectral camera

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Highlights

- ER classification and LNC estimation with UAV platform.
- Feasibility of using a low-cost RGB camera.
- RGB-percentile-CSMs for ER classification.
- RGB-average-NDTI for LNC estimation.



4.1. Introduction

The global population is projected to reach 9.7 billion by 2050, representing a 20% increase from the current population of 8 billion (United Nations, 2022). This underlines the significance of enhancing crop yield, particularly for rice, which is the second leading source of calories among staple crops (Tilman et al., 2011). In recent years, rice yield has remained stagnant (Ray et al., 2012). The inefficient use of nitrogen fertilizer is considered as a key constraint to rice production, and it also results in resource wastage, cost inefficiencies, and environmental pollution (Vitousek et al., 2009; Good and Beatty, 2011).

The timing and amount of nitrogen fertilizer application are crucial factors in reducing costs and minimizing the environmental impact. Before determining the required amount of nitrogen supplement, it is important to consider the appropriate timing, as the nitrogen absorption and utilization behavior of rice plants varies across different growth stages. The peak demand for nitrogen occurs during the panicle initiation stage that is a critical point between the late vegetative and early reproductive phases (Sheehy et al., 2004). Moreover, the late vegetative phase is the interval between maximum tillering and panicle initiation, while the early reproductive phase encompasses the panicle initiation (R0) and panicle differentiation (R1) stages (Counce et al., 2000). Numerous studies reported that applying nitrogen at the early reproductive phase could increase rice yield (Carreres et al., 2000; Kamiji et al., 2011; Zhou et al., 2017a; Liu et al., 2020), improve rice quality (Xiong et al., 2008), enhance nitrogen use efficiency (Muhammad et al., 1974; Liang et al., 2019), reduce lodging risk (Quang Duy et al., 2004; Pan et al., 2019), and minimize pollution (Liang et al., 2019). However, assessing the transition from the late vegetative phase to the early reproductive phase of the rice plant through human observation can be challenging. Therefore, the development of an efficient approach to detect the occurrence of the early reproductive phase is of critical importance.

Several studies have focused on monitoring rice growth stages, including the entire or partial transition event, utilizing various technologies such as heat-units (Lai and Lin, 2021a), ground-based detection (Lai and Lin, 2021b; Sheng et al., 2022), and aerial-based remote sensing (Yang et al., 2020; Zeng et al., 2020). Among these technologies, aerial-based monitoring, particularly using unmanned aerial vehicles (UAVs), appears to be a competent approach for large-scale and site-specific practices. The wide application of UAVs is mainly due to its high data resolution and flexible revisit windows. The data acquired by UAVs can generally be divided into point-cloud based and index-based data. Point-cloud based data provides three-dimensional information about the architecture of the detected area, while index-based data typically provides reflectance information from a horizontal view. The data grids were conventionally averaged into a point-based data prior to modelling process. This manipulation could probably neglect obscure changes of the rice plant during the transition event, such as reductions in leaf area and leaf length (Tivet et al., 2001; Lai and Lin, 2021b), degradation of leaf chlorophyll and nitrogen translocation (Peng and Cassman, 1998; Lai and Lin, 2021b), and retardation of leaf dry weight (Lai and Lin, 2021b). To extract details of plant properties, other than obtaining average values of the region of interest or utilizing deep learning technologies, the extraction of percentile data could be a potential option. By extracting percentile data from a two-dimensional array to a one-dimensional array, the computation cost can be lesser than the convolutional operations.

As the rice early reproductive phase is detected, the next step is to estimate the plant nitrogen status. Leaf nitrogen content (LNC) is a crucial indicator that are closely related to yield potential (Kropff et al., 1993). In the purpose of field application, SPAD meter (SPAD-502, Minolta Camera Co., Osaka, Japan) is a convenient tool for non-destructive measurement of LNC either in leaf area basis or dry matter basis. Peng et al. (2010) had promoted a site-specific nitrogen management (SSNM) strategy that successfully reduces 30% N usage without yield penalty by controlling N input according to the SPAD readings

(Peng et al., 2010). However, monitoring N demand with SPAD in scattered fields can be time-consuming and labor-intensive. Moreover, the SPAD value was found to be incapable of determining the need for N topdressing at panicle initiation stage (Carreres et al., 2000). On the other hand, Ryu et al. (2009) represented the potential of aerial-based hyperspectral remote sensing on estimating rice LNC during panicle initiation stage (Ryu et al., 2009). Nevertheless, the high-cost hyperspectral sensor could be rather difficult on practical usage.

The primary objective of this study is to investigate the reliability of assessing rice growth phase transition and LNC with low-cost aerial-based sensor. The second is to evaluate the potential of constructing models with percentile data extracted from different layers to assess growth phase transition and leaf nitrogen estimation Finally, several competent approaches are expected to be revealed for the rice growth phase discrimination and LNC estimation.

4.2. Materials and method

4.2.1. Field experiments

A two-year field experiment was conducted during 2021 – 2022 at the National Pingtung University of Science and Technology's (NPUST) experimental field (120°37’E, 22°38’N (WGS 1984)) (Figure 4-1). A semi-dwarf japonica rice cultivar, Kaohsiung 147 (KH147), was manually transplanted when the seedlings had reached three-leaf age, with approximately seven seedlings per hill. The nitrogen fertilizer was applied in four splits (20% at basal, 20% at 50%-tillering stage, 30% at active tillering stage, and 30% at panicle initiation stage). The total amount of nitrogen applied was varied among three levels (50 kg/ha, 100 kg/ha, and 200 kg/ha) within three regions of the field (LN, MN, and HN) to induce variation in LNC of the plants (Table 4-1). The other farming practices followed the local standard protocol. Non-destructive data collection was weekly conducted, while destructive sampling was conducted twice when the rice plants reached late vegetative phase and

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early reproductive phase. The late vegetative phase is determined according to the lag of tillers number development, while the early reproductive phase is evaluated referring to the preceding studies (Lai and Lin, 2021b).

4.2.2. Aerial data acquisition

4.2.2.1. Ground control points (GCPs)

Four ground control points (GCPs) were installed at the four corners of the experimental field using REACH RS+ and REACH RS2+ devices (Emlid, Budapest, Hungary). The Reach RS+ was used as the base, while the RS2+ was used as the rover. The capture area of each flight mission always included the four GCPs. After the GCPs were installed, the geolocation of 30 randomly selected hills (ten for each nitrogen region) was recorded for further data extraction. These geolocation points were then buffered into 30 sampling grids (approximately 0.25 m × 0.25 m) using the QGIS geographical information system (QGIS Association, http://www.qgis.org).

4.2.2.2. Image capture

Two UAVs, which are DJI Mavic 2 Pro (Mavic) and DJI Phantom 4 Multispectral RTK (P4M) (DJI Technology Co., Shenzhen, China), were used to capture field imagery (Table 4-2). When capturing images, the above ground level was set at 20 meters. The camera angle was set at 90 and 70 degrees to obtain vertical and oblique views of the plant images, respectively. Both the front and side overlap were set at 50%. Images were captured between 10:00 a.m. and 2:00 p.m., when the sun is nearly directly overhead, to minimize shadow occlusion.

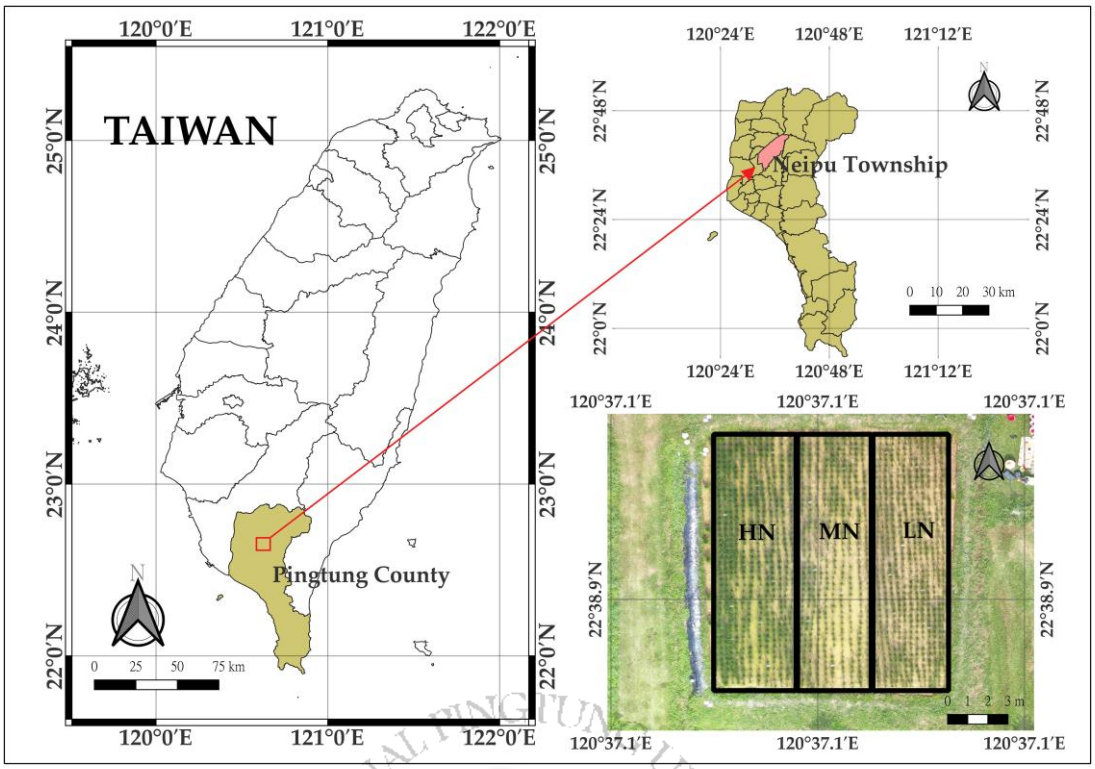


Figure 4-1. Experimental field in National Pingtung University of Science and Technology (NPUST).

Table 4-1. Field experiment

CS	Date of events		Nitrogen treatment	Flight mission
	Transplanting	Sampling	(kg/ha)	(times)
1	2021-Jan-19	2021-Mar-18 2021-Mar-25	50, 100, 200	9
2	2021-Sep-27	2021-Dec-04 2021-Dec-11		14
3	2022-Mar-14	2022-May-08 2022-May-20		10
4	2022-Aug-31	2022-Oct-14 2022-Oct-28		12

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Table 4-2. Parameters of the UAVs

	Mavic	P4M
Sensor type	Color	Multispectral
Bands	RGB	B (450 ± 16 nm), G (560 ± 16 nm), R (650 ± 16 nm), RE (730 ± 16 nm), NIR (840 ± 26 nm)
Resolution	5472×3648	1600×1300
Field of view (FOV)	77°	62.7°
Equivalent focal length	28 mm	40 mm
Aperture	f/2.8 – f/11	f/2.2
ISO range	100 – 3200	200 – 800
Electronic shutter	8 – 1/8000s	1/100 – 1/20000 s

4.2.3. Image processing

4.2.3.1. Image alignment and raw-layers generation

The images were preprocessed with the Pix4Dmapper (version 4.6.3, Pix4D SA, Lausanne, Switzerland). There are four steps should be manipulated before obtaining the band layers and elevation layers: (1) GCPs marking: at least five images that included the GCPs were marked for georeferencing, (2) initial processing: the key-points image scale were set to full scale and the geometrically verified matching option were selected, (3) point cloud and mesh: all options remain default except for selecting the option of point cloud classification, and (4) digital surface model (DSM), orthomosaic, and index layers output: the raster DSM is generated by using the triangulation method and selecting the option of digital terrain model (DTM) generation.

4.2.3.2. Feature layers generation

The DSM, DTM, orthomosaic, and index layers were used to generate feature layers according to the following formulas (Table 4-3). After generating

the feature layers, the raster data of the 30 individuals were clipped with shapefiles that were generated in section 4.2.2.1., while the percentile and average data were also simultaneously extracted from the data grids. This process was done with *geopandas* (Jordahl et al., 2020), *rioxarray* (Dutrieux et al., 2019), and *numpy* (Harris et al., 2020) libraries in python language (Van Rossum and Drake, 2009).

Table 4-3. Layer type used in this study

Abbreviation	Layer type	Formula	References
CSMs	Spatial crop surface model	$DSM_n - DTM_n$	(Bendig et al., 2014)
CSMt	Temporal crop surface model	$DSM_n - DSM_0$	(Tilly et al., 2014; Malambo et al., 2018; Peprah et al., 2021)
DATT	DATT	$\frac{NIR - RE}{NIR - R}$	(Datt, 1999; Zheng et al., 2018b)
GNDVI	Green normalized difference vegetation index	$\frac{NIR - G}{NIR + G}$	(Gitelson et al., 1996; Zheng et al., 2018)
NDTI [1]	Normalized difference texture index	$\frac{G_{mea} - R_{mea}}{G_{mea} + R_{mea}}$	(Li et al., 2019)
NDVI	Normalized difference vegetation index	$\frac{NIR - R}{NIR + R}$	(Rouse et al., 1974; Murata et al., 2019; Lai and Lin, 2021a)
NExG	Normalized excess green index	$\frac{2G - R - B}{R + G + B}$	(Woebbecke et al., 1995; Zheng et al., 2018)
NGRDI	Normalized green-red difference vegetation index	$\frac{G - R}{G + R}$	(Tucker, 1979; Gitelson et al., 2002a; Zheng et al., 2018)
VARI [2]	Visible atmospherically resistant index	$\frac{g - r}{g + r - b}$	(Gitelson et al., 2002a; Li et al., 2019)

[1] The details of NDTI manipulation are illustrated in Appendix 4-A. The G_{mea} is referring to GLCM-based mean texture in green band; the R_{mea} is referring to GLCM-based mean texture in red band. The code was implemented from https://github.com/tzm030329/GLCM.

[2] The r, g, and b are normalized red, green, and blue bands, which is calculated as R / [(R + G + B)], G / [(R + G + B)], and B / [(R + G + B)], respectively.

4.2.4. Estimation of map discrepancy

The map discrepancy is evaluated through the differences of X, Y, and Z axes between the bare soil layer (the first flight) and the subsequent layers (from the second to the last flight mission) (Figure 4-2). The X and Y axes represent horizontal shift of the maps among each flight, which could affect the similarity of the extracted plant data with the same sampling grids. The Z axis represent vertical shift of the maps among each flight, which could affect the stability and reliability of the elevation data.

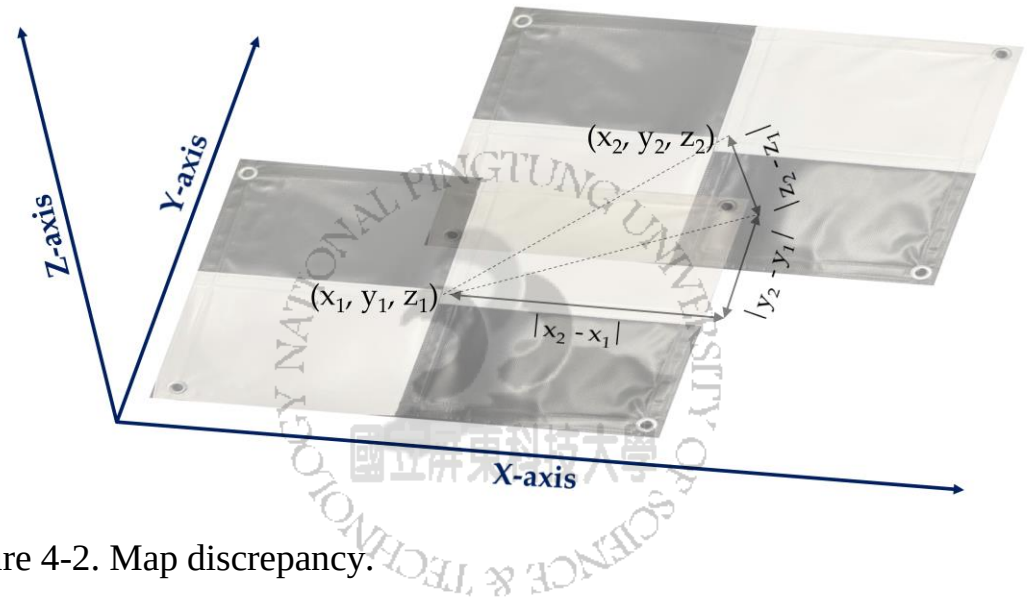


Figure 4-2. Map discrepancy.

4.2.5. Laboratory measurements

Destructive sampling was conducted twice within a crop season when the rice plants reach late vegetative phase and early reproductive phase. The late vegetative phase is determined according the lag of tillers number development, while the early reproductive phase is evaluated referring to the preceding studies (Lai and Lin, 2021b). Three hills were sampled from each nitrogen regions, with a total of nine individuals. The leaves were separated and immediately oven-dried at 60°C for five days. The LNC was determined by the Kjeldahl method.

4.2.6. Data analysis

4.2.6.1. Modelling procedure

After extracting the percentile and average values from the layers, the data analysis and modelling procedures were conducted in R language (R Core Team, 2022). The data were primarily split into two parts: internal dataset (data that were collected in the 1st, 2nd, and 3rd crop season) and external dataset (data that were collected in the 4th crop season) (Figure 4-3). The internal dataset was partitioned into calibration and testing dataset with 7:3 ratio. Next, the calibration dataset was used to train classification models for growth phase discrimination and regression models for LNC estimation by using the *caret* package (Kuhn, 2008). The details of the algorithms were listed in Appendix 4-B. During the calibration process, the parameters of each algorithm were tuned by a ten-fold cross-validation cycle. The ten-fold cross validation was automatically done with the *train* function that is implemented in the *caret* package. The calibrated models were then validated by the testing dataset to obtain internal testing result. The performance of each method was compared according to the testing result. Consequently, the entire internal dataset was re-modelled and finally tested by the external dataset.

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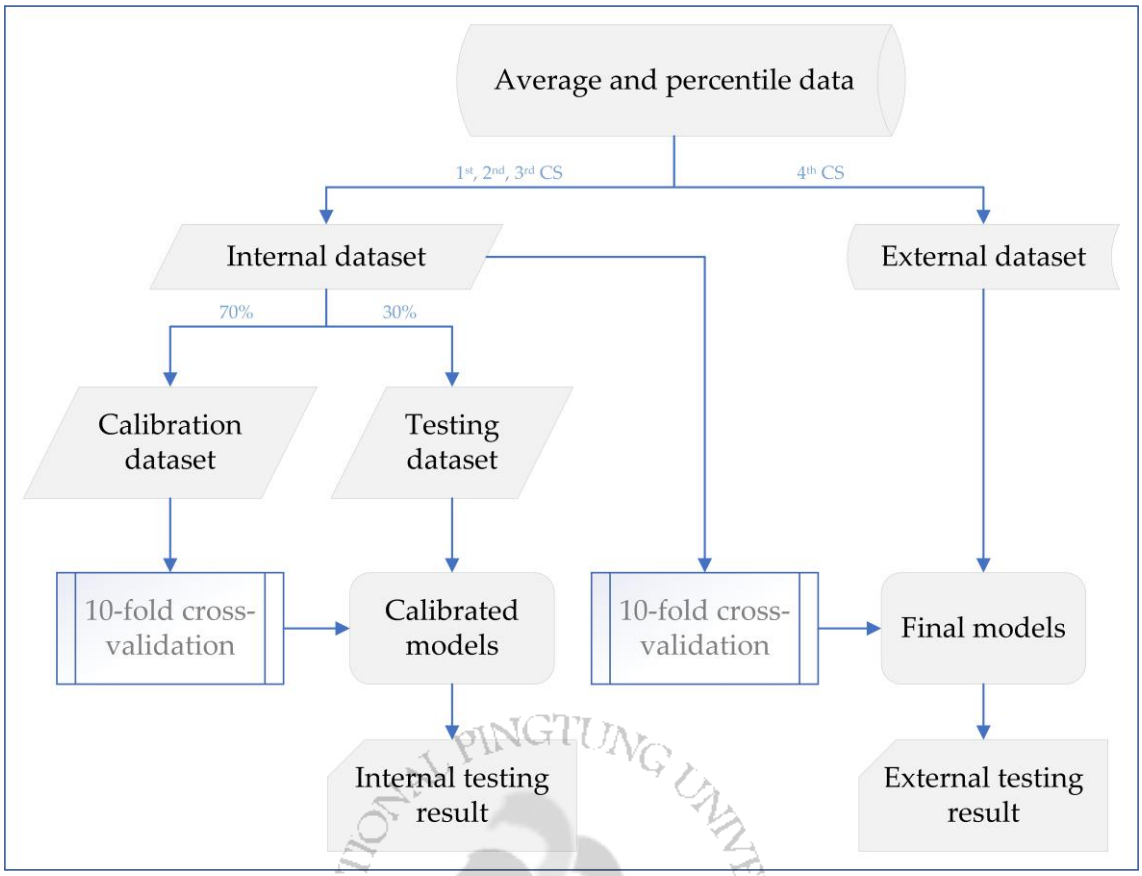


Figure 4-3. Modelling procedure.

4.2.6.2. Model performance metrics

The performance of the classification and regression models were respectively evaluated by the F1-score (Equation 4-1) and relative root mean square error (RRMSE) (Equation 4-2) metrics.

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \qquad \text{(Equation 4-1)}$$

$$\text{RRMSE} = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{x}_i)^2}}{\bar{x}} \times 100 \qquad \text{(Equation 4-2)}$$

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The precision is calculated as: TP / (TP + FP); the recall is calculated as: TP / (TP + FN). The x_i is the actual observation, the $\hat{x}_i$ is the estimated value, the $\bar{x}$ is the mean of observations, and the N is the number of observations.

4.2.6.3. Variable importance

The variable importance of each percentile during classification for the growth phase discrimination and regression for the estimation of LNC is inferenced with partial least square algorithms that are implemented by the *VarImp* function in the *caret* package. Although dimensionality reduction of the percentile data is out of the scope in this study, these inferences could provide valuable information for future research.

4.3. Results and discussion

4.3.1. Field records

The timing of the rice growth phase transition was recorded using days after transplanting (DAT) and growing degree days (GDD) (Table 4-4). When an over 50% green-ring was observed, the LNC of the samples was also measured. The interval of the LNC was expanded by approximately 1.4% (from 3.02% to 4.44%) across the three levels of nitrogen treatment. The variation in DAT between N regions during warm seasons (CS2 and CS4) was up to four days, while the variation during cold seasons (CS1 and CS3) was only two days. The variation in DAT across all crop seasons was quite high (ranging from 43 to 65 for LN, 46 to 69 for MN, and 42 to 68 for HN). Even though GDD was used to minimize the variation of DAT, a significant amount of fluctuation was still observed among crop seasons. The rice cultivar and GDD_{10} calculation in this work is as the same with the preceding study (Lai and Lin, 2021a). The GDD_{mc2} and GDD_{trial} calculation are referring to the Chapter 1. This finding contradicts previous research that found GDD of the panicle initiation event to be stable across crop seasons in outdoor pot experiments (Lai and Lin, 2021a). This contradictory result could be resulted from the differences in the micro-environment between the pot and the field. In other words, the result indicates

that GDD is inappropriate to be apply as a site-specific monitoring tool. Therefore, an alternative technique is needed for the detection of rice growth phase transition.

4.3.2. Map discrepancy

The center points of four GCPs were extracted from the DSM layers generated by the Mavic and P4M to obtain the values for the X, Y, and Z axes. The DSM layer of the bare soil was denoted as DSM_0, while the subsequent DSM layers were denoted as DSM_n. The difference between DSM_n and DSM_0 was then extracted to investigate map discrepancy over time for each crop season. The map discrepancy on the X and Y axes for the Mavic and P4M in each crop season was negligible (less than 10 centimeters), indicating stability and homogeneity of the data across different layers (Table 4-5). The shift on the Z-axis between the maps was generally less than 10 centimeters, except for the Mavic in the first crop season (over 10 meters). The temporal uncertainty on the Z-axis would not affecting the spatial crop surface model (CSMs) but can lead to unreliable temporal crop surface models (CSMt). Therefore, the CSMt layers generated by the Mavic in the first crop season were excluded from further analysis.

Table 4-4. Records of the growth phase transition events and leaf nitrogen content (LNC) in each crop season with different nitrogen treatment

CS	Nitrogen level	Growth phase transition				LNC (%)
		DAT (days)	GDD_{10} (°C)	GDD_{mc2} (°C)	GDD_{trial} (°C)	
1	LN	56	647.5	610.7	610.7	4.05 ± 0.26
	MN	55	633.4	597.0	597.0	4.03 ± 0.31
	HN	57	662.4	624.9	625.1	4.40 ± 0.24
2	LN	65	1051.1	985.9	988.0	3.85 ± 0.09
	MN	69	1088.2	1021.0	1023.1	4.02 ± 0.13
	HN	68	1078.7	1012.0	1014.1	4.40 ± 0.30
3	LN	53	814.5	763.0	764.1	3.02 ± 0.07
	MN	51	781.1	730.6	731.7	3.38 ± 0.07
	HN	52	797.7	746.7	747.8	4.03 ± 0.20
4	LN	43	783.7	726.5	730.9	4.12 ± 0.09
	MN	46	835.8	777.0	781.5	4.14 ± 0.15
	HN	42	767.4	710.7	715.1	4.44 ± 0.07

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Table 4-5. Map discrepancy

UAV	CS	Map discrepancy (meter)		
		X-axis	Y-axis	Z-axis
Mavic	1	0.0093 ± 0.0015	0.0233 ± 0.0022	17.8996 ± 1.6167
	2	0.0071 ± 0.0006	0.0041 ± 0.0004	0.0080 ± 0.0013
	3	0.0023 ± 0.0004	0.0028 ± 0.0004	0.0058 ± 0.0010
	4	0.0040 ± 0.0004	0.0022 ± 0.0002	0.0041 ± 0.0005
P4M	1	0.0068 ± 0.0013	0.0067 ± 0.0011	0.0392 ± 0.0032
	2	0.0046 ± 0.0005	0.0089 ± 0.0007	0.0295 ± 0.0026
	3	0.0027 ± 0.0004	0.0036 ± 0.0004	0.0168 ± 0.0021
	4	0.0049 ± 0.0006	0.0072 ± 0.0009	0.0179 ± 0.0018

4.3.3. Growth phase discrimination

This study evaluated the ability to distinguish rice growth phases using RGB and multispectral (MS) UAV imagery. The raw maps were processed into several feature maps, and the feature maps were extracted as either an average value or a set of percentile data (P_0 – P_{100}). The performance was compared within internal testing, using a pooled data set of the first, second, and third crop seasons. External testing was used to validate the stability of the reconstructed model after internal testing, rather than comparing the performance of each combination. Ten algorithms were applied for the growth phase discrimination: regularized discriminant analysis (RDA), stochastic gradient boosting (GBM), naïve bayes (NB), penalized logistic regression (PLR), C5.0, gaussian process with radial kernel (GP), support vector machine with radial kernel (SVM), random forest (RF), k-nearest neighbor (KNN), and multilayer perceptron (MLP).

4.3.3.1. RGB camera

Six types of layers, including CSMs, CSMt, NDTI, NExG, NGRDI, and VARI, were generated using an RGB camera. These layers only consist of the red, green, and blue bands and were analyzed to determine their differences

between growth phases (Figure 4-4). The percentile and average data in the CSMs, CSMt, NDTI, and VARI show substantial variations between the growth stages (Figure 4-4a–c, f), while the differences in NExG and NGRDI were insignificant (Figure 4-4d, e). The most informative data was found to be at the upper end of the percentile data, except for NGRDI (Figure 4-5). The highest variable importance was observed at either P_{99} or P_{100}, but other percentile values, such as $P_{75} – P_{99}$ in CSMs and $P_5 – P_{50}$ in NGRDI, were also informative (Figure 4-5a, e).

In the internal testing, the models constructed using percentile data demonstrated a higher F1-score (mean = 0.76 – 0.90) compared to models built from average data (mean = 0.66 – 0.86) (Table 4-6). The highest overall F1-score of 0.90 ± 0.03 was achieved by extracting percentile data from the CSMs layer. This combination also performed well in external testing, with an F1-score of 0.85 ± 0.01. The top-performing algorithms in internal testing were Random Forest (RF), Gradient Boosting Machine (GBM), C5.0, Support Vector Machine (SVM), and Multi-layer Perceptron (MLP) with F1-scores of 0.96, 0.92, 0.92, 0.92, and 0.92, respectively. However, during external testing, a decline in the F1-score was observed for RF (from 0.96 to 0.86), GBM (from 0.92 to 0.85), C5.0 (from 0.92 to 0.84), MLP (from 0.92 to 0.85), and SVM (from 0.92 to 0.87), with a decline of 10%, 7%, 8%, 7% and 5% respectively. Among the algorithms, the SVM which decline only 5% demonstrated relatively stability in the external testing.

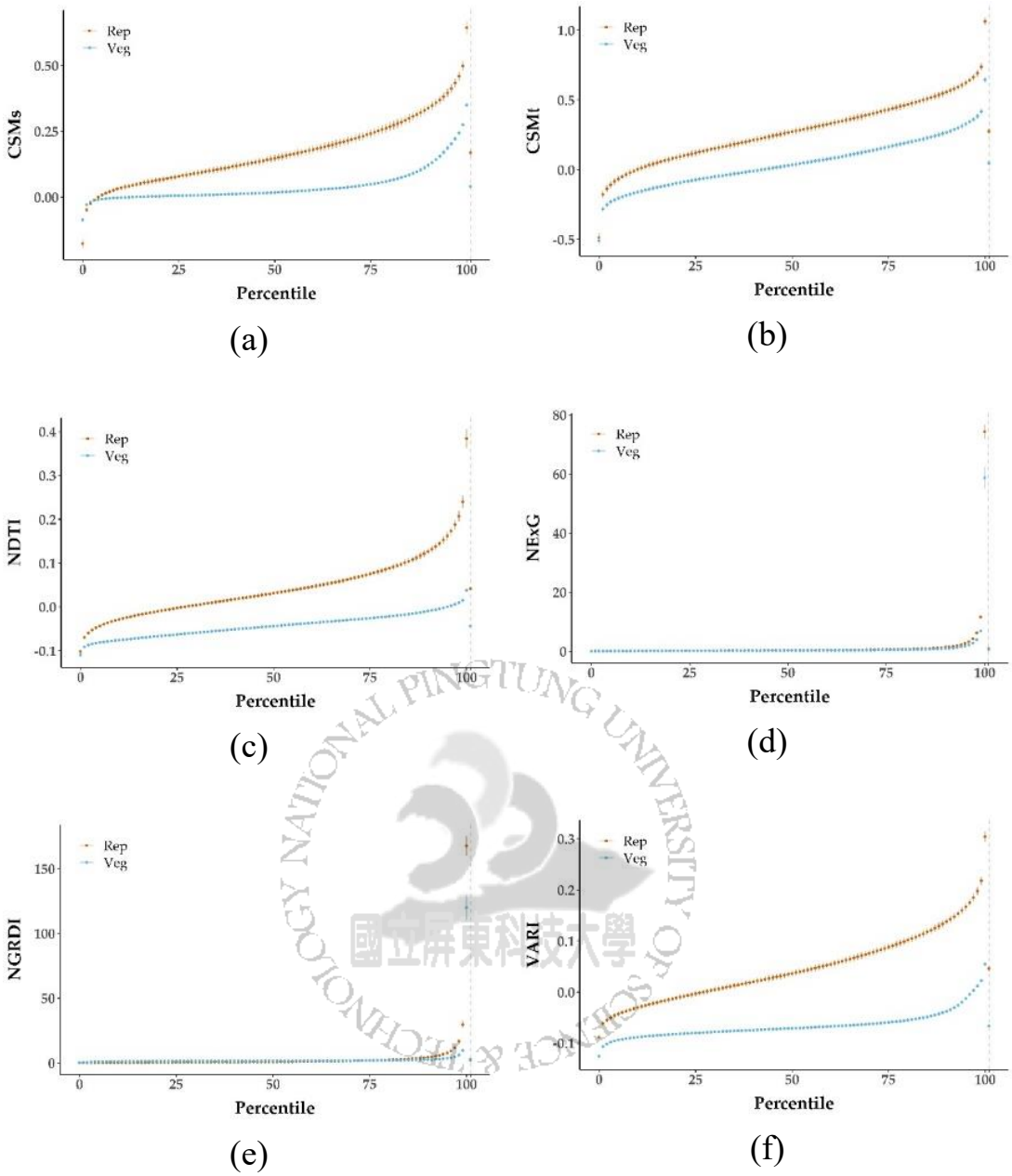


Figure 4-4. Percentile and average value of the RGB-generated layers. The layers are (a) CSMs, (b) CSMt, (c) NDTI, (d) NExG, (e) NGRDI, and (f) VARI. The stroked-points represent mean ± se. The late vegetative phase (Veg) are colored in blue, while the early reproductive phase (Rep) are colored in red. The vertical grey-dashed-line indicates the average value.

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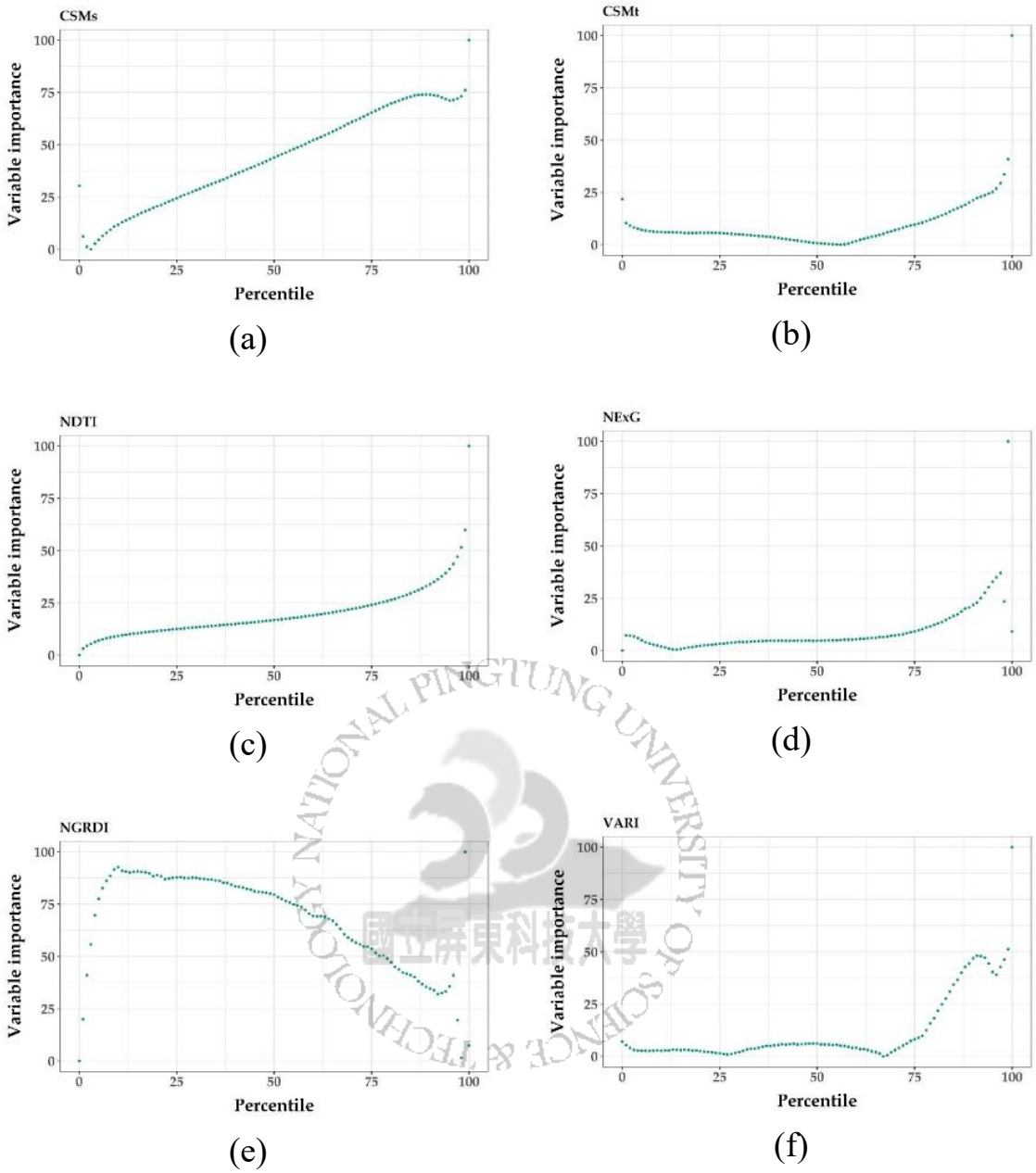


Figure 4-5. The importance of each percentile extracted from RGB imagery on growth phase discrimination. The layers are (a) CSMs, (b) CSMt, (c) NDTI, (d) NExG, (e) NGRDI, and (f) VARI.

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Table 4-6. The F1-score of growth phase discrimination by using RGB camera with different combinations of data type and layer type.

Data Type	Layer Type	Internal testing				External testing			
		min	mean	max	sd	min	mean	max	sd
Avg[†]	CSMs	0.84	0.86	0.89	0.02	0.78	0.85	0.87	0.03
	CSMt	0.57	0.66	0.71	0.04	0.65	0.75	0.78	0.04
	NDTI	0.62	0.74	0.80	0.06	0.47	0.62	0.86	0.11
	NExG	0.65	0.71	0.78	0.05	0.43	0.64	0.74	0.10
	NGRDI	0.47	0.72	0.82	0.10	0.66	0.73	0.76	0.03
	VARI	0.60	0.72	0.78	0.06	0.56	0.69	0.82	0.07
Perc[§]	CSMs	0.85	0.90	0.96	0.03	0.83	0.85	0.87	0.01
	CSMt	0.73	0.82	0.92	0.06	0.76	0.81	0.83	0.02
	NDTI	0.52	0.79	0.87	0.10	0.30	0.65	0.79	0.14
	NExG	0.72	0.76	0.80	0.03	0.31	0.63	0.78	0.16
	NGRDI	0.32	0.79	0.88	0.17	0.64	0.70	0.78	0.04
	VARI	0.76	0.83	0.87	0.04	0.28	0.58	0.70	0.12

[†] The F1-scores resulted from each algorithm were listed in Appendix 4-C.

[§] The F1-scores resulted from each algorithm were listed in Appendix 4-D.

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4.3.3.2. Multispectral camera

The results show that using the percentile and average values of the layers generated from the multispectral imagery of the P4M (CSMs, CSMt, DATT, GNDVI, NDTI, NDVI, NExG, and VARI) were useful in discriminating growth phases (Figure 4-6). The CSMs and CSMt were generated from points cloud while the others were composed of either R, G, B, or RE and/or NIR bands. The most informative data was found to be at the upper end of the percentile data (Figure 4-7), except for the DATT where significant differences were found in P_0 and P_{100}. The highest variable importance was observed at P_{100} for most layers, except for GNDVI (peak at P_{99}), NDTI (peak at P_{97}), NDVI (peak at P_{91}), and VARI (peak at P_{97}).

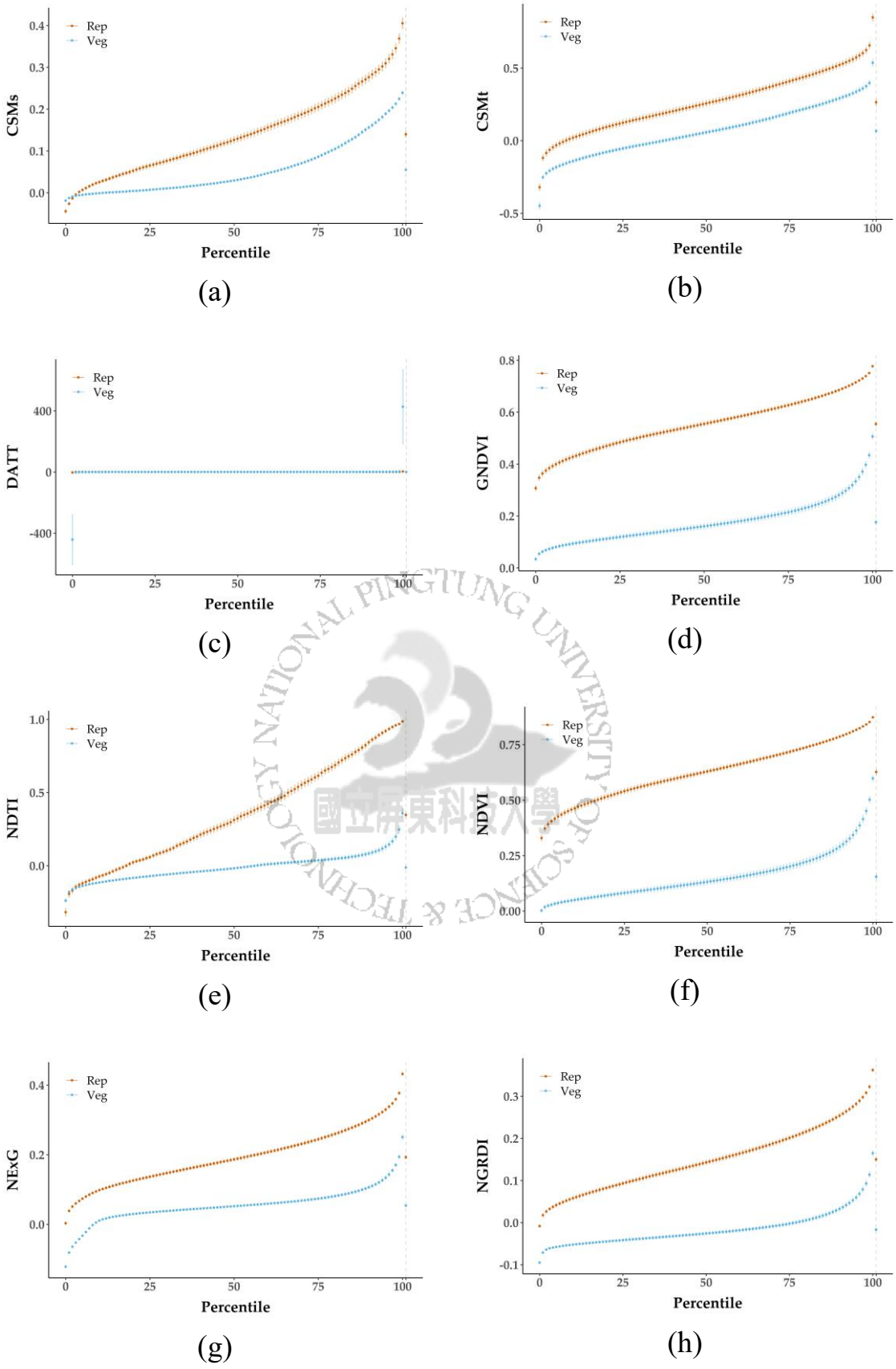
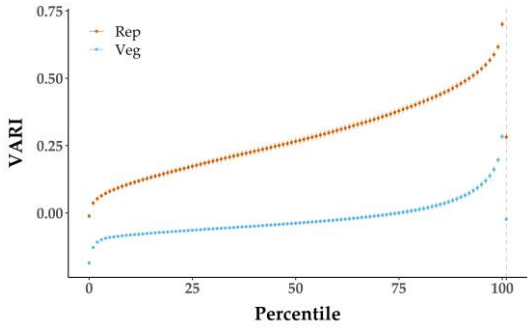


Figure 4-6. Cont.

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(i)

Figure 4-6. Percentile and average value of the MS-generated layers. The layers are (a) CSMs, (b) CSMt, (c) DATT, (d) GNDVI, (e) NDTI, (f) NDVI, (g) NExG, (h) NGRDI, and (i) VARI. The stroked-points represent mean ± se. The late vegetative phase (Veg) are colored in blue, while the early reproductive phase (Rep) are colored in red. The vertical grey-dashed-line indicates the average value.

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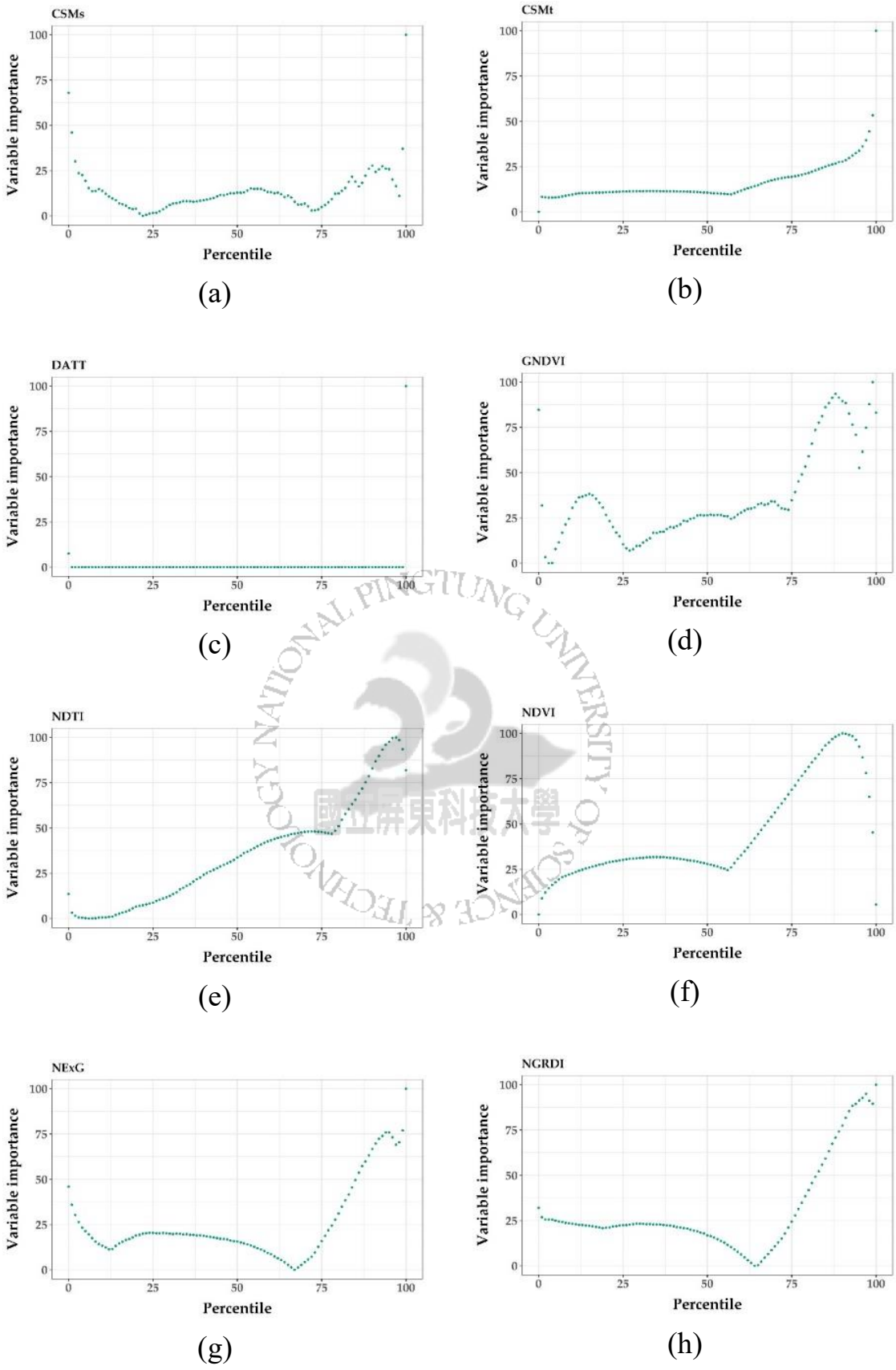
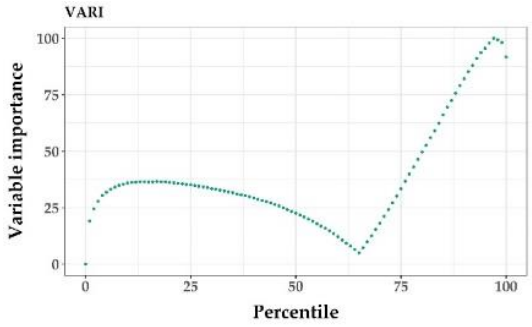


Figure 4-7. Cont.

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(i)

Figure 4-7. The importance of each percentile extracted from MS imagery on growth phase discrimination. The layers are (a) CSMs, (b) CSMt, (c) DATT, (d) GNDVI, (e) NDTI, (f) NDVI, (g) NExG, and (h) NGRDI, (i) VARI.

Models that were constructed with percentile data (with a mean accuracy range of 0.79 – 0.88) had slightly better performance than those constructed with average data (with a mean accuracy range of 0.70 – 0.87) during internal testing (Table 4-7). Additionally, the elevation layers (CSMs and CSMt) performed better than the vegetation index layers. In overall, the percentile CSMt had the most promising performance, while other approaches, such as average CSMs and percentile CSMs, were overfitted during external testing. The top-5 F1-scores by modelling the percentile CSMt in internal testing were obtained from GBM, RF, PLR, KNN, and GP algorithms, with scores of 0.92, 0.91, 0.88, 0.88, and 0.88 respectively. During external testing, GBM, RF, and KNN still performed well, maintaining F1-scores of 0.81. Additionally, the KNN model was relatively well-fitted during external testing, with a decrease from 0.88 to 0.81 in F1-score. On the other hand, GBM and RF models were found to be overfitted, with a decrease from 0.92 to 0.81 and 0.91 to 0.81 respectively.

Table 4-7. The F1-score of growth phase discrimination by using multispectral camera with different combinations of data type and layer type.

Data Type	Layer type	Internal testing				External testing			
		min	mean	max	sd	min	mean	max	sd
Avg†	CSMs	0.82	0.87	0.93	0.03	0.67	0.74	0.77	0.03
	CSMt	0.83	0.86	0.91	0.02	0.68	0.75	0.77	0.03
	DATT	0.47	0.70	0.76	0.09	0.42	0.80	0.89	0.14
	GNDVI	0.66	0.74	0.78	0.03	0.71	0.82	0.93	0.06
	NDTI	0.58	0.74	0.79	0.07	0.70	0.86	0.91	0.06
	NDVI	0.68	0.74	0.79	0.05	0.67	0.78	0.92	0.08
	NExG	0.72	0.78	0.81	0.03	0.62	0.75	0.89	0.10
	NGRDI	0.62	0.80	0.88	0.09	0.68	0.72	0.90	0.07
	VARI	0.64	0.80	0.87	0.08	0.67	0.76	0.90	0.07
Perc§	CSMs	0.78	0.88	0.92	0.04	0.58	0.72	0.77	0.05
	CSMt	0.77	0.87	0.92	0.04	0.76	0.80	0.81	0.02
	DATT	0.72	0.79	0.83	0.04	0.79	0.85	0.89	0.03
	GNDVI	0.71	0.84	0.88	0.05	0.77	0.83	0.92	0.04
	NDTI	0.74	0.80	0.84	0.03	0.68	0.82	0.94	0.09
	NDVI	0.63	0.79	0.86	0.06	0.49	0.75	0.88	0.10
	NExG	0.78	0.83	0.86	0.02	0.37	0.70	0.89	0.17
	NGRDI	0.79	0.84	0.89	0.03	0.47	0.65	0.93	0.12
	VARI	0.74	0.84	0.89	0.04	0.33	0.62	0.74	0.12

† The F1-scores resulted from each algorithm were listed in Appendix 4-E.

§ The F1-scores resulted from each algorithm were listed in Appendix 4-F.

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4.3.4. Leaf nitrogen concentration (LNC) estimation

When the growth phase transition event had been observed, the LNC of the rice plant was subsequently estimated when the growth phase transition event had been observed. Ten algorithms were applied for the regression purpose: generalized linear model (GLM), k-nearest neighbor (KNN), independent component regression (ICR), principal component regression (PCR), partial least square regression (PLSR), ridge regression (RIDGE), multilayer perceptron (MLP), support vector machine (SVM), random forest (RF), and extreme gradient boosting (XGB). The ICR, PCR, PLSR, and RIDGE were not applied on the average data, while the GLM was not applied on the percentile data.

4.3.4.1. RGB camera

Six types of layers were generated using the RGB imagery to estimate the LNC (Figure 4-8). Among the layers, the CSMs, CSMt, NDTI and VARI depict obvious between different nitrogen-treated regions (Figure 4-8a–c, f). In the NExG and NGRDI, only the values upon P_{95} show obvious discriminative ability (Figure 4-8d, e). In the CSMt, NDTI and VARI, it is reasonable that the values between the LN and MN are mostly overlapping as the difference of the LNC between the LN and the MN is insignificant (Table 4-4). By extension, the large differences between the LN and the HN and MN in the CSMs, NExG, and NGRDI could be inconsistent with the observed LNC (Figure 4-8a, d, e). The most informative data was found to be at the upper end of the percentile data (Figure 4-9), except for the CSMs where great importance was found around median (Figure 4-9a). The highest variable importance was observed at P_{100} for CSMt, NExG, and VARI (Figure 4-9b, d, f); and at P_{99} for NDTI and NGRDI (Figure 4-9c, e).

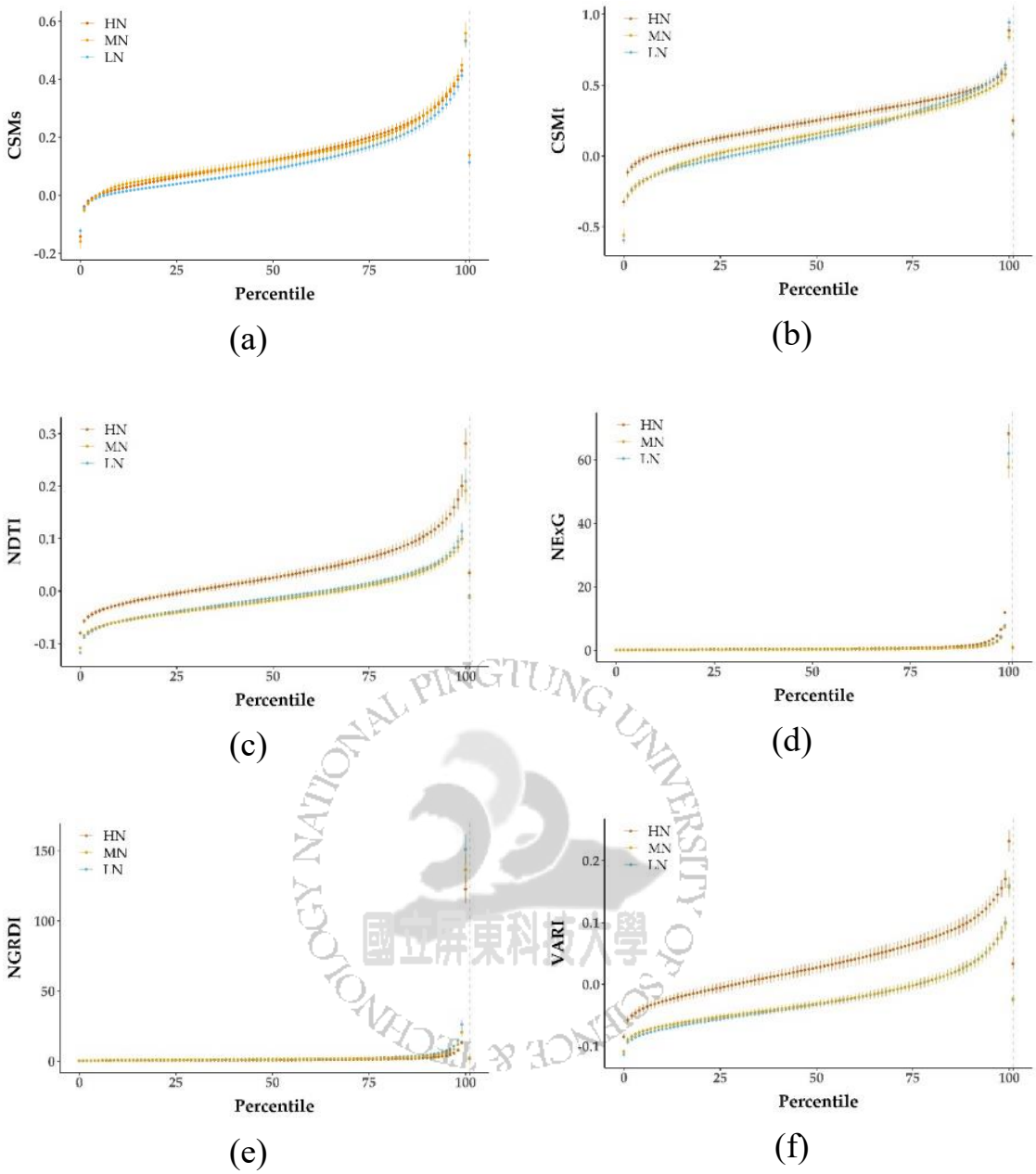


Figure 4-8. Percentile and average value of the RGB-generated layers. The layers are (a) CSMs, (b) CSMt, (c) NDTI, (d) NExG, (e) NGRDI, and (f) VARI. The stroked-points represent mean ± se. The LN, MN, and HN are respectively colored in blue, orange, and red. The vertical grey-dashed-line indicates the average value.

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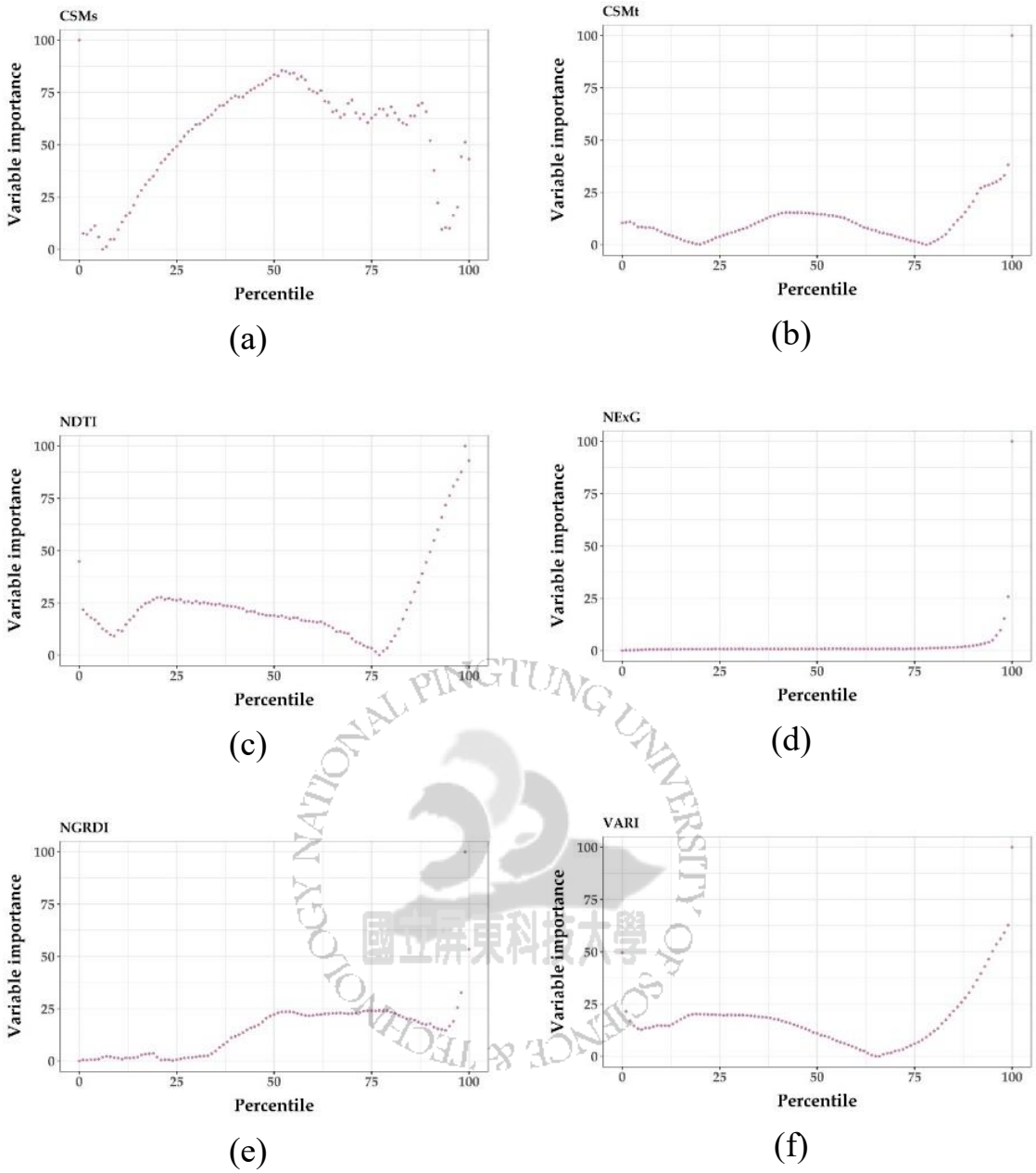


Figure 4-9. The importance of each percentile extracted from RGB imagery on LNC estimation. The layers are (a) CSMs, (b) CSMt, (c) NDTI, (d) NExG, (e) NGRDI, and (f) VARI.

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The relative root-mean-square error (RRMSE) of average data in all layers are generally lower than that of percentile data (except for the CSMs layer) (Table 4-8). Therefore, percentile data were not utilized in this section. The NDTI layer was found to be the best among the layers, with RRMSE of 12.62 ± 2.49% and 15.64 ± 3.80% in internal and external testing, respectively. No overfitting was observed from the NDTI layer. Among the algorithms, the three lowest RRMSE values in internal testing were observed from the KNN, GLM, and SVM algorithms (8.13%, 11.7%, and 12.9%, respectively). In external testing, the RRMSE increased by up to 5% for KNN (from 8.13% to 14.0%), 3% for GLM (from 11.7% to 14.8%), and 3% for SVM (from 12.9% to 16.3%). The GLM and SVM algorithms are relatively stable during external testing.

Table 4-8. Estimation of the leaf nitrogen content (LNC) with RGB camera. The values represent the minimum, mean, maximum, and standard deviation of the relative root mean square error (RRMSE (%)) of the models' prediction.

Data	Layer	Internal testing				External testing			
Type	type	min	mean	max	sd	min	mean	max	sd
Avg†	CSMs	11.65	17.83	25.73	4.98	3.39	7.86	11.53	3.23
	CSMt	8.30	13.66	19.36	4.13	8.23	10.08	11.58	1.40
	NDTI	8.13	12.62	15.12	2.49	9.62	15.64	20.28	3.80
	NExG	8.03	12.87	19.36	4.30	12.61	15.45	19.78	2.56
	NGRDI	7.96	13.40	20.73	4.69	6.14	10.94	17.44	4.32
	VARI	6.95	12.95	23.66	5.93	14.00	17.11	19.49	2.18
Perc§	CSMs	9.17	15.17	20.50	3.70	4.24	7.05	10.14	1.93
	CSMt	7.69	14.18	18.98	4.03	7.12	8.99	12.39	2.95
	NDTI	6.30	13.60	19.70	4.55	12.90	16.17	23.12	3.17
	NExG	7.33	14.26	20.30	4.80	11.42	16.42	21.43	2.96
	NGRDI	10.07	14.48	22.82	4.13	5.24	13.13	22.80	6.81
	VARI	5.43	14.75	22.47	4.78	9.56	14.59	19.00	2.88

† The RRMSE resulted from each algorithm were listed in Appendix 4-G.

§ The RRMSE resulted from each algorithm were listed in Appendix 4-H.

4.3.4.2. Multispectral camera

Nine types of layers were generated using the MS imagery to estimate the LNC (Figure 4-10). All layers show obvious between-groups differences in both percentile and average values (Figure 4-10a–b, d–i), except the DATT (Figure 4-10c). The most informative data was found to be at the lower end of the percentile data (Figure 4-11), except for the VARI. The highest variable importance was observed at P_0 for most layers, except for CSMt (peak at P_{100}), NExG (peak at P_{35} and P_{99}), and VARI (peak at P_{100}).

In internal testing, the percentile NDTI (RRMSE = 10.73 ± 3.04%) performed exceptionally well among the other combinations (Table 4-9). However, during external testing, the RRMSE increased dramatically from 10.73% to 20.37%, indicating significant overfitting. The average VARI is the next competent approach with a RRMSE of 13.79 ± 3.64% in internal testing and 19.01 ± 4.21% in external testing. Among the algorithms, the three lowest RRMSE values in internal testing were observed from the MLP, GLM, and RF algorithms (7.97%, 12.2%, and 12.9%, respectively). In external testing, the RRMSE increased by up to 15% for MLP (from 7.97% to 23.8%), 4% for GLM (from 12.2% to 16.9%), and 8% for RF (from 12.9% to 21.5%). The GLM is relatively stable during external testing.

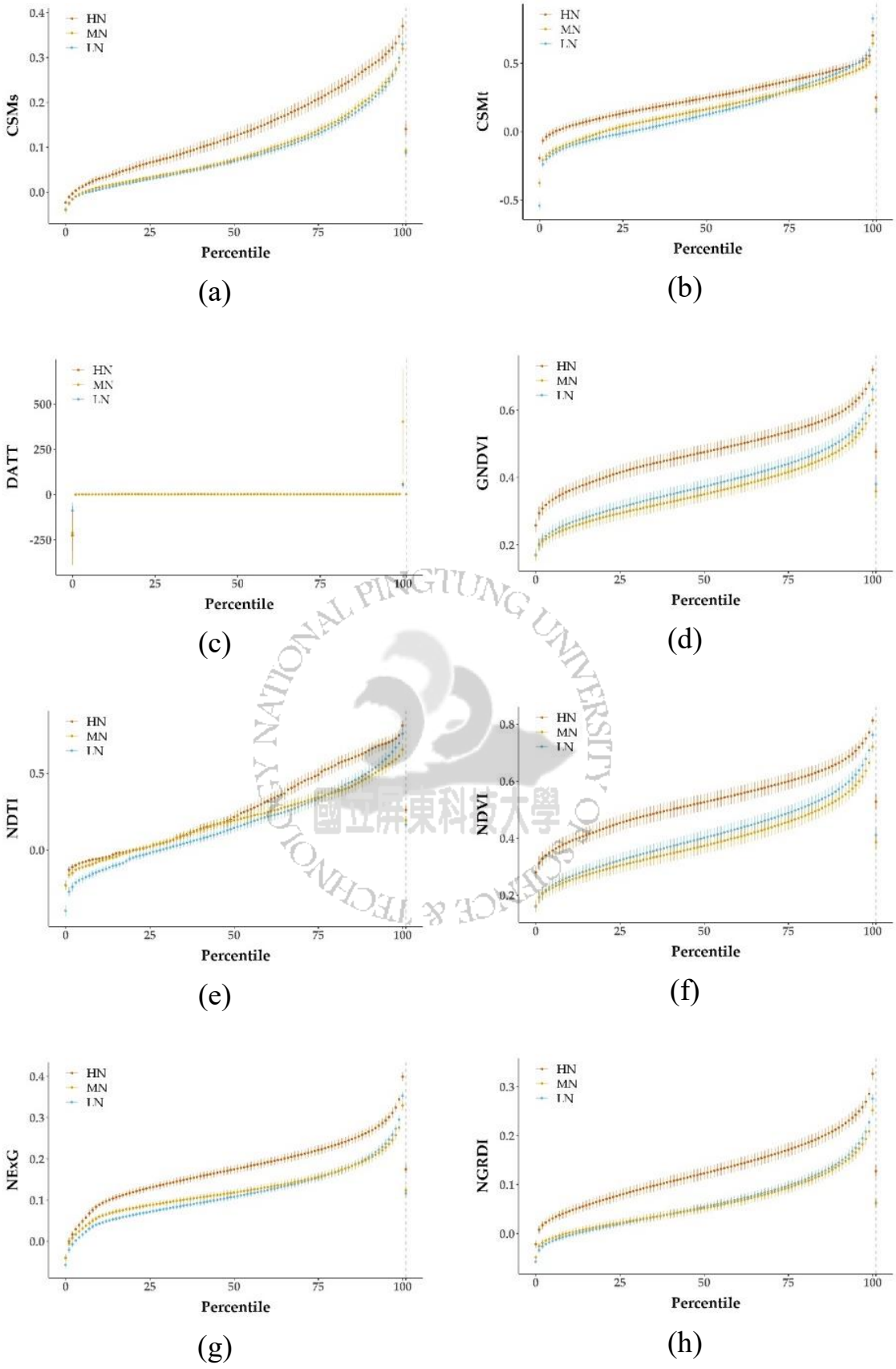
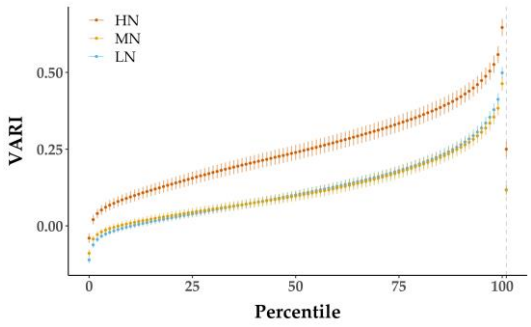


Figure 4-10. Cont.

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(i)

Figure 4-10. Percentile and average value of the MS-generated layers. The layers are (a) CSMs, (b) CSMt, (c) DATT, (d) GNDVI, (e) NDTI, (f) NDVI, (g) NExG, (h) NGRDI, and (i) VARI. The stroked-points represent mean ± se. The LN, MN, and HN are respectively colored in blue, orange, and red. The vertical grey-dashed-line indicates the average value.

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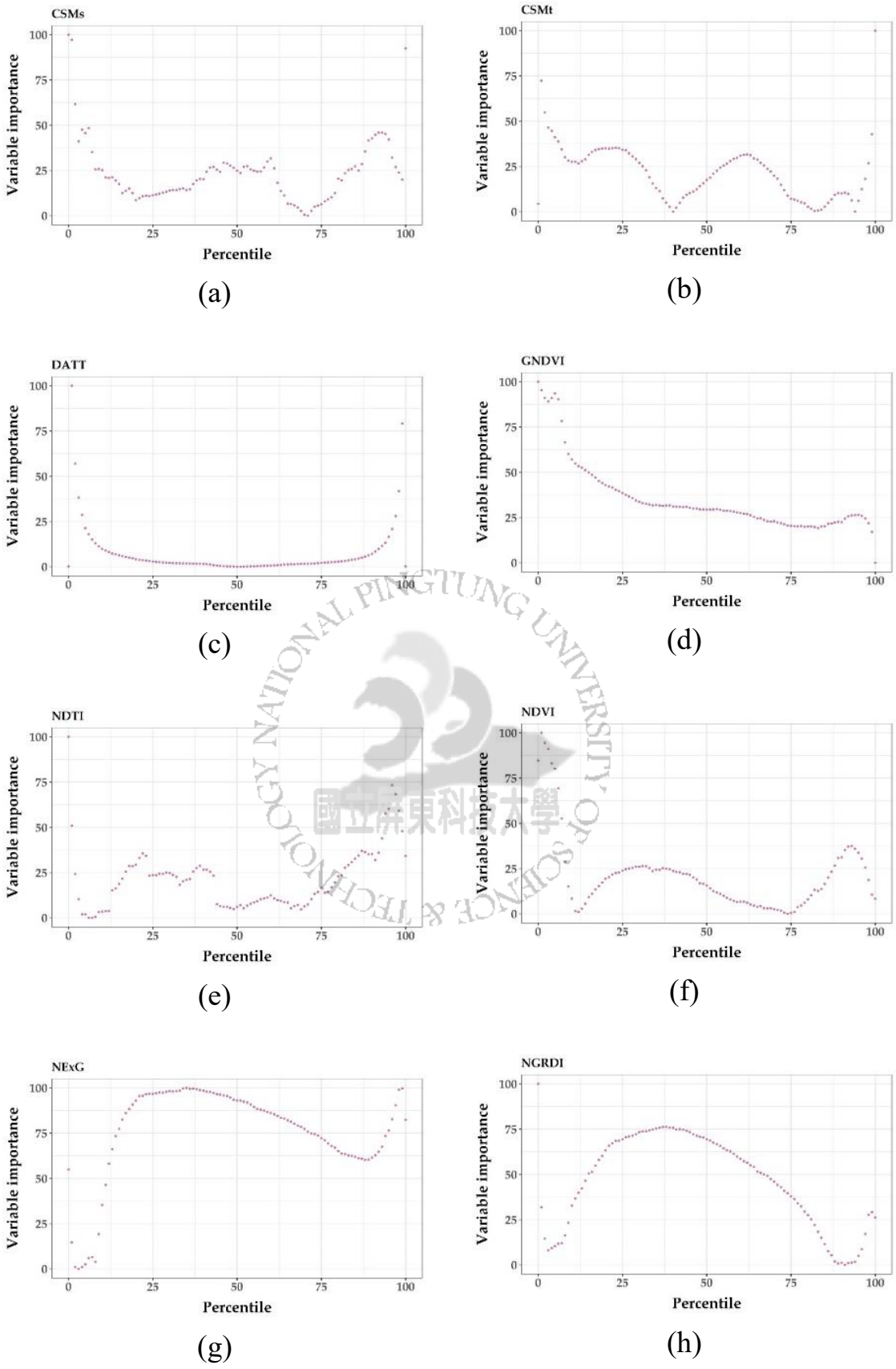
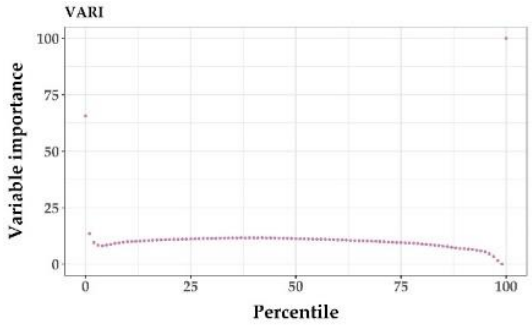


Figure 4-11. Cont.

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(i)

Figure 4-11. The importance of each percentile extracted from MS imagery on LNC estimation. The layers are (a) CSMs, (b) CSMt, (c) DATT, (d) GNDVI, (e) NDTI, (f) NDVI, (g) NExG, and (h) NGRDI, (i) VARI.

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Table 4-9. Estimation of the leaf nitrogen content (LNC) with multispectral camera. The values represent the minimum, mean, maximum, and standard deviation of the relative root mean square error (RRMSE (%)) of the models' prediction.

Data Type	Layer type	Internal testing				External testing			
		min	mean	max	sd	min	mean	max	sd
Avg[†]	CSMs	12.38	14.98	17.38	1.68	7.63	9.33	11.21	1.31
	CSMt	11.36	14.49	21.06	3.37	6.60	9.65	14.59	2.87
	DATT	16.49	17.50	20.23	1.55	8.55	11.78	15.83	2.64
	GNDVI	12.43	16.60	22.20	3.66	10.58	16.57	23.90	4.79
	NDTI	11.49	16.04	21.10	3.06	15.42	19.24	25.11	3.77
	NDVI	8.56	15.80	19.36	4.27	15.41	19.33	24.63	3.55
	NExG	10.78	14.42	20.31	3.86	11.43	15.78	17.78	2.56
	NGRDI	10.10	14.91	22.05	3.99	13.52	18.24	25.85	4.18
	VARI	7.97	13.79	18.05	3.64	14.20	19.01	23.84	4.21
Perc[§]	CSMs	10.96	16.30	21.33	3.24	5.88	10.90	13.71	2.63
	CSMt	8.27	15.36	18.37	3.44	7.28	8.94	11.77	1.99
	DATT	7.53	14.78	19.55	4.24	8.85	12.17	22.15	4.17
	GNDVI	8.50	15.70	21.40	4.10	11.96	15.72	22.38	4.58
	NDTI	6.02	10.73	15.64	3.04	14.15	20.37	28.42	5.52
	NDVI	10.77	16.04	24.82	3.86	10.58	14.26	16.56	2.64
	NExG	9.01	14.64	21.19	4.13	11.30	14.47	16.73	1.88
	NGRDI	8.98	14.62	21.96	4.28	11.08	16.27	21.39	3.88
	VARI	10.21	16.87	36.59	7.98	9.37	15.01	18.76	3.85

[†] The RRMSE resulted from each algorithm were listed in Appendix 4-I.

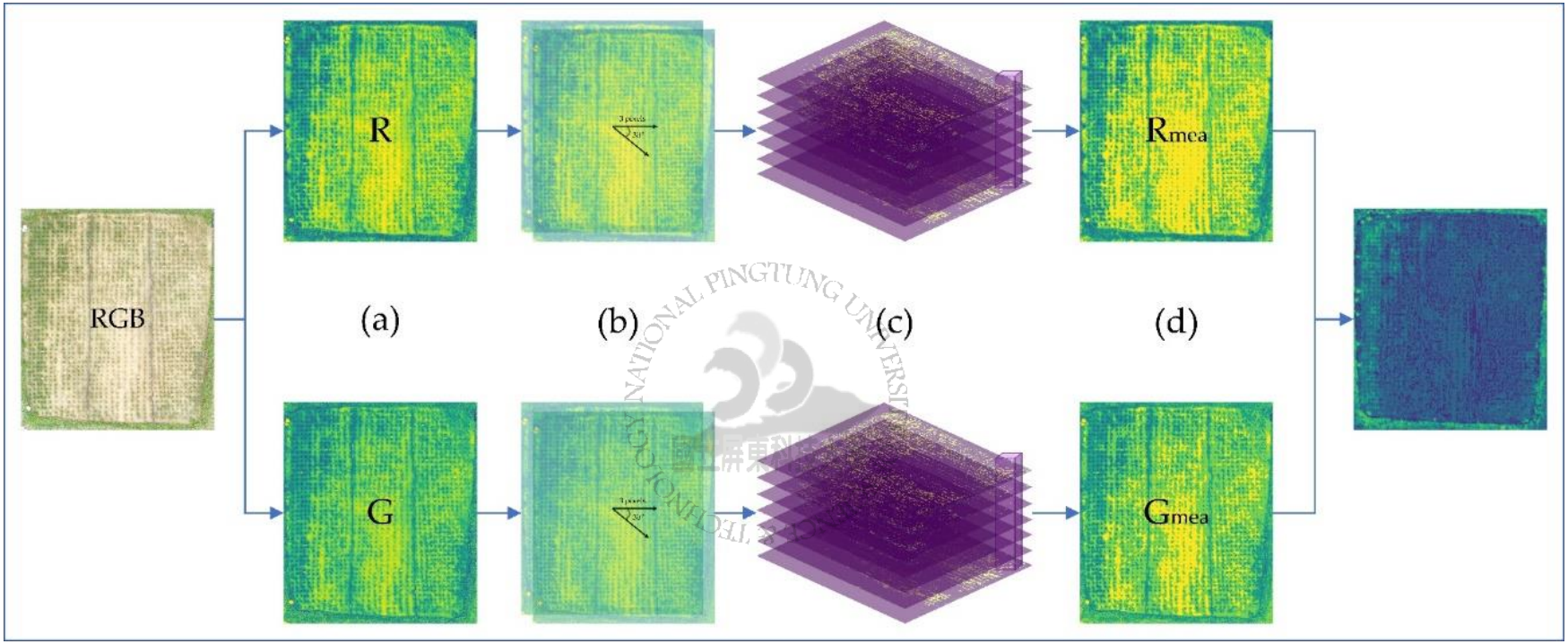
[§] The RRMSE resulted from each algorithm were listed in Appendix 4-J.

4.4. Conclusions

In growth phase discrimination, elevation layers outperformed vegetation index layers. Meanwhile, percentile data exhibit higher F1-score than average data. When using an RGB camera, the CSMs layer is preferable; while using a multispectral camera, the CSMt is more reliable. For LNC estimation, vegetation index layers performed better than elevation layers. When adopting an RGB camera, the NDTI layer is outperforming; while using a multispectral camera, the VARI layer is more reliable. If considering the complexity and time-cost of generating a NDTI layer, estimating LNC via the VARI layer seems to be competent when a multispectral camera is available.

This work attempted to evaluate the feasibility of detecting the rice growth phase transition and LNC via UAV imagery for precise nitrogen application in rice cultivation. A significant progress has been made by combining different types of layers, and data extraction methods. The current work is currently not applicable for infrastructure use unless a recommendation system that could simultaneously considering other critical factors, such as the leaf nitrogen content threshold in different cultivars and the field conditions, is available. Further efforts are still required such as investigating the leaf nitrogen content threshold for different cultivars, developing a practical protocol for applying the required amount of nitrogen according to the estimated leaf nitrogen content, and integrating the field conditions and weather variability into these approaches.

4.5. Appendix



Appendix 4-A. Illustration of the NDTI layer manipulation. (a) The R and G band layers were extracted from an RGB image; (b) copy of the layers were respectively shifted by a distance of three pixels with a degree of 30°; (c) a set of 64 array-layers (8 bits × 8 bits) was mean-filtered with a 5×5 kernel; (d) the grey level covariance matrix (GLCM) were generated, and the NDTI layer was could be calculated according to the formula in Table 4-3 as aforementioned.

Appendix 4-B. Libraries for regression and classification in the R language

Algorithm	Abbreviation	Method	Dependencies
	Regression		
Generalized linear model	GLM*	glm	–
k-Nearest neighbors	KNN	knn	–
Independent Component Regression	ICR*	icr	fastICA
Principal component regression	PCR*	pcr	pls
Partial least square regression	PLSR*	pls	pls
Ridge regression	RIDGE*	ridge	elasticnet
Multiple layer perceptron	MLP	mlpWeightDecayML	RSNNS
Support vector machine (radial kernel)	SVM	svmRadial	kernlab
Random forest	RF	rf	randomForest
eXtreme gradient boosting	XGB	xgbLinear	xgboost
	Classification		
Regularized Discriminant Analysis	RDA	rda	klaR
Stochastic Gradient Boosting	GBM	gbm	gbm, plyr
Naïve bayes	NB	nb	klaR
Penalized Logistic Regression	PLR	plr	stepPlr
C5.0	–	C5.0	C50, plyr
Gaussian process with radial kernel	GP	gausprRadial	kernlab
Support vector machine (radial kernel)	SVM	svmRadial	kernlab
Random forest	RF	rf	randomForest
k-Nearest neighbors	KNN	knn	–
Multilayer perceptron	MLP	mlpWeightDecayML	RSNNS

* The ICR, PCR, PLSR, and RIDGE are not applicable for a single independent variable. The GLM is not applied to retrieve multiple independent variables

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Appendix 4-C. Internal and external testing result of growth phase discrimination. The models were respectively constructed with ten algorithms using average data each of six feature layers generated by RGB imagery

Algorithms	F1-score					
	CSMs	CSMt	NDTI	NExG	NGRDI	VARI
			Internal testing			
C5.0	0.88	0.71	0.71	0.76	0.69	0.76
GP	0.86	NA	0.78	0.78	0.77	0.75
GBM	0.85	0.67	0.80	0.65	0.82	0.78
KNN	0.85	0.65	0.78	0.72	0.71	0.74
MLP	0.84	0.65	0.62	0.65	0.76	0.73
NB	0.86	0.69	0.79	0.76	0.80	0.76
PLR	0.89	0.65	0.70	0.66	0.73	0.63
RDA	0.86	0.67	0.69	0.68	0.47	0.60
RF	0.85	0.57	0.76	0.65	0.69	0.70
SVM	0.87	0.70	0.78	0.74	0.75	0.77
			External testing			
C5.0	0.86	0.78	0.47	0.72	0.71	0.56
GP	0.87	NA	0.59	0.73	0.74	0.70
GBM	0.86	0.78	0.58	0.73	0.75	0.69
KNN	0.86	0.76	0.60	0.62	0.74	0.68
MLP	0.86	0.78	0.86	0.43	0.74	0.82
NB	0.85	0.65	0.63	0.72	0.76	0.67
PLR	0.86	0.76	0.71	0.55	0.73	0.72
RDA	0.86	0.76	0.51	0.58	0.66	0.63
RF	0.78	0.74	0.61	0.62	0.75	0.71
SVM	0.86	0.78	0.63	0.74	0.74	0.70

NA: not available

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Appendix 4-D. Internal and external testing result of growth phase discrimination. The models were respectively constructed with ten algorithms using percentile data each of six feature layers generated by RGB imagery

Algorithms	F1-score					
	CSMs	CSMt	NDTI	NExG	NGRDI	VARI
			Internal testing			
C5.0	0.92	0.80	0.77	0.77	0.87	0.80
GP	0.89	NA	0.82	0.75	0.87	0.87
GBM	0.92	0.83	0.79	0.80	0.88	0.86
KNN	0.88	0.92	0.87	0.76	0.82	0.83
MLP	0.92	0.84	0.86	0.73	0.78	0.86
NB	0.85	0.74	0.52	0.72	0.82	0.76
PLR	0.89	0.73	0.84	0.74	0.87	0.85
RDA	0.88	0.81	0.77	0.72	0.32	0.82
RF	0.96	0.85	0.80	0.80	0.87	0.82
SVM	0.92	0.84	0.82	0.79	0.80	0.87
			External testing			
C5.0	0.84	0.81	0.79	0.75	0.72	0.66
GP	0.86	NA	0.66	0.60	0.64	0.60
GBM	0.85	0.81	0.65	0.78	0.73	0.66
KNN	0.85	0.82	0.64	0.61	0.68	0.54
MLP	0.85	0.82	0.77	0.80	NA	0.58
NB	0.83	0.76	0.30	0.31	0.78	0.28
PLR	0.86	0.83	0.73	0.77	0.68	0.57
RDA	0.86	0.83	0.71	0.48	0.65	0.70
RF	0.86	0.78	0.59	0.77	0.73	0.68
SVM	0.87	0.82	0.64	0.60	0.72	0.55

NA: not available

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Appendix 4-E. Internal and external testing result of growth phase discrimination. The models were respectively constructed with ten algorithms using average data each of nine feature layers generated by MS imagery

Algorithms	F1-score								
	CSMs	CSMt	DATT	GNDVI	NDTI	NDVI	NExG	NGRDI	VARI
					Internal testing				
C5.0	0.85	0.85	0.72	0.78	0.79	0.78	0.79	0.88	0.87
GP	0.90	0.91	0.74	0.75	0.77	0.79	0.76	0.83	0.86
GBM	0.85	0.85	0.73	0.76	0.75	0.79	0.80	0.86	0.83
KNN	0.93	0.87	0.72	0.72	0.77	0.79	0.81	0.85	0.86
MLP	0.85	0.86	0.74	0.74	0.74	0.68	0.73	0.73	0.82
NB	0.86	0.86	0.73	0.73	0.79	0.77	0.79	0.82	0.82
PLR	0.90	0.88	0.76	0.75	0.73	0.70	0.80	0.62	0.69
RDA	0.82	0.83	0.47	0.73	0.67	0.69	0.79	0.70	0.64
RF	0.87	0.83	0.68	0.66	0.58	0.71	0.72	0.80	0.74
SVM	0.90	0.85	0.74	0.74	0.79	0.73	0.79	0.85	0.87
					External testing				
C5.0	0.73	0.74	0.87	0.93	0.91	0.92	0.62	0.68	0.75
GP	0.77	0.77	0.86	0.83	0.90	0.73	0.70	0.70	0.74
GBM	0.77	0.76	0.86	0.81	0.90	0.75	0.66	0.70	0.76

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KNN	0.77	0.76	0.84	0.78	0.82	0.74	0.69	0.69	0.76
MLP	0.75	0.74	0.80	0.90	0.89	0.83	0.82	0.90	0.90
NB	0.73	0.68	0.88	0.83	0.89	0.74	0.86	0.71	0.80
PLR	0.75	0.77	0.89	0.82	0.86	0.82	0.89	0.70	0.67
RDA	0.67	0.77	0.42	0.82	0.86	0.86	0.86	0.69	0.67
RF	0.73	0.76	0.72	0.71	0.70	0.69	0.68	0.72	0.73
SVM	0.77	0.77	0.83	0.77	0.90	0.67	0.69	0.71	0.78

Appendix 4-F. Internal and external testing result of growth phase discrimination. The models were respectively constructed with ten algorithms using percentile data each of nine feature layers generated by MS imagery

Algorithms	F1-score								
	CSMs	CSMt	DATT	GNDVI	NDTI	NDVI	NExG	NGRDI	VARI
					Internal testing				
C5.0	0.91	0.87	0.81	0.87	0.83	0.80	0.84	0.82	0.89
GP	0.87	0.88	0.79	0.85	0.79	0.82	0.83	0.89	0.85
GBM	0.92	0.92	0.83	0.86	0.81	0.84	0.81	0.87	0.87
KNN	0.89	0.88	0.80	0.79	0.84	0.80	0.83	0.85	0.86
MLP	0.91	0.86	0.79	0.85	0.82	0.75	0.81	0.86	0.87
NB	0.78	0.85	0.80	0.71	0.75	0.63	0.85	0.81	0.82
PLR	0.90	0.88	0.74	0.88	0.74	0.83	0.78	0.80	0.83
RDA	0.87	0.77	0.73	0.86	0.79	0.80	0.81	0.79	0.74
RF	0.89	0.91	0.83	0.85	0.83	0.81	0.86	0.85	0.83
SVM	0.89	0.87	0.72	0.86	0.77	0.86	0.84	0.85	0.86
					External testing				
C5.0	0.77	0.79	0.89	0.82	0.68	0.74	0.89	0.70	0.71
GP	0.73	0.81	0.85	0.83	0.88	0.77	0.75	0.65	0.70
GBM	0.74	0.81	0.87	0.84	0.75	0.79	0.83	0.59	0.56

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KNN	0.74	0.81	0.79	0.80	0.74	0.82	0.75	0.68	0.65
MLP	0.76	0.81	0.87	0.92	0.81	0.74	0.37	0.61	0.33
NB	0.58	0.76	0.85	0.80	0.87	0.49	0.72	0.59	0.66
PLR	0.73	0.80	0.83	0.83	0.91	0.80	0.45	0.47	0.55
RDA	0.71	0.81	0.80	0.88	0.94	0.88	0.88	0.93	0.63
RF	0.71	0.81	0.86	0.81	0.76	0.77	0.64	0.65	0.72
SVM	0.76	0.80	0.88	0.77	0.87	0.76	0.73	0.66	0.74

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Appendix 4-G. Internal and external testing result of LNC estimation. The models were respectively constructed with six algorithms using average data of each six feature layers generated by RGB imagery

Algorithms	RRMSE (%)					
	CSMs	CSMt	NDTI	NExG	NGRDI	VARI
			Internal testing			
GLM	20.71	8.30	11.65	12.55	15.14	6.95
KNN	11.65	14.23	8.13	11.00	20.73	11.73
MLP	18.57	9.48	15.12	8.03	7.96	12.12
RF	15.56	14.61	14.08	9.75	8.65	8.51
SVM	14.78	15.98	12.92	16.55	14.28	14.70
XGB	25.73	19.36	13.83	19.36	13.66	23.66
			External testing			
GLM	9.57	8.23	14.82	15.08	7.60	15.23
KNN	8.72	9.92	13.97	13.60	6.14	14.00
MLP	11.53	8.67	9.62	19.78	17.44	16.94
SVM	4.39	10.63	18.87	14.77	11.33	19.49
RF	9.58	11.44	16.26	12.61	8.74	17.85
XGB	3.39	11.58	20.28	16.88	14.37	19.17

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Appendix 4-H. Internal and external testing result of LNC estimation. The models were respectively constructed with nine algorithms using percentile data of each six feature layers generated by RGB imagery

Algorithms	RRMSE (%)					
	CSMs	CSMt	NDTI	NExG	NGRDI	VARI
			Internal testing			
ICR	11.98	7.69	11.36	11.93	17.05	17.95
KNN	16.65	13.14	9.66	14.26	12.95	13.17
MLP	13.49	17.71	6.30	7.33	22.82	10.73
PCR	11.92	16.52	19.70	19.38	12.62	22.47
PLSR	18.42	15.90	18.52	20.30	11.13	14.63
RF	20.50	9.32	12.73	8.69	10.07	16.66
RIDGE	16.66	18.98	11.36	NA	NA	5.43
SVM	9.17	11.08	18.55	18.05	16.44	16.79
XGB	17.69	17.26	14.22	14.13	12.78	14.89
			External testing			
ICR	10.14	NA	14.78	15.64	NA	15.38
KNN	8.97	12.39	13.95	15.63	5.24	14.38
MLP	7.04	NA	23.12	NA	16.27	19.00
PCR	6.78	NA	14.88	16.63	NA	15.00
PLSR	6.32	NA	14.90	16.63	NA	15.08
RF	4.24	7.46	17.64	17.56	12.66	16.64
RIDGE	8.41	NA	12.90	NA	NA	NA
SVM	7.00	7.12	14.57	11.42	8.69	11.72
XGB	4.56	NA	18.76	21.43	22.80	9.56

NA: not available

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Appendix 4-I. Internal and external testing result of LNC estimation. The models were respectively constructed with six algorithms using average data of each nine feature layers generated by MS imagery

Algorithms	RRMSE (%)								
	CSMs	CSMt	DATT	GNDVI	NDTI	NDVI	NExG	NGRDI	VARI
					Internal testing				
GLM	14.94	13.06	NA	12.43	16.19	8.56	10.78	15.98	14.58
KNN	16.00	11.36	16.71	22.20	15.99	18.59	11.47	14.16	12.97
MLP	14.24	13.77	20.23	15.66	11.49	12.94	11.31	22.05	12.63
RF	12.38	13.39	17.25	19.69	21.10	19.36	16.97	13.38	14.98
SVM	17.38	14.31	16.49	15.66	15.40	18.89	15.67	13.80	14.04
XGB	14.97	21.06	16.83	13.94	16.09	16.45	20.31	10.10	14.34
					External testing				
GLM	10.26	7.37	8.55	18.98	20.16	19.90	16.59	17.23	14.12
KNN	11.21	6.60	11.89	15.51	15.42	16.05	17.41	13.52	13.52
MLP	9.50	10.88	9.21	12.53	NA	24.63	17.78	16.75	33.03
RF	9.20	8.77	12.78	17.90	18.90	18.08	15.67	19.46	17.17
SVM	7.63	9.67	12.44	10.58	16.59	15.41	11.43	16.62	15.21
XGB	8.20	14.59	15.83	23.90	25.11	21.94	NA	25.85	23.23

NA: not available

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Appendix 4-J. Internal and external testing result of LNC estimation. The models were respectively constructed with nine algorithms using percentile data of each nine feature layers generated by MS imagery

Algorithms	RRMSE (%)								
	CSMs	CSMt	DATT	GNDVI	NDTI	NDVI	NExG	NGRDI	VARI
					Internal testing				
ICR	15.64	17.98	NA	14.86	11.02	16.72	9.01	13.28	21.41
KNN	21.33	13.39	13.57	17.17	8.53	17.41	15.21	9.73	12.67
MLP	18.11	NA	7.53	21.40	15.64	15.45	19.68	21.96	13.48
PCR	14.52	13.86	12.59	14.03	6.02	13.31	13.95	8.98	36.59
PLSR	15.77	18.05	14.01	8.50	11.83	15.04	17.01	19.79	14.83
RF	13.76	17.12	19.44	18.01	12.10	14.35	10.00	12.12	14.36
RIDGE	10.96	18.37	NA	10.66	NA	16.46	11.76	15.94	13.52
SVM	16.17	15.82	16.79	18.07	12.62	10.77	21.19	15.10	14.80
XGB	20.39	8.27	19.55	18.61	8.11	24.82	13.93	14.67	10.21
					External testing				
ICR	13.71	NA	8.85	NA	28.42	NA	16.73	17.75	NA
KNN	11.34	7.87	11.49	13.42	14.21	15.58	15.44	14.20	14.27
MLP	5.88	11.77	10.29	22.38	27.72	12.37	14.69	21.39	17.79
PCR	13.44	NA	11.90	NA	18.90	NA	NA	NA	9.37

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PLSR	13.04	NA	11.70	NA	17.69	NA	NA	NA	18.24
RF	10.17	7.28	9.69	18.57	19.03	16.56	11.30	16.95	18.76
RIDGE	12.44	NA	NA	NA	NA	NA	NA	NA	NA
SVM	8.23	8.83	11.29	12.26	14.15	10.58	13.49	11.08	10.29
XGB	9.83	NA	22.15	11.96	22.88	16.19	15.16	NA	16.38

NA: not available

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Conclusion and future perspectives

In Chapter 1, the relationship between air temperature and rice leaf appearance had been investigated to promote the application of growing degree days (GDD) on the detection of rice panicle initiation. First, a novel measure modified from Haun's scale (HAUN), which is termed as leaf appearance scale (LAS), was proposed to solve the non-continuity of HAUN. Besides, the LAS also possesses advantages compares to the HAUN, such as consistency of the scale range among different cultivars and weather conditions. Accordingly, the daily growth rate (d.LAS) could be easily defined as the difference of LAS between two consecutive days. The base temperature (T_{base}), optimum temperature (T_{opt}), and upper limit temperature (T_{up}) were evaluated by retrieving d.LAS with six types of daily thermal units (DTU) in different growth stages. The linearity between d.LAS and DTU in each growth stages was confirmed through simple linear regression analysis. Although the optimal T_{base}, T_{opt}, and T_{up} for each growth stages were inferenced with statistical aspect and physiological inferences, fine-tuning the DTUs calculation with these parameters according to different growth stages failed to unify the GDDs between cold and warm season. On the other hand, when the growth stages were directly classified with GDDs without parameters tuning for each growth stages, GDD_{mc2} and GDD_{trial} unified their values between growing seasons.

In Chapter 2, remote sensing technique was introduced to solve the limitation of GDD method ― lack of site-specific availability. A commonly used vegetation index, normalized difference vegetation index (NDVI), was applied with quadratic polynomial regression model and first-order differentiation of the function. Several NDVIs, which were calculated using different combinations of wavelength, were found to be competent for PI stage determination, such as NDVI (700, 720), NDVI (700, 730), NDVI (708, 730),

NDVI (700, 760), NDVI (708, 760), and NDVI (708, 800). Although the exact values of NDVI had great variabilities among different planting dates, the trend is stable — PI occurred when the slope of NDVI equal to zero. According to the relationship between PI and NDVI slope, the PI estimation for a specific plot could be achieved via determining the proportion of zeros. However, the accuracy and stability of NDVI were insufficient for real-time detection due to fluctuations in NDVI values that resulted in uncertainty in the current day's slope. In conclusion, while this approach is suitable for site-specific and historical screening, it is difficult if not impossible to use for real-time PI detection purposes.

In Chapter 3, machine learning technique was introduced to address the bottleneck. The classification result showed that applying the feature extraction-support vector machine with radial bases kernel (FE-SVM) to the full spectrum of hyperspectral data provided accurate and precise discrimination of growth phases. The extracted features (PC1, PC2, and PC3) appear to correlate globally with both physical and biochemical traits, primarily with leaf cover area (LCA) and leaf nitrogen accumulation (LNA). LCA and LNA increased gradually before PI, peaked during PI, and decreased progressively after PI. Although this approach has high instrumental costs and is not suitable for large-scale data collection, it may become more feasible in the future as hyperspectral instruments and ground-based automation screening platforms become more affordable.

In Chapter 4, unmanned aerial vehicles mounting with either RGB camera or a multispectral camera were tested for large-scale screening purpose. This study also works on leaf nitrogen content (LNC) estimation at the same time of PI detection. The results suggested that a low cost RGB camera is sufficient for both purposes. When discriminating growth phases, using the percentile data of spatial crop surface model (CSMs) manifested greater accuracy and reliability. When estimating LNC, using average data of normalized difference texture index (NDTI) is preferable.

When considering the cost and ease of promoting high-throughput phenotyping (HTP) technologies to farmers, the GDD method is relatively more feasible compared to remote sensing approaches. By introducing the critical GDD to farmers, nitrogen topdressing timing can be systematically conducted by referring to the temperature records of the nearest meteorological station. However, when advanced practices such as site-specific management and precise resource input are required, remote sensing techniques become essential for detecting the conditions of specific plots or even the status of individual plants. By promoting HTP practices to the infrastructure, experience-driven field practices can be incrementally shifted to data-driven management. This type of systematic field management reduces reliance on the farming experience of older generations and promotes the development and sustainability of agricultural production.

Further efforts are required to investigate the leaf nitrogen content threshold in different cultivars, where to maximize the grain yield or to compromise the yield and quality. Besides, it is necessary to develop a practical protocol for applying appropriate amount of nitrogen according to the estimated LNC. Finally, integrating with the on-site scenario, such as irrigation schedule, chemical and fertilizer applications, planting density, and soil properties, into these approaches may be unavoidable for infrastructure application in the future.

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